The sharp iid Berry–Esseen inequality: the mathematical argument and its certificates*

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Abstract

We give a computer-assisted proof of the sharp iid Berry–Esseen inequality with constant $(3+\sqrt{10})/(6\sqrt{2\pi})$. The argument excludes a first failing index using a finite sharp base and two local comparisons. Extremal contact and a nonnegative support-gap defect force a possible maximizer near a two-point law; conditional binomial counts then measure the effect of its remaining fluctuations. Fourier bounds and exact certificates establish the quantitative inequalities on complete parameter regions. We prove their analytic reductions and explain certificate construction, acceptance and independent reproduction. Equality in the uniform local comparison also gives a strict bound for each sample size at least 47.

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1 The problem and the proof strategy

Let $\mathcal{P}$ be the probability laws on $\mathbb{R}$ with mean zero, variance one and finite third absolute moment $\rho(P)=\mathbb{E}_P|X|^3$. For independent copies with this same law, write $S_n=X_1+\cdots+X_n$ and define

$$\Delta_n(P)=\sup_{H\in\mathbb{R}}|\mathbb{P}_P(S_n\le H)-\Phi(H/\sqrt{n})|,\qquad C_n=\sup_{P\in\mathcal{P}}\frac{\sqrt{n}\Delta_n(P)}{\rho(P)}.$$

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The threshold H is in the units of the raw sum; $H/\sqrt{n}$ is its normal-standardized value. The Berry–Esseen inequality [1–3] asserts a universal finite upper bound on the C_n. The sharp-constant problem asks whether this bound can be taken to be

$$C_{\mathrm{E}} = \frac{3+\sqrt{10}}{6\sqrt{2\pi}} = 0.409732\ldots.$$

The lower bound comes from one fixed skew Bernoulli law, rather than from a sequence of progressively more complicated distributions. The difficulty is to bound every possible marginal, uniformly in the sample size.

Theorem 1.1 (Sharp iid Berry–Esseen bound)**.** *For every $P \in \mathcal{P}$ and every integer $n \geq 1$,*

$$\Delta_n(P) \leq \frac{C_{\mathrm{E}}\rho(P)}{\sqrt{n}}.$$

The constant is optimal: $\sup_{n\geq 1} C_n = C_{\mathrm{E}}$. *Moreover $C_n < C_{\mathrm{E}}$ for every $n \geq 47$.*

The proof combines the analytic arguments below with the finite inequalities specified by their certificates. Section 8 assembles these results and proves strictness using the equality case of the uniform local exclusion.

1.1 The three parts of the upper-bound argument

The proof first establishes $C_j \leq C_{\mathrm{E}}$ for $1 \leq j \leq 42$. Suppose a later failure exists and let n be its least index. Then both deleted predecessors satisfy $C_{n-1}, C_{n-2} \leq C_{\mathrm{E}}$. A Gaussian one-step estimate, proved in Section 3, gives

$$0 < n(C_n - C_{\mathrm{E}}) < \frac{3003}{10250} < 0.3, \qquad C_n < \frac{73}{175}, \qquad n(C_{n-1} - C_n)_+ = 0.$$

Two local comparisons exclude precisely such an extremizer: one covers the finite range $43 \leq n \leq 48$, and one is uniform for $n \geq 49$. Thus the finite base and these two comparisons prove the upper bound at every index. The role of the recurrence is to supply the hypotheses of the local comparisons, rather than to propagate the sharp constant by itself.

The mathematical object shared by the comparisons is a nonnegative support-gap defect. At a fixed index, the optimal ratio is attained. Its first variation gives a piecewise cubic function which bounds the conditional probability $y \mapsto \mathbb{P}(S_{n-1} \leq H - y)$ and agrees with it on the support. Integrating this contact relation against the zero-bias density gives a conservation identity. Its nonnegative terms measure the score above C_{E}, the departure from the extremal two-point geometry, and the probability entering support gaps. Its remainder is bounded by Fourier estimates.

Small remainder therefore forces the same extremizing law near two clusters. Conditioning on the signs of its summands gives an exact binomial count plus centered conditional noise. The local comparison keeps this noise and shows that it cannot improve the score enough to violate the sharp bound. At zero noise, the actual law is Bernoulli and a published binomial theorem applies. At positive noise, the errors must remain proportional to the noise, however small it is.

The smallest indices require different quantitative estimates, but use the same logic: first acquire a valid support and moment domain, then majorize a probability on that domain. No step assumes that an unrestricted extremizer has two atoms, or replaces the iid marginal by an optimizer of a moment relaxation.

1.2 Where the computer enters

The reductions leave explicit inequalities on bounded parameter regions. These include polynomial majorants, enclosures of Fourier integrals, and inequalities in the probability, noise and phase of a binomial count. A certificate supplies finite data allowing these inequalities to be checked on the whole region. Typical data are rational polynomial coefficients and a finite tree of closed boxes. The checker proves a valid inequality or an exclusion on each leaf and verifies that no part of the root domain was lost.

Two mathematical questions precede this computation: why every possible witness belongs to the specified domain, and why the checked inequality gives the desired probability bound. Both are proved in the text. The certificates supply the remaining finite arithmetic. They are not samples from a class of probability laws. Nor does numerical precision alone make them rigorous: rational arithmetic and proved outward enclosures control every rounding error.

Finding a certificate and verifying it are different tasks. For some majorants, floating-point linear programming proposes coefficients; for enclosure and count bounds, adaptive subdivision proposes a finite proof tree. These procedures may fail and need not be trusted. The acceptance program reconstructs the precise inequality, checks all of its cells using exact arithmetic, and rejects unresolved cases. Section 4 explains these constructions and the common soundness argument. The reproduction package contains the large coefficient arrays and complete checking programs. Their size does not enlarge the mathematical assumptions of the proof.

1.3 Published inputs and sharpness

For clarity, the external probability estimates used below are stated here with their full law class.

Proposition 1.2 (Affine Berry–Esseen estimates)**.** *For every $P \in \mathcal{P}$ and integer $n \geq 1$,*

$$\sqrt{n}\Delta_n(P) \leq \min\left\{0.4690\rho,\ 0.3322(\rho + 0.429),\ 0.3031(\rho + 0.646)\right\}.$$

These are the iid estimates of Shevtsova [9]. Standardizing the summands converts the total-variance normalization there into the displayed one. Only a finite third absolute moment is assumed. A putative large normalized score consequently has an explicit upper bound on ρ, which starts several support arguments. All terminating decimals in this paper denote the corresponding rational numbers.

Proposition 1.3 (Finite binomial comparison and the lower limit)**.** *For $0 < p < 1$, let $X = (B - p)/\sqrt{p(1-p)}$ with $B \sim \mathrm{Bernoulli}(p)$. At every finite $n \geq 1$ its normalized discrepancy is strictly smaller than C_{E}. For $p_* = (4 - \sqrt{10})/2$ and this fixed marginal,*

$$\lim_{n\to\infty} \frac{\sqrt{n}\Delta_n(P_*)}{\rho(P_*)} = C_{\mathrm{E}}.$$

The finite assertion is the binomial result of Schulz [7, Theorem 1]; its fixed-law limit is Esseen's lattice asymptotic [4], in the normalization recorded in Schulz [7, Remarks 4.1–4.2]. Indeed, with $q_* = 1 - p_*$,

$$\rho(P_*) = \frac{p_*^2 + q_*^2}{\sqrt{p_* q_*}}, \qquad \frac{3 + q_* - p_*}{6\sqrt{2\pi}(p_*^2 + q_*^2)} = C_{\mathrm{E}}.$$

Thus every coefficient smaller than C_{E} fails for a sufficiently large sum of this one law. No interchange of a supremum over laws with a limit is needed. Together with the upper bound, this proves that the universal constant is sharp.

The present argument reorganizes the earlier certificate proof [6]. Its purpose is to give each analytic reduction once and to expose the precise interfaces between those reductions and the finite checks. The classical zero-bias transform [5] explains the conservation identity; explicit contact and same-law domain control are needed to apply it here. The final result concerns the optimal iid constant, not eventual exact two-point optimality.

2 Extremizers, two-point geometry and conservation

Throughout, $X, X_1, X_2, \dots$ are iid, $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $\rho = \mathbb{E}|X|^3 < \infty$, and $S_n = \sum_{i=1}^n X_i$. Write $\beta = \mathbb{E}[X|X|]$, $\gamma = \mathbb{E}X^3$, and $\phi = \Phi'$. A threshold H always refers to the raw sum; $z = H/\sqrt{n}$ is its standardized value. Define the ratio and an auxiliary excess by

$$r(P,H) = \frac{\sqrt{n}}{\rho}\{\mathbb{P}(S_n \le H) - \Phi(z)\}, \qquad a_c(P,H) = \sqrt{n}\{\mathbb{P}(S_n \le H) - \Phi(z)\} - c\rho.$$

We maximize the ratio. The auxiliary quantity a_c is used only to state an identity for arbitrary coefficients and off-contact laws.

2.1 Ratio attainment and contact, including atoms

Theorem 2.1. *For every $n \ge 2$, C_n is attained as a positive closed-CDF ratio by a compactly supported standardized law and a real raw threshold H. At such a pair put $c = C_n$, $F = \mathbb{P}(S_n \le H)$, $h(y) = \mathbb{P}(S_{n-1} \le H - y)$ and $\phi_z = \phi(z)$. Then*

$$I_c(y) = n^{3/2}(h(y) - F) + ny\phi_z + \frac{H\phi_z + 3c\rho}{2}(y^2 - 1) - c(|y|^3 - \rho - 3y\beta) \le 0, \tag{1}$$

$$Q(y) = F + \frac{H\phi_z + c\rho}{2n^{3/2}} - \frac{n\phi_z + 3c\beta}{n^{3/2}}y - \frac{H\phi_z + 3c\rho}{2n^{3/2}}y^2 + \frac{c}{n^{3/2}}|y|^3. \tag{2}$$

Thus $\Delta = Q - h \ge 0$ everywhere and $\Delta = 0$ on the entire closed support. No sign assumption on H is needed.

Proof. Reflection takes a negative error to a positive closed error because

$$\mathbb{P}(-S_n \le -H) - \Phi(-z) = \Phi(z) - \mathbb{P}(S_n < H) \ge \Phi(z) - \mathbb{P}(S_n \le H).$$

Hence $C_n = \sup r(P,H)$. Since $\rho \ge 1$, $C_n \le \sqrt{n}$; symmetric Bernoulli at $H = n$ shows $C_n > 0$. Choose $r_j \uparrow C_n$ with $r_j \ge C_n/2$. Then $\rho_j \le 2\sqrt{n}/C_n$, and the positive discrepancies are bounded away from zero. Chebyshev on the negative tail and the normal upper tail on the positive side bound H_j. Pass to $P_j \Rightarrow P$, $H_j \to H$, $\rho_j \to b$. A common third-moment ceiling M bounds the first- and second-moment tails by M/K^2 and M/K, so the limit retains mean zero and variance one. Lower semicontinuity gives $\rho(P) \le b$. For the sum laws, the closed-set inequality, first at $H + t$ and then as $t \downarrow 0$, gives

$$\mathbb{P}(S_n \le H) - \Phi(H/\sqrt{n}) \ge \limsup_j \{\mathbb{P}(S_{n,j} \le H_j) - \Phi(H_j/\sqrt{n})\} = C_n b/\sqrt{n}.$$

Thus $r(P,H) \ge C_n b/\rho(P) \ge C_n$, proving attainment and $\rho(P) = b$. This does not assume uniform integrability of third moments.

Contaminate the raw law by $(1-\epsilon)P + \epsilon\delta_y$, then recenter and rescale. Its mean is ϵy, variance $v_\epsilon = 1 + \epsilon(y^2 - 1) - \epsilon^2 y^2$, and centered third absolute moment has derivative $|y|^3 - \rho - 3y\beta$. Dominated differentiation uses $3|y|(|x| + |y|)^2$. Evaluate the standardized law at the image of the *fixed raw event* $S_n \le H$. Its probability is a polynomial in ϵ, with derivative $n(h(y) - F)$ even at an atom collision. The normal argument $(H - n\epsilon y)/\sqrt{nv_\epsilon}$

has derivative $-\sqrt{n}y - H(y^2-1)/(2\sqrt{n})$. The ratio derivative is $I_c(y)/\rho$, and maximality makes it nonpositive. Furthermore $\int I_c\,dP = 0$. Since the closed-event function h is upper semicontinuous, strict negativity at a support point would hold on a neighborhood of positive mass. Thus $I_c = 0$ on the closed support. Finally $I_c(y) \le -c|y|^3 + O(y^2 + |y| + 1)$, with constants depending on this pair, excludes both sufficiently distant rays. □

At a ratio maximizer, put $D_0 = F - \Phi(z)$ and $\lambda = D_0/\rho = C_n/\sqrt{n}$. The majorant used in the finite calculations is simply

$$nQ(y) = \lambda|y|^3 - \frac{z\phi_z + 3D_0}{2}y^2 - (\sqrt{n}\phi_z + 3\lambda\beta)y + nF + \frac{z\phi_z + D_0}{2}. \tag{3}$$

Thus its equality on support is $nh = nQ$; it is not a majorant for a different marginal. If $A \ge C_{n-1}$, this equality implies, at every support point t,

$$0 \le \mathcal{H}_n(H,t) + C_n\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} + \rho s_n n(A - C_n), \tag{4}$$

where

$$s_n = \sqrt{n/(n-1)}, \quad e_n = n(s_n - 1) - \tfrac{1}{2},$$
$$\mathcal{H}_n(H,t) = n^{3/2}\left\{\Phi\left(\frac{H-t}{\sqrt{n-1}}\right) - \Phi\left(\frac{H}{\sqrt{n}}\right)\right\} + nt\phi_z + \tfrac{H}{2}(t^2-1)\phi_z.$$

Indeed substitute $h(t) \le \Phi((H-t)/\sqrt{n-1}) + A\rho/\sqrt{n-1}$ in (1) and use $\sqrt{n}(F - \Phi(z)) = C_n\rho$. The signed difference $A - C_n$ may be negative; replacing it by a nonnegative upper bound gives a weaker permitted inequality.

For later induction, if $C_j \le c$ for $j < N$ and some $n \ge N$ has $C_n \ge c$, the least such n has an attained ratio pair with $C_n \ge c$ and $C_j \le c$ for every $j < n$. This includes equality at the selected index; it requires no supremum over varying sample sizes.

2.2 One sign normalization and its moment constraints

Put $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $v = pq$, and $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$. Assign a zero atom to the nonpositive class. For $B = \mathbf{1}_{\{X>0\}}$, let $V = \mathbb{E}[X \mid B]$, $W = X - V$, and $\theta = \mathbb{E}W^2$. Then $V = -m/q$ or m/p, and

$$\ell = \frac{m}{pq} = \sqrt{\frac{1-\theta}{pq}}, \quad U = W/\ell, \quad X = \ell(B + U - p) = \frac{B + U - p}{\sqrt{pq + s}}, \quad s = \mathbb{E}U^2 = \frac{pq\theta}{1-\theta}. \tag{5}$$

Here $0 < p, q < 1$, $0 \le \theta < 1$ and $\mathbb{E}[U \mid B] = 0$. Write U_i for the conditional laws, $w_i = \mathbb{E}U_i^2$, $\mu_i = \mathbb{E}U_i^3$, and $\lambda = ns$. Thus $s = qw_0 + pw_1$; if $s > 0$, $\alpha = pw_1/s \in [0,1]$ gives $w_0 = s(1-\alpha)/q$, $w_1 = s\alpha/p$. If $s = 0$, both conditional laws are zero and X is standardized Bernoulli. The exact common-marginal identities are

$$\begin{aligned}
\rho/\ell^3 &= v(1-2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1,\\
\gamma/\ell^3 &= v(q-p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1,\\
\beta/\ell^2 &= v(q-p) + pw_1 - qw_0.
\end{aligned} \tag{6}$$

They follow by expanding $\ell^3(p - U_0)^3$ and $\ell^3(q + U_1)^3$ on the two sign classes and using conditional centering. In particular the two cubic moments in the first two lines cannot be selected independently. Higher moments require an established support bound or another explicit integrability hypothesis.

Lemma 2.2. *Every standardized law with finite third moment satisfies*

$$\rho \geq \frac{\gamma^2+2}{\sqrt{\gamma^2+4}}, \qquad |\beta|<1, \qquad \rho \geq \frac{1+\beta^2}{\sqrt{1-\beta^2}}.$$

Proof. For the first inequality set $s_0=\sqrt{\gamma^2+4}$, $a_0=(s_0-\gamma)/2$, $b_0=(s_0+\gamma)/2$ and $c_0=\gamma(\gamma^2+6)/s_0^3$. The polynomial on each half-line

$$w(y)=|y|^3-c_0y^3-12y^2/s_0^3+6\gamma y/s_0^3+4/s_0^3$$

has the factorizations, for $y,t\geq 0$,

$$w(y)=\frac{2a_0^2}{s_0^3}(y-b_0)^2[2+(a_0+3b_0)y], \quad w(-t)=\frac{2b_0^2}{s_0^3}(t-a_0)^2[2+(3a_0+b_0)t].$$

Taking expectations gives $0\leq \mathbb{E}w=\rho-(\gamma^2+2)/s_0$. For the second inequality let $a_\pm=\mathbb{E}X_\pm^2=(1\pm\beta)/2>0$. Cauchy gives $m^2/a_+ + m^2/a_-\leq 1$, hence $2m\leq\sqrt{1-\beta^2}$, and $m\rho\geq a_+^2+a_-^2=(1+\beta^2)/2$. These imply the result; a zero atom only strengthens the first inequality. □

2.3 A defect measuring departure from two points

Let $K=\operatorname{supp}P$ and $\tau(t)=\mathbb{E}[X\mathbf{1}_{\{X\geq t\}}]$. Centering and signwise Tonelli give

$$\tau\geq 0, \qquad \int\tau=1, \qquad 2\int|t|\tau(t)\,dt=\rho.$$

On every bounded gap (u,v) of K, τ has a constant interior value $\tau_{u,v}$. Set

$$J=\sum_{(u,v)}\tau_{u,v}(v-u)^2, \qquad \sigma=\sqrt{10}, \qquad D=(3+\sigma)\rho-\gamma-3J. \tag{7}$$

The gaps are countable; the next bounds justify their sum even for unbounded support.

Theorem 2.3. *For every such law,*

$$\gamma^2+4m^2\leq\rho^2, \qquad J\leq\rho+2m, \qquad D=(\sigma\rho-\gamma-6m)+3(\rho+2m-J)\geq 0.$$

Equality $D=0$ occurs exactly for $X=(B-p_*)/\sqrt{p_*q_*}$, where $p_*=(4-\sqrt{10})/2$ and $q_*=1-p_*$. If $0\in K$, then $D\geq(\sigma-1)\rho$. Otherwise write the central gap as $(-A_-,A_+)$ and define

$$T_\pm=\mathbb{E}X_\pm^3-mA_\pm^2, \quad R_\pm=\mathbb{E}X_\pm^2-mA_\pm, \quad J_o=J-m(A_-+A_+)^2.$$

The quantities $T_\pm,R_\pm,J_o$ are nonnegative, and

$$J_o^\pm\leq T_\pm-2A_\pm R_\pm, \tag{8}$$

$$D=m\{\sqrt{\sigma+1}A_- -\sqrt{\sigma-1}A_+\}^2+(\sigma-1)T_+ +(\sigma+1)T_- +3(T_++T_--J_o). \tag{9}$$

All four terms in the last expression are nonnegative. Moreover $1/(2\rho)\leq m\leq 1/2$; if $D<2$, then $T_++T_-\leq D/2$ and $A_-+A_+\geq\sqrt{2-D}$.

Proof. Put $a_\pm=\mathbb{E}X_\pm^2$, $b_\pm=\mathbb{E}X_\pm^3$. The signwise Cauchy inequalities give $a_\pm^2\leq mb_\pm$ and $m^2\leq a_+a_-$. Thus $b_+b_-\geq m^2$, proving the circle inequality. A gap lying on one side of zero satisfies $(v-u)^2\leq 2\int_u^v|t|\,dt$. A central gap has weight m and excess $2mA_-A_+$ over this integral; since $0\leq\mathbb{E}[(X+A_-)(X-A_+)]=1-A_-A_+$, summation gives $J\leq\rho+2m$. Without a central gap it gives $J\leq\rho$. Cauchy in $\mathbb{R}^2$ now gives $\gamma+6m\leq\sigma\rho$. Equality requires

equality in both signwise Cauchy bounds and in $m^2 \le a_+ a_-$: each sign is a single nonzero atom and there is no zero atom. Writing its positive probability as p, the remaining equality is $3(1-2p) = 2p(1-p)$, whose unique solution in $(0,1)$ is p_*. Conversely this two-point law gives equality.

For a positive outer gap, $(v-u)^2 \le 2\int_u^v (t - A_+)\,dt$; summing yields $J_o^+ \le \mathbb{E}[X_+(X_+ - A_+)^2] = T_+ - 2A_+R_+$. The negative side is identical after reflection. Expansion using $\sqrt{\sigma+1}\sqrt{\sigma-1} = 3$ proves (9). Finally Cauchy gives $1 \le 2m\rho$ and $2m \le 1$. Since $\sigma - 1 > 2$, $T_+ + T_- \le D/2$; hence $m(A_-^2 + A_+^2) = \rho - T_+ - T_- \ge 1 - D/2$, which gives the last claim. All gap summations use nonnegative Tonelli; endpoint atoms change no Lebesgue integral. □

For the standardized Bernoulli law itself,

$$J = 1/\sqrt{pq}, \quad \rho_B = (1-2pq)/\sqrt{pq}, \quad \gamma_B = (q-p)/\sqrt{pq}, \quad D_B = (6+2\sigma)(p-p_*)^2/\sqrt{pq}.$$

Thus the defect charges a quadratic displacement of p as well as conditional noise. The following bound charges them simultaneously.

Lemma 2.4 (Joint defect budget)**.** *In a central gap put $\bar{A}_- = \ell p$, $\bar{A}_+ = \ell q$, $\Delta_\pm = \bar{A}_\pm - A_\pm \ge 0$, $a_0 = \Delta_-/\ell$, $a_1 = \Delta_+/\ell$, $d = a_0 + a_1 < 1$, and $w = 1 - \theta = pq\ell^2$. Then $U_0 \le a_0$, $U_1 \ge -a_1$, and*

$$\begin{aligned} D \ge{}& \ell w(6+2\sigma)(p-p_*)^2 + 6\ell w\{d(1-d) + a_0 a_1\} \\ &+ \mathbb{E}[\{(\sigma-1)(X + 2\bar{A}_+) + 6A_+\}W^2; X > 0] \\ &+ \mathbb{E}[\{(\sigma+1)(-X + 2\bar{A}_-) + 6A_-\}W^2; X < 0]. \end{aligned} \tag{10}$$

In particular, with $\theta_\pm = \mathbb{E}[W^2; \pm X > 0]$,

$$\begin{aligned} D \ge{}& \frac{(6+2\sigma)(1-\theta)^{3/2}}{\sqrt{pq}}(p-p_*)^2 + 6m\{(A_- + A_+)(\Delta_- + \Delta_+) + \Delta_-\Delta_+\} \\ &+ 3(\sigma+1)A_+\theta_+ + 3(\sigma+3)A_-\theta_-. \end{aligned} \tag{11}$$

Proof. Insert (8) in the definition of D. The result is

$$D \ge (\sigma-1)M_{3,+} + (\sigma+1)M_{3,-} + 6A_+M_{2,+} + 6A_-M_{2,-} - 6m(A_-^2 + A_-A_+ + A_+^2),$$

where $M_{j,\pm} = \mathbb{E}X_\pm^j$. Conditional centering gives $M_{2,+} = m\bar{A}_+ + \mathbb{E}[W^2; X > 0]$ and $M_{3,+} = m\bar{A}_+^2 + \mathbb{E}[(X + 2\bar{A}_+)W^2; X > 0]$, and the reflected identities. Substitute $A_- = \ell(p - a_0)$, $A_+ = \ell(q - a_1)$; the deterministic part is precisely the first line of (10). All kernels are nonnegative. Bounding them below by $3(\sigma+1)A_+$ and $3(\sigma+3)A_-$ proves (11). Finite third moments suffice throughout. □

2.4 Conservation: one identity, with and without contact

For a compactly supported law retain $a = a_c$, Q, h, Δ from above, and put $\nu = \mathcal{L}(H - S_{n-1})$. Over its bounded support gaps define

$$J_1 = \sum \tau_{u,v} \int_u^v (t-u)t\,dt, \quad J_2 = \sum \tau_{u,v} \int_u^v (t-u)t|t|\,dt, \quad L = \sum \tau_{u,v} \int_{[u,v)} (t-u)\,d\nu(t) \ge 0.$$

The half-open interval is essential: an atom at v belongs to $h(v)$, whereas an atom at u receives zero weight. If $K \subset [-R, R]$, then $J \le 2R$, $|J_1| \le RJ/2$, $|J_2| \le R^2 J/2$ and $L \le 2R$. Let $M_4 = \mathbb{E}X^4$, $M_5 = \mathbb{E}|X|^5$, and set

$$K_q = M_5/4 - \rho M_4/2 - 3\beta\gamma/2 + \rho/2, \quad K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2, \quad \Omega = J_1 + (M_4 - 3)/6.$$

For $Z = S_n/\sqrt{n}$, a standard normal Z_0, and $q_z(x) = (x-z)|x-z|$, put

$$\varepsilon = \mathbb{E}q_z(Z) - \mathbb{E}q_z(Z_0) - 2\gamma\phi_z/(3\sqrt{n}), \qquad E_n = -n^{3/2}\varepsilon/2,$$

$$\mathcal{R}_c = c(3\beta J/2 + 3\rho J_1 - 3J_2 - K_q) + H(cK_1 + \phi_z\Omega) - H^2(c\rho + \gamma\phi_z/2) + E_n. \tag{12}$$

Define the residuals

$$\xi = \mathbb{E}[X\Delta(X)], \qquad R_K = \int_K \tau(t)\Delta(t)\,dt + \sum_{(u,v)} \tau_{u,v}(v-u)\Delta(v).$$

The integral on K is a Lebesgue integral, not a P-integral. All terms converge absolutely since Δ is bounded on the convex hull and $\sum \tau_{u,v}(v-u) \le 1$.

Theorem 2.5. *For every compactly supported standardized law and every $c > 0$, $H \in \mathbb{R}$,*

$$(n+H^2)a + nc\rho - \frac{n\phi_z}{6}(\gamma + 3J) + n^{3/2}L = \mathcal{R}_c + n^{3/2}(R_K - H\xi). \tag{13}$$

At a ratio maximizer set $c = C_n$, so $a = 0$. With $e = C_n - C_{\mathrm{E}}$ and $w = H^2/(2n)$, the residuals vanish and

$$\mathcal{R}_{C_n} = ne\rho + nC_{\mathrm{E}}\rho(1-e^{-w}) + n\phi(0)e^{-w}D/6 + n^{3/2}L. \tag{14}$$

All four terms are nonnegative when $C_n \ge C_{\mathrm{E}}$, including equality.

Proof. Write $\mathcal{A} = \int \tau(Q-h)$. On a gap, $h(t) = h(v) + \nu([t,v))$. Fubini and integration of Q' therefore give

$$\mathcal{A} = R_K + \frac{\phi_z J}{2\sqrt{n}} + \frac{3c\beta J/2 + (H\phi_z + 3c\rho)J_1 - 3cJ_2}{n^{3/2}} - L. \tag{15}$$

For absolutely continuous g, signwise Fubini gives $\int \tau g' = \mathbb{E}[Xg(X)]$. Thus $\int t\tau = \gamma/2$, $\int t^2\tau = M_4/3$, $\int |t|^3\tau = M_5/4$. Applying the same identity conditionally to $g_s(x) = \min(s+x, H)$ and using exchangeability gives

$$\int \tau h = \frac{1}{n}\mathbb{E}[S_n \min(S_n, H)] = T, \qquad T = \mathbb{E}[(Z^2 - zZ)\mathbf{1}_{\{Z \le z\}}].$$

The integrand at the threshold atom is zero. Consequently

$$\mathcal{A} = F - \frac{\gamma\phi_z}{2\sqrt{n}} + \frac{H\phi_z(3-M_4)/6 + cK_q}{n^{3/2}} - T.$$

If $M = \mathbb{E}[Z\mathbf{1}_{\{Z\le z\}}]$, integrating $X(Q-h)$ against P gives

$$M + \phi_z = -\gamma H\phi_z/(2n) + cK_1/n - \sqrt{n}\xi.$$

Finally $q_z(x) = (x-z)^2 - 2(x-z)^2\mathbf{1}_{\{x\le z\}}$, for both Z and Z_0, implies

$$T - \Phi(z) = -\varepsilon/2 - \gamma\phi_z/(3\sqrt{n}) + z(M+\phi_z) - z^2(F - \Phi(z)).$$

Substitute these two identities, use $\sqrt{n}(F - \Phi(z)) = c\rho + a$, and equate with (15); collecting terms yields (13). At contact $\Delta = 0$ on the entire support, so $R_K = \xi = 0$. Substituting $a = 0$, $c = C_n$ and $C_{\mathrm{E}} = (3+\sqrt{10})\phi(0)/6$ gives (14). □

The off-contact identity permits checks on arbitrary bounded laws; it is not a positivity statement there. In particular the support integral in R_K cannot be dropped for a continuous law. The proof uses contact only when setting the two residuals to zero. Section 5 proves uniform bounds for this exact E_n, including changing laws and thresholds.

Corollary 2.6 (The sign license)**.** *At a ratio maximizer with $C = C_n \geq C_{\mathrm{E}}$, suppose a first pass proves $\mathcal{R}_C \leq T < nC\rho$. Then $\gamma + 3J > 0$ and*

$$n(C - C_{\mathrm{E}})\rho + n\phi(0)D/6 + n^{3/2}L \leq \mathcal{R}_C. \tag{16}$$

Proof. The contact identity before splitting the defect gives $\phi_z(\gamma + 3J)/6 = C\rho + \sqrt{n}L - \mathcal{R}_C/n > 0$. Thus $M = C_{\mathrm{E}}\rho - \phi(0)D/6 > 0$. Regrouping (14) writes its right side as the left side of (16) plus $n(1 - e^{-w})M \geq 0$. □

The initial bound is on the raw remainder. Only after it licenses the positive phase term may that term improve a later resource bound. No $D < 2$ assumption or positive lower bound on $C_n - C_{\mathrm{E}}$ was used.

For context, on a lattice with maximal span $h > 0$, its gaps have lengths $k_g h$, $k_g \geq 1$. Since τ vanishes outside the convex hull and the support has Lebesgue measure zero, $\sum_g \tau_g k_g h = 1$ and $J = \sum_g \tau_g k_g^2 h^2 \geq h$. Reflection preserves J, ρ, h and reverses γ, whence $|\gamma| + 3h \leq (3 + \sqrt{10})\rho$. Equality is precisely the zero-defect Bernoulli law or its reflection. Thus the gap functional extends the lattice span to arbitrary supports; no lattice hypothesis enters the proof.

3 What a first failing index must satisfy

A simple recurrence supplies the same small excess at any first failure of the sharp bound. This lets us use local exclusions directly, without first proving a separate sequence of positive upper barriers.

Lemma 3.1 (Gaussian derivative bounds)**.** *The standard normal density satisfies*

$$\|\phi''\|_\infty = \phi(0) < \frac{2}{5}, \qquad \|\phi'''\|_\infty \leq \frac{2}{\pi}, \qquad \frac{1}{4\pi} < \frac{2}{25}, \qquad C_{\mathrm{E}} < \frac{41}{100}.$$

Proof. Since $\phi''(x) = (x^2 - 1)\phi(x)$, its absolute maximum is at zero: the other candidates $x = \pm\sqrt{3}$ have value $2e^{-3/2}\phi(0) < \phi(0)$. The absolutely convergent Gaussian Fourier integral gives

$$|\phi'''(x)| \leq \frac{1}{2\pi}\int_{\mathbb{R}} |t|^3 e^{-t^2/2}\, dt = \frac{2}{\pi}.$$

The bounds for $\phi(0)$ and $1/(4\pi)$ follow from $\pi > 25/8$. For the last comparison one may use $\sqrt{10} < 31623/10000$ and $\pi > 314159/100000$, square the positive sides, and compare rational numbers. □

Theorem 3.2 (One-step smoothing)**.** *For every integer $n \geq 2$,*

$$C_n \leq \sqrt{\frac{n}{n-1}}\, C_{n-1} + \frac{\sqrt{n}}{15(n-1)^{3/2}} + \frac{2\sqrt{n}}{25(n-1)^2}. \tag{17}$$

Proof. Fix one standardized marginal P, write $v = n - 1$, and put $G_v(x) = \Phi(x/\sqrt{v})$. Conditioning on the last summand yields

$$|\mathbb{P}(S_n \leq H) - \mathbb{E}G_v(H - X)| \leq C_{n-1}\rho/\sqrt{v}.$$

Taylor expansion of the smooth Gaussian CDF, followed by expectation under this same marginal, gives

$$|\mathbb{E}G_v(H - X) - G_v(H) - \tfrac{1}{2}G_v''(H)| \leq \frac{\phi(0)\rho}{6v^{3/2}} < \frac{\rho}{15v^{3/2}}.$$

Here $\mathbb{E}X = 0$ and $\mathbb{E}X^2 = 1$ cancel the first two Taylor terms. For the change of variance, $\partial_v G_v = G_v''/2$ and $\partial_v^2 G_v = G_v^{(4)}/4$. Thus

$$G_{v+1}(H) - G_v(H) - \tfrac{1}{2}G_v''(H) = \frac{1}{4}\int_0^1 (1-t)G_{v+t}^{(4)}(H)\,dt.$$

Since $G_w^{(4)}(H) = w^{-2}\phi'''(H/\sqrt{w})$, this remainder has absolute value at most $1/(4\pi v^2) < 2/(25v^2)$. Combine the three estimates, multiply by $\sqrt{n}/\rho$, and use $\rho \geq 1$. The bound is uniform in P and H and treats both signs of the discrepancy, so it passes to the two suprema defining C_n. □

Corollary 3.3 (Entry from a sharp predecessor)**.** *If $n \geq 40$ and $C_{n-1} \leq C_{\mathrm{E}} \leq C_n$, then*

$$0 \leq n(C_n - C_{\mathrm{E}}) < \frac{1433}{4875} < \frac{3}{10}, \qquad C_n < \frac{167}{400} < 0.4226, \qquad n(C_{n-1} - C_n)_+ = 0.$$

For $n \geq 42$ the excess bound improves to $3003/10250$ and the score bound to $73/175$. For every standardized marginal P, the same-law predecessor loss $n(\sqrt{n-1}\Delta_{n-1}(P)/\rho(P) - C_n)_+$ is zero. At a first failing index $n \geq 43$ above a sharp base, both deleted constants C_{n-1} and C_{n-2} are below 0.4253.

Proof. Put $s = \sqrt{n/(n-1)}$. The recurrence and $n(s-1) = \{n/(n-1)\}/(s+1)$ give

$$n(C_n - C_{\mathrm{E}}) \leq \frac{n}{n-1}\left\{\frac{C_{\mathrm{E}}}{s+1} + \frac{s}{15} + \frac{2s}{25\sqrt{n-1}}\right\}.$$

For $n \geq 40$, $n/(n-1) \leq 40/39$, $\sqrt{n-1} > 6$, and $1 < s < 51/50$, since $39 \cdot 51^2 - 40 \cdot 50^2 = 1439 > 0$. The right side is therefore less than

$$\frac{40}{39}\left(\frac{41}{200} + \frac{17}{250} + \frac{17}{1250}\right) = \frac{1433}{4875} < \frac{3}{10}.$$

This gives $C_n < 0.41 + 0.3/40 = 167/400$. For $n \geq 42$, $n/(n-1) \leq 42/41$ and $1 < s < 81/80$; the last comparison follows from $41 \cdot 81^2 - 42 \cdot 80^2 = 201 > 0$. Using $C_{\mathrm{E}} < 41/100$ and $s + 1 > 2$, the right side is less than

$$\frac{42}{41}\left(\frac{41}{200} + \frac{27}{400} + \frac{27}{2000}\right) = \frac{3003}{10250} < \frac{3}{10}.$$

Consequently $C_n < 0.41 + 0.3/42 = 73/175$. Every law's predecessor ratio is bounded by $C_{n-1} \leq C_{\mathrm{E}} \leq C_n$, so both loss assertions follow. At a first failure the two deleted indices are sharp and $C_{\mathrm{E}} < 0.41 < 0.4253$. □

3.1 The loose ceilings needed after index twenty-four

Lemma 3.4. *If $n \geq 25$ and $C_{n-1} \leq C_{\mathrm{E}} \leq C_n$, then*

$$n(C_n - C_{\mathrm{E}}) < \frac{4177}{13824} < \frac{303}{1000}, \qquad C_n < \frac{145873}{345600} < \frac{423}{1000}.$$

For each marginal the same-law predecessor allowance is zero.

Proof. Set $s = \sqrt{n/(n-1)}$. Theorem 3.2 gives

$$n(C_n - C_{\mathrm{E}}) \leq \frac{n}{n-1}\left(\frac{C_{\mathrm{E}}}{s+1} + \frac{s}{15} + \frac{2s}{25\sqrt{n-1}}\right).$$

For $n \geq 25$ one has $n/(n-1) \leq 25/24$, $1 < s < 49/48$ and $\sqrt{n-1} > 24/5$. The latter strict comparisons follow from $24 \cdot 49^2 - 25 \cdot 48^2 = 24 > 0$ and $24 > 576/25$. Together with $C_{\mathrm{E}} < 41/100$, these imply

$$n(C_n - C_{\mathrm{E}}) < \frac{25}{24}\left(\frac{41}{200} + \frac{49}{720} + \frac{49}{2880}\right) = \frac{4177}{13824}.$$

Dividing by $n \geq 25$ proves the score bound. The exact positive comparison margins are $303/1000 - 4177/13824 = 1459/1728000$ and $423/1000 - 145873/345600 = 1579/1728000$. Finally each marginal's predecessor score is at most $C_{n-1} \leq C_{\mathrm{E}} \leq C_n$. □

The entry bounds concern the same marginal as the attained ratio. They license weaker own-score and predecessor-loss inputs used in the finite calculations. Transferring a calculation from a floor index to a later index additionally requires the actual-index support and weighted resource statements proved below; the scalar recurrence alone does not transfer a support bound.

4 What a certificate proves

A certificate in this proof is finite data from which a small collection of exact inequalities can be checked. It may specify coefficients of a polynomial majorant, a tree of parameter boxes, or a schedule of bounds which successively restricts an extremizer. The numerical statements are about whole regions, not sampled distributions. The mathematical reduction must first show that every still possible extremizer maps into one of those regions.

Here is a typical example. Suppose a preceding argument has confined the support of the same marginal to K and its moment vector to M. For a threshold interval $[a, b]$, a certificate specifies a function T such that

$$T(x_1, \ldots, x_n) \geq \mathbf{1}_{\{\sum_i x_i \leq b\}} \quad (x_1, \ldots, x_n \in K), \qquad \mathbb{E}T(X_1, \ldots, X_n) \leq L_a + \frac{c\rho}{\sqrt{n}},$$

where the second inequality holds for every moment vector in M and $L_a \leq \Phi(a/\sqrt{n})$. For every $H \in [a, b]$, monotonicity gives

$$\mathbb{P}(S_n \leq H) - \Phi(H/\sqrt{n}) \leq \mathbb{E}T - L_a \leq c\rho/\sqrt{n}.$$

Thus verification proves a probability inequality, including threshold atoms. Choosing a for the normal lower bound and b for the closed event is essential. This is how the small-summand certificates work; the count comparisons apply the same principle after conditioning on the signs of the summands.

4.1 A common acceptance argument

Lemma 4.1 (Expectation under one iid marginal)**.** *Let $f_0 = 1, f_1, \ldots, f_d$ be absolutely integrable under P, and put $m_j = \mathbb{E}f_j(X)$. For a finite tensor sum*

$$T(x_1, \ldots, x_n) = \sum_{\alpha} t_\alpha \prod_{i=1}^{n} f_{\alpha_i}(x_i), \qquad M_T(m) = \sum_{\alpha} t_\alpha \prod_i m_{\alpha_i},$$

one has $\mathbb{E}T(X_1, \ldots, X_n) = M_T(m)$. If T majorizes the closed sum event on the support product, a uniform upper bound on M_T gives the corresponding upper bound on its probability.

Proof. Independence factors the absolute expectation of each product into a product of finite absolute moments. Fubini and linearity therefore evaluate every term. Integrating the pointwise majorant proves the last assertion. □

The same m occurs in every factor. In particular, a factor with zero mean makes its tensor term have zero expectation. Such terms can improve the pointwise inequality without changing its expected value. Features of degree above three are used only after support has been bounded, so their integrability does not strengthen the theorem's moment assumption.

Lemma 4.2 (Exact positivity on a box)**.** *Let $p(t) = \sum_\alpha a_\alpha t^\alpha$ have coordinate degrees at most d_i. Its tensor Bernstein coefficients on $[0,1]^r$ are*

$$b_\nu = \sum_{\alpha \le \nu} a_\alpha \prod_{i=1}^{r} \frac{\binom{\nu_i}{\alpha_i}}{\binom{d_i}{\alpha_i}}.$$

If all $b_\nu \ge 0$, then $p \ge 0$ on the entire closed box.

Proof. The binomial theorem gives

$$t^k = \sum_{j=k}^{d} \frac{\binom{j}{k}}{\binom{d}{k}} \binom{d}{j} t^j (1-t)^{d-j}.$$

Apply this in each coordinate. The resulting basis functions are nonnegative and sum to one, so $p(t)$ is a convex combination of the b_ν. □

An affine rational substitution treats any rational box. On a simplex, the analogous multinomial basis in barycentric coordinates is nonnegative and sums to one, giving the same criterion after an exact change of basis. Subdivision can improve an inconclusive coefficient test. It never turns that test into sampling.

Proposition 4.3 (Coverage and exclusion)**.** *Suppose an analytic argument maps every law and threshold under consideration into a finite union of closed root cells. At each root, let a finite tree subdivide a parent into closed children whose union contains the parent. On each leaf suppose either a proved necessary condition excludes every actual mapped point, or a valid inequality establishes the desired score bound. Then the bound holds for every mapped law and threshold. If all leaves exclude a putative violating extremizer, no such extremizer exists.*

Proof. A point in a root belongs to a child at each subdivision, including on a shared face. Finiteness leads to a leaf, where its conclusion applies. The root cover includes every actual point. □

There are three distinct obligations in this proposition: the map from laws to parameters, validity of the leaf inequalities, and coverage. A relaxed moment domain may contain unrealizable points; this only makes verification harder. Missing an actual point invalidates the argument. In particular, we split sign orthants and event planes explicitly: on $\sum x_i = b$ the required inequality is $T - 1 \ge 0$, not just $T \ge 0$. Non-strict inequalities in the domain retain endpoint atoms. Resource limits, unresolved leaves and failed divisions cannot be interpreted as acceptance.

4.2 Exact evaluation and rigorous enclosures

The acceptance programs evaluate rational numbers by integer arithmetic. Irrational constants are enclosed between rationals with proved error bounds. For example, alternating arctangent series in Machin's identity

$$\frac{\pi}{4} = 4\arctan(1/5) - \arctan(1/239)$$

bound π; integer square roots then bound its required square roots. Integrating the Taylor expansion of $e^{-t^2/2}$ gives rational bounds for the Gaussian CDF at rational endpoints. Taylor's remainder, rather than a printed decimal approximation, supplies the error sign and size. The same principle bounds exponential, trigonometric and binomial expressions used below. Each application states its finite interval or separate tail estimate. Products and quotients use outward bounds, with denominator positivity established on the whole cell. Multiplying a negative expression by a lower bound requires the corresponding reversal of inequality; the signed comparisons retain this distinction.

These statements are proved inequalities in ordinary analysis. Their finite arithmetic evaluation is performed by the programs. Exact arithmetic removes round-off uncertainty, but does not by itself prove coverage, correctness of the formulas, or their applicability to a particular marginal. The preceding lemmas and the applications below supply those mathematical obligations. This separation is also used in other computer-assisted inequality proofs; formal interval verification provides a stronger implementation assurance when available [10]. The certificate format here is specific to this argument, and is not a standard proof-assistant format.

4.3 How the certificates are found and checked

For the two-summand majorants, the unknown tensor coefficients enter linearly. The construction program solves floating-point linear programs imposing the proposed majorant and expectation inequalities on an initial finite set of points. It searches for violated constraints, adds them, and solves again. Coefficients are converted to rational numbers and passed to the exact checker. The three-summand construction uses the same linear-programming and separation approach, with extra zero-expectation tensor features. Success of the sampled optimization is only a way of proposing a polynomial; the whole-domain check is the proof. If rationalization loses a margin, the candidate must be improved or rejected.

The resource and count certificates are obtained by adaptive subdivision. Starting with the analytically licensed root domains, construction applies valid bounds, discards infeasible cells, and splits cells whose inequality is still undecided. A successful finite forest records a complete cover and enough information to re-evaluate every leaf. Some modules recompute their entire enclosure schedule during acceptance rather than trusting recorded outputs. The schedules may have been chosen by experimentation; their acceptance depends only on the stated inequalities and coverage. The method does not require a theorem that this search always terminates. The particular finite accepted objects suffice.

The distributed candidates are treated as untrusted data. Acceptance reconstructs the defined polynomials or bound expressions, checks their identities and complete domains, and evaluates the exact inequalities. It does not accept a stored success flag or an optimizer's declaration of infeasibility. The public count candidates were verified in this way; their historical search was not rerun. This has no bearing on the logical validity of a verified certificate, just as finding a factor and checking its product are different tasks.

4.4 Proof data and reproduction

The companion reproduction package identifies each certificate family, its mathematical conclusion, checking source and complete execution. The fixed reference is commit `c31d0cbe2af867d26871fbae37a3a7f4066f4a55` of the public certificate repository, in `manuscript/extended-v17/reproduction`. Its `REPRODUCE.md` gives the restoration and acceptance commands; `COMPLETION_LEDGER.json` identifies the proof objects and completed result records. The mathematical interfaces used in this revision are stated in the numerical propositions below. It preserves the exact accepted candidates and their domain interfaces.

A reader can inspect the analytic arguments independently of running the programs, and can verify the finite arithmetic independently of repeating the search. Both parts are necessary for the whole proof. The completed executions use Python with assertions enabled, with integer and rational arithmetic; the three-summand kernel uses C++17 and Boost.Multiprecision integers. The compiler, language runtimes and faithful execution are part of the trusted computational environment. The implementation is not a proof-assistant formalization.

The full execution history, command lines and machine identities belong to that package rather than the mathematical argument. They distinguish complete native calculations from later checks of their input interfaces, and preserve failed or interrupted attempts. All selected numerical obligations in the reference proof have completed; the separate 114-command recipe has not been rerun as one relocated campaign. This revision changes the exposition, not those certificate bytes or acceptance targets. Any new mathematical extension of a certificate's scope must be justified explicitly and cannot follow from its previous success alone.

5 Fourier bounds for one marginal, uniformly in the index

Two transformations are needed. A compensated inversion formula bounds the smooth-payoff error E_n in conservation; Prawitz's inequality bounds the CDF itself. Both use characteristic functions of the same marginal. The formulas below are pointwise inequalities with explicit moment and support hypotheses, so they also cover triangular arrays of changing laws. The finite certificates evaluate these formulas on closed cells; they do not supply an asymptotic or infinite-domain analytic argument.

Lemma 5.1 (Gaussian normalizations)**.** *One has* $\|\phi''\|_\infty = \phi(0) < 2/5$, $\|\phi'''\|_\infty \le 2/\pi$, *and*

$$\frac{333}{106} < \pi < \frac{22}{7}, \quad \sqrt{\pi} > \frac{4431}{2500}, \quad \sqrt{\pi/2} < \frac{251}{200},$$
$$\phi(0) > 0.39886 > \frac{997}{2500}, \quad \frac{819}{2000} < C_{\mathrm{E}} < \frac{20487}{50000}.$$

Proof. Differentiation and integration by parts in the Gaussian integral give $\phi(x) = (2\pi)^{-1}\int_{\mathbb{R}} e^{-t^2/2-itx}dt$; its normalization follows by squaring the integral and using polar coordinates. Absolute convergence gives

$$\|\phi^{(k)}\|_\infty \le \frac{1}{2\pi}\int_{\mathbb{R}} |t|^k e^{-t^2/2}dt.$$

This proves the third derivative bound. For the second derivative, $(x^2-1)e^{-x^2/2}$ has absolute maximum one (the other candidate is $2e^{-3/2} < 1$). For the rational constants put $s_k(x) = \sum_{j=0}^{k-1}(-1)^j x^{2j+1}/(2j+1)$. Tangent addition, with $0 < 4\arctan(1/5) - \arctan(1/239) < \pi/2$, gives Machin's identity $\pi = 16\arctan(1/5) - 4\arctan(1/239)$. The geometric remainder in $1/(1+t^2)$, integrated from zero, gives

$$\pi_- = 16s_6(1/5) - 4s_1(1/239) < \pi < 16s_5(1/5) - 4s_2(1/239) = \pi_+.$$

Exact rational comparison gives $333/106 < \pi_- < \pi_+ < 22/7$ and $\pi_- > (4431/2500)^2$, $\pi_+ < 2(251/200)^2$. Also $(0.39886)^2 < 7/44$. Finally, for $r_- = 3162277660/10^9$, $r_+ = 3162277661/10^9$, $r_-^2 < 10 < r_+^2$ and

$$(819/2000)^2 < \frac{19+6r_-}{72\pi_+} < C_{\mathrm{E}}^2 < \frac{19+6r_+}{72\pi_-} < (20487/50000)^2.$$

All denominators are positive; the displayed finite sums specify the comparisons as integer cross-multiplications. □

For $n \geq 2$, write

$$f(t) = \mathbb{E}e^{itX}, \quad e_n(t) = e^{-nt^2/2}(1 - in\gamma t^3/6), \quad V_n(t) = f(t)^n - e_n(t),$$

$$I_0(H) = \int_0^{1/2} \frac{|\Im(e^{-itH}V_n(t))|}{t^3}\, dt, \qquad I_\infty = \int_{1/2}^{\infty} \frac{|V_n(t)|}{t^3}\, dt.$$

5.1 Compensated inversion, including atoms

The compensation below makes absolute Fubini valid, including at atoms.

Lemma 5.2. *If a finite signed measure μ has mass and first moment zero and $\int x^2\, d|\mu|(x) < \infty$, then, for every $H \in \mathbb{R}$,*

$$\int (x - H)|x - H|\, d\mu(x) = -\frac{4}{\pi} \int_0^{\infty} \frac{\Im(e^{-itH}\widehat{\mu}(t))}{t^3}\, dt.$$

The integral on the right is absolutely convergent.

Proof. For real y, substitution and integration by parts give

$$\int_0^{\infty} \frac{ty - \sin(ty)}{t^3}\, dt = \frac{\pi}{4} y|y|.$$

The boundary terms vanish: near zero the numerator is $O(t^3)$, and at infinity it is $O(t)$. The integrand has the sign of y, since $v - \sin v \geq 0$ for $v \geq 0$. To calculate the constant, nonnegative Tonelli and elementary Laplace integration give

$$\int_0^{\infty} \frac{1 - \cos t}{t^2}\, dt = \int_0^{\infty} s\left(\frac{1}{s} - \frac{s}{s^2 + 1}\right) ds = \frac{\pi}{2}.$$

The absolute double integral of the compensated expression against $d|\mu|(x)$ equals

$$\frac{\pi}{4} \int (x - H)^2 d|\mu|(x) < \infty.$$

Absolute Fubini is therefore valid. The linear compensation disappears because $\int (x - H) d\mu = 0$, proving both assertions. No continuity of a CDF at H is involved. □

Apply this to the sum law minus the scale-$\sqrt{n}$ pushforward of the signed density

$$\phi(x)\{1 + \gamma(x^3 - 3x)/(6\sqrt{n})\}.$$

This signed density has mass one, first moment zero and finite absolute second moment. Its transform after scaling is e_n. Gaussian integration by parts, using $q_z(x) = (x - z)|x - z|$ and $q_z''' = 4\delta_z$, yields

$$\varepsilon_n := \mathbb{E}q_{H/\sqrt{n}}(S_n/\sqrt{n}) - \mathbb{E}q_{H/\sqrt{n}}(Z) - \frac{2\gamma\phi(H/\sqrt{n})}{3\sqrt{n}} = -\frac{4}{\pi n} \int_0^{\infty} \frac{\Im(e^{-itH}V_n(t))}{t^3}\, dt.$$

Consequently the error $E_n = -n^{3/2}\varepsilon_n/2$ satisfies

$$|E_n| \leq \frac{2\sqrt{n}}{\pi}\{I_0(H) + I_\infty\}. \tag{18}$$

The identity and bound hold for each n and H separately.

5.2 Characteristic-function bounds from common moments

Let X be centered with $\mathbb{E}X^2 = 1$, $\rho = \mathbb{E}|X|^3 < \infty$, $\gamma = \mathbb{E}X^3$, and $f(u) = \mathbb{E}e^{iuX}$. In this section put

$$h = \mathbb{E}|X|, \quad m = \mathbb{E}X_+ = \mathbb{E}X_- = h/2, \quad A = \mathbb{E}X_+^2, \quad B = \mathbb{E}X_-^2,$$
$$\zeta = 2AB/m, \qquad b(r) = \frac{\sqrt{r^2+8} - r}{2}.$$

Here $X_- = \max\{-X, 0\}$, so $A + B = 1$ and all of h, m, A, B are positive. The letter b is used for this root function to avoid confusion with the signed second moment $\beta = \mathbb{E}[X|X|]$.

Lemma 5.3 (Anchored characteristic function)**.** *For $0 \le u \le \pi$,*

$$\cos u \le \Re f(u) \le \cos u + (\rho - 1)u^3/6, \qquad |\Im f(u)|^2 \le \frac{u^6}{36}(\rho - 1)(\rho + 5/3).$$

Proof. Write $\operatorname{sinc} t = \int_0^1 \cos(st)\, ds$. For real a, b,

$$|\operatorname{sinc} a - \operatorname{sinc} b| \le |a - b|/2, \qquad |\operatorname{sinc} a - \operatorname{sinc} b| \le |a^2 - b^2|/6.$$

The first follows from the Lipschitz bound for cosine. For the second, $v \mapsto \cos\sqrt{v}$ has derivative of absolute value at most $1/2$ on $v \ge 0$, including its limit at zero; integrate with respect to s. For $r \ge 0$, integration from 1 to r of the first estimate gives

$$\left|\cos(ur) - \cos u + \frac{u \sin u}{2}(r^2 - 1)\right| \le \frac{u^3}{12}(r-1)^2(2r+1).$$

Since $\mathbb{E}(|X| - 1)^2(2|X| + 1) = 2(\rho - 1)$, averaging proves the upper real bound. For the lower bound, when $0 < u \le \pi$ set $a = \sin u/(2u)$ and $J(v) = \cos v + av^2$. The function $\operatorname{sinc} v$ decreases on $[0, \pi]$, so J decreases up to u and increases from u to π. For $v \ge \pi$, $J(v) \ge -1 + a\pi^2 = J(\pi) \ge J(u)$. Evenness gives the same conclusion for negative v. Average $J(uX) \ge J(u)$ and use $\mathbb{E}X^2 = 1$.

The second sinc estimate, together with $\mathbb{E}X = 0$, gives $|\Im f(u)| \le u^3\mathbb{E}[|X|\,||X|^2 - 1|]/6$. Put $W(r) = (r-1)^2(2r+1)$. Cauchy's inequality applies to the factorization $r|r^2 - 1| = \sqrt{W(r)}\, r(r+1)/\sqrt{2r+1}$. The exact identity

$$r^3 + \frac{29r^2 + 1}{18} - \frac{2r^2(r+1)^2}{2r+1} = \frac{(r-1)^2(4r+1)}{18(2r+1)} \ge 0$$

bounds the second squared factor in expectation by $(\rho + 5/3)/2$; the first has expectation $2(\rho - 1)$. All expectations are finite under the stated third-moment hypothesis. The value at $u = 0$ is immediate. □

Lemma 5.4 (One common moment surrogate)**.** *For an independent copy X', put $\eta = \mathbb{E}|X - X'|^3/2 - \rho$. Then*

$$\eta \le h \le 1, \qquad \zeta \ge h, \qquad \rho + \zeta \ge 2/h, \qquad \zeta \ge b(\rho), \qquad \gamma^2 + \zeta^2 \le \rho^2.$$

For every real u one also has $36|\Im f(u)|^2 \le u^6(\rho^2 - \zeta^2)$. If $|X| \le R$ almost surely, then additionally

$$\eta \le b(\rho) + \frac{(R - \sqrt{\rho^2 + 8})_+^2}{4R}.$$

In particular $R \le \sqrt{\rho^2 + 8}$ implies $\eta \le b(\rho)$.

Proof. The scalar inequality $|x-y|^3 \le (x-y)^2(|x|+|y|)$, independence and centering give $\mathbb{E}|X-X'|^3 \le 2\rho + 2h$; Cauchy gives $h \le 1$. Signwise Cauchy gives

$$A^2 \le m\mathbb{E}X_+^3, \quad B^2 \le m\mathbb{E}X_-^3, \quad \frac{m^2}{A} + \frac{m^2}{B} \le \mathbb{P}(X \ne 0) \le 1.$$

Thus $AB \ge m^2$, $\zeta \ge 2m = h$, and $\rho + \zeta \ge (A+B)^2/m = 2/h$. Multiplication yields $\zeta(\rho+\zeta) \ge 2$, proving $\zeta \ge b(\rho)$.

For completeness, the sine-circle bound uses an exact mixture, not an extreme-point assertion about convolution. Let μ_+ and μ_- be the restrictions of the law of $|X|$ to its positive and negative sign classes. On $a, b > 0$ define

$$d\nu(a,b) = \frac{a+b}{m}\, d\mu_+(a) d\mu_-(b).$$

Integrating the centered law with atoms $a, -b$ and probabilities $b/(a+b), a/(a+b)$ against ν recovers the nonzero part of the original law; the zero atom is added separately. For this two-point law the second sinc estimate implies

$$\frac{6|\Im f_{a,b}(u)|}{|u|^3} \le \frac{ab|a^2-b^2|}{a+b}.$$

Together with $e_{a,b} = 2a^2b^2/(a+b)$, the right side is one coordinate of a vector of length $ab(a^2+b^2)/(a+b)$. Minkowski's inequality in $\mathbb{R}^2$ therefore bounds the vector $(\zeta, 6\Im f(u)/u^3)$ by ρ. The third-moment assumption justifies all integrals. Letting $u \to 0$ proves the assertion for γ: Taylor's remainder is dominated by $|X|^3/6$, so dominated convergence applies after using $\mathbb{E}X = 0$.

For the final assertion the exact same-sign deficit is

$$h - \eta = \mathbb{E}\Big[\min(|X|,|X'|)(X-X')^2 \mathbf{1}_{\{XX'>0\}}\Big].$$

On a common sign class, $\min(a,b) \ge ab/R$. Expansion of the two independent class integrals gives

$$h - \eta \ge \frac{h\rho - 2(A^2+B^2)}{R} = \frac{h(\rho+\zeta) - 2}{R}.$$

If $h \le b(\rho)$, the already proved $\eta \le h$ suffices. Otherwise, write $b = b(\rho)$, use $\zeta \ge h$ and $b(\rho+b) = 2$ to obtain

$$\eta - b \le \frac{(h-b)(R-\rho-h-b)}{R} \le 0,$$

when $R \le \rho + 2b = \sqrt{\rho^2+8}$. For general R, the quadratic $y(R - \sqrt{\rho^2+8} - y)$ on $y \ge 0$ is at most $(R-\sqrt{\rho^2+8})_+^2/4$. This proves the stated remainder too. In particular radius three suffices for every $\rho \ge 1$. □

For a universal calculation put $H_* = \max\{h, b(\rho)\}$. Then $b(\rho) \le H_* \le 1$, $\eta \le H_*$, $\zeta \ge H_*$ and $\rho + \zeta \ge 2/H_*$. On a closed cell $\rho \in [\ell, r]$, $H_* \in [a,b]$, it is consequently safe to use

$$\eta \le b, \qquad \zeta \ge \max\{a, 2/b - r, b(r)\}.$$

Every cell uses both bounds from the same marginal law. A lower bound on H_* is not a lower bound on h; no later sign-probability argument may substitute it for one. If support $|X| \le R \le \sqrt{\ell^2+8}$ has independently been acquired, the last part of Lemma 5.4 instead permits $\eta \le b(\ell)$ and $\zeta \ge b(r)$. For larger R its displayed nonnegative remainder must be retained.

5.3 Keeping the cubic compensation in the high frequencies

Suppose now that $|X| \le R$ and $\rho \le r$. We may use

$$M_4 = \min\left\{ Rr, 1 + \frac{(R+1)^2}{R+1/2}(r-1) \right\},$$
$$M_5 = \min\left\{ RM_4, 1 + \frac{R^3 + 2R^2 + 3R + 3/2}{R+1/2}(r-1) \right\}$$

as upper bounds on $\mathbb{E}X^4$ and $\mathbb{E}|X|^5$ of the same law. To verify the less immediate branches, set $y = |X|$ and expand

$$\rho - 1 = \mathbb{E}(y-1)^2(y+1/2), \quad \mathbb{E}X^4 - 1 = \mathbb{E}(y-1)^2(y+1)^2,$$
$$\mathbb{E}|X|^5 - 1 = \mathbb{E}(y-1)^2(y^3 + 2y^2 + 3y + 3/2).$$

The ratios of the last two factors to $y+1/2$ increase on $y \ge 0$: their derivatives are respectively $y(y+1)/(y+1/2)^2$ and $y(4y^2+7y+4)/(2(y+1/2)^2)$. The other branches follow from $y \le R$. These moment bounds require no bound on the product-gap defect.

Put $G = \min\{(r-1)(r+5/3), r^2 - Z^2\}$, where $Z > 0$ is a proved lower bound for ζ on the whole cell; $Z = b(r)$ always suffices. Then $\gamma^2 \le G$: the first branch follows by letting $u \to 0$ in Lemma 5.3, and the second from Lemma 5.4. Assume that a proved nonnegative envelope $m(u)$ bounds $|f(u)|$ on the interval being integrated. Set

$$g = e^{-u^2/2}, \quad v = -i\gamma u^3/6, \quad b_u = \sqrt{G}\, u^3/6, \quad W = g\sqrt{1 + b_u^2},$$

and, for $0 \le u \le \pi$,

$$A_u = \max\{|\cos u - g|, |\cos u + (r-1)u^3/6 - g|\},$$
$$B_u = \min\{M_4 u^4/24, M_5 u^5/120\} + b_u(1-g).$$

Taylor's theorem gives

$$|\Im f(u) + \gamma u^3/6| \le \min\{M_4 u^4/24, M_5 u^5/120\}.$$

The anchored real estimate therefore gives $|f - g(1+v)| \le \sqrt{A_u^2 + B_u^2}$. Telescoping the nth powers and then applying the integral Taylor remainder to $(1+sv)^n$, $0 \le s \le 1$, proves, for every integer $n \ge 2$,

$$|f(u)^n - g^n(1+nv)| \le \sqrt{A_u^2 + B_u^2} \sum_{j=0}^{n-1} m(u)^j W^{n-1-j} + \binom{n}{2} b_u^2 g^n (1+b_u^2)^{(n-2)/2},$$
$$|f(u)^n - g^n(1+nv)| \le m(u)^n + g^n\sqrt{1 + n^2 b_u^2}.$$

One may take their pointwise minimum. These formulas use no quotient of $m(u) - W$, so coincident and zero values cause no exceptional cases.

On a closed frequency cell $[a,b] \subset [1/2, T]$, a rigorous upper bound for this minimum is multiplied by the exact weight $\int_a^b u^{-3}\,du = (a^{-2} - b^{-2})/2$. The infinite remainder is bounded without a periodicity assumption: for $T > 0$,

$$\int_T^\infty \frac{|f(u)^n - g(u)^n(1+nv(u))|}{u^3}\,du \le \frac{1}{2T^2} + \frac{e^{-nT^2/2}}{nT}\left(T^{-3} + \frac{n\sqrt{G}}{6}\right).$$

Indeed $|f| \le 1$, $\sqrt{1+x^2} \le 1 + |x|$, $u^{-3} \le T^{-3}$ on the normal part, and $\int_T^\infty e^{-nu^2/2}\,du \le e^{-nT^2/2}/(nT)$.

To preserve cancellation in the real residual on a closed frequency cell, put $d_0(u) = \cos u - e^{-u^2/2}$ and $d_1(u) = d_0(u) + (r-1)u^3/6$. Their derivatives are

$$d_0'(u) = -\sin u + ue^{-u^2/2}, \qquad d_1'(u) = d_0'(u) + (r-1)u^2/2.$$

If D_j encloses d_j' throughout a cell of width w and midpoint u_0, the fundamental theorem of calculus gives $d_j(u) \in d_j(u_0) + [-w/2, w/2]D_j$. Intersecting this with a direct interval evaluation remains valid.

5.4 The compatible modulus profile

The scalar certificate uses $\kappa = 62/625$ and proves

$$\cos x \le 1 - x^2/2 + \kappa|x|^3 \qquad (x \in \mathbb{R}). \tag{19}$$

Here is the entire reduction to its finite check. For $x \ge 625/124$ the right side is at least one. On the remaining interval, the degree-sixteen Taylor polynomial P_{16} bounds cosine above: the sixteenth derivative of $\cos - P_{16}$ is $\cos - 1 \le 0$, and its lower derivatives vanish at zero. The degree-thirteen polynomial

$$Q(x) = \kappa + \sum_{j=2}^{8} \frac{(-1)^{j+1}x^{2j-3}}{(2j)!}$$

satisfies $x^3Q(x) = 1 - x^2/2 + \kappa x^3 - P_{16}(x)$. On the eighty cells $[i/16, (i+1)/16]$, $0 \le i < 80$, and the last cell $[5, 625/124]$, the exact verifier checks the inverse Bernstein identity and all fourteen coefficients. The smallest coefficient is $2479189/65585520000 > 0$. Lemma 4.2 proves positivity on every closed cell; evenness completes (19).

Put $a = 1/2 - 4\kappa = 129/1250$, $C = 1 - \cos 4$ and $A_0 = 16a/C$. The profile

$$q(z) = \begin{cases} 1/2 - \kappa z, & 0 \le z \le 4, \\ A_0(1 - \cos z)/z^2, & 4 \le z \le 2\pi, \\ 0, & z \ge 2\pi \end{cases}$$

is nonnegative, convex and nonincreasing on $[0, \infty)$, and satisfies $\cos z \le 1 - z^2 q(z)$. Indeed (19) gives $0 < A_0 \le 1$. The pieces agree at their endpoints. On the middle interval the first derivative is nonpositive and the second has the sign of

$$B_0(z) = 6 + (z^2 - 6)\cos z - 4z \sin z.$$

On $[4, 3\pi/2]$, $B_0'(z) = -2z\cos z + (2 - z^2)\sin z \ge 0$ and $B_0(4) \ge 4$: indeed $7\pi/6 \le 4 \le 3\pi/2$ implies $\sin 4 \le -1/2$, and $\cos 4 \ge -1$, so $6 + 10\cos 4 - 16\sin 4 \ge 4$. On $[3\pi/2, 2\pi]$, $B_0(z) \ge 6$. The derivative at 2π is zero. The derivative jump at 4 has the correct sign because $258\sin 4 + 119(1 - \cos 4) > 0$. Its exact lower bound is obtained by replacing sine by its degree-fifteen Taylor lower bound and cosine by P_{16}. The same scalar checker verifies this rational inequality and the needed $25/8 < \pi < 22/7$ by Machin's formula.

To apply the profile, weight the distribution of $X - X'$ by $(X - X')^2/2$. This is a probability measure whose first absolute moment is $\rho + \eta$. Jensen's inequality gives, for $u \ge 0$,

$$|f(u)|^2 = \mathbb{E}\cos(u(X - X')) \le 1 - 2u^2 q(u(\rho + \eta)).$$

If $\rho + \eta \le \mathcal{D}$, monotonicity permits $\mathcal{D}$ on the right. Thus the linear part gives $1 - u^2 + 2\kappa\mathcal{D}u^3$; when $4 \le \mathcal{D}u \le 2\pi$ the curved part gives $1 - 2A_0(1 - \cos(\mathcal{D}u))/\mathcal{D}^2$. Take the minimum with one and with the coordinate-disk estimate from Lemmas 5.3–5.4. The upper square of a real interval is the larger of the squares of its two endpoints, even when that interval crosses zero or lies below it.

5.5 A Fourier majorant of a closed half-line

Lemma 5.5. *For every integrable real random variable Y, every $h \in \mathbb{R}$ and $A > 0$,*

$$\mathbb{P}(Y \le h) \le \frac{1}{2} + 2 \lim_{\epsilon \downarrow 0} \Re \int_\epsilon^1 K(t) e^{-iAth} \mathbb{E} e^{iAtY}\, dt, \tag{20}$$

where

$$K(t) = \frac{1-t}{2} + \frac{i}{2}\left((1-t)\cot(\pi t) + \frac{1}{\pi}\right).$$

The limit concerns the real part; the complex integral itself may diverge.

Proof. Put $W(x) = (\sin(\pi x)/(\pi x))^2$, with $W(0) = 1$, and

$$H_B(x) = \sum_{m \ge 1} \{W(x-m) - W(x+m)\} + 2xW(x).$$

The series converges absolutely and locally uniformly. It is odd, vanishes at zero, and takes value $\operatorname{sgn}(j)$ at every nonzero integer. The partial-fraction identity

$$\sum_{m \in \mathbb{Z}} (x-m)^{-2} = \pi^2 \csc^2(\pi x) \quad (x \notin \mathbb{Z})$$

follows by integrating $\pi \cot(\pi z)/(z-x)^2$ around the square with vertices $\pm(N+1/2) \pm i(N+1/2)$: the boundary integral is $O(N^{-1})$, and the residues are $(m-x)^{-2}$ and $-\pi^2 \csc^2(\pi x)$. For positive noninteger x this identity gives

$$H_B(x) - 1 = \left(\frac{\sin \pi x}{\pi}\right)^2 \left(\frac{2}{x} - \frac{1}{x^2} - 2\sum_{m \ge 1} (x+m)^{-2}\right).$$

Integral comparison of the decreasing function $(x+s)^{-2}$ yields

$$\sum_{m \ge 1} (x+m)^{-2} \le \frac{1}{x} \le \frac{1}{x^2} + \sum_{m \ge 1} (x+m)^{-2}.$$

Thus $|H_B(x) - \operatorname{sgn}(x)| \le W(x)$; oddness and the integer values supply the remaining points. Consequently

$$M(x) := \frac{1 - H_B(x) + W(x)}{2} \ge \mathbf{1}_{\{x \le 0\}}, \qquad M(0) = 1, \qquad 0 \le M(x) - \mathbf{1}_{\{x \le 0\}} \le W(x).$$

Elementary integration gives

$$W(x) = 2\int_0^1 (1-t)\cos(2\pi t x)\, dt, \qquad 2xW(x) = \frac{2}{\pi}\int_0^1 \sin(2\pi t x)\, dt.$$

Inserting the first formula into the shifted series with Abel weights r^m, $0 < r < 1$, gives

$$\sum_{m \ge 1} r^m \{W(x-m) - W(x+m)\} = 4\int_0^1 (1-t)\sin(2\pi t x) \frac{r \sin(2\pi t)}{1 - 2r\cos(2\pi t) + r^2}\, dt.$$

The fraction tends to $\frac{1}{2}\cot(\pi t)$ and is bounded in absolute value by $\frac{1}{2}|\cot(\pi t)|$, since its denominator is $(1-r)^2 + 4r\sin^2(\pi t)$. The resulting majorant is integrable: the sine vanishes at zero and $(1-t)\cot(\pi t)$ is bounded at one. Dominated convergence therefore proves

$$H_B(x) = 2\int_0^1 \left((1-t)\cot(\pi t) + \frac{1}{\pi}\right)\sin(2\pi t x)\, dt.$$

Combining the formulas yields

$$M(x) = \frac{1}{2} + 2 \lim_{\epsilon \downarrow 0} \Re \int_\epsilon^1 K(t) e^{2\pi i t x}\, dt = 1 + 2\Re \int_0^1 K(t)(e^{2\pi i t x} - 1)\, dt.$$

The last integral is absolutely convergent. For integrable U, $\mathbb{E}|e^{2\pi i t U} - 1| \le 2\pi t \mathbb{E}|U|$; together with $K(t) = O(t^{-1})$ at zero and boundedness at one, this permits absolute Fubini. Apply the identity to $U = A(Y - h)/(2\pi)$ and take expectations. The value $M(0) = 1$ proves the claimed closed-event convention even at atoms. □

For $Y = (X_1 + \cdots + X_4)/2$ and $A = 6$, independence gives $\mathbb{E}e^{6itY} = f(3t)^4$. Thus the lemma supplies exactly the factor six, positive imaginary kernel, and boundary value used in the four-summand comparison below.

5.6 From characteristic functions to the CDF

We use Prawitz's smoothing inequality in the form recorded in [8, Lemma 4.1, p. 263]. For $n \ge 2$, $c > 0$, $0 < \tau \le 1$, put

$$K_2(t) = (1 - t) + i\{(1 - t)\cot(\pi t) + 1/\pi\}, \qquad 0 < t < 1.$$

With $T = c\sqrt{n}$, that inequality gives

$$\begin{aligned}\Delta_n \le & \int_0^\tau |K_2(t)||f(ct)^n - e^{-nc^2t^2/2}|\, dt + \int_\tau^1 |K_2(t)|\, |f(ct)|^n\, dt \\ & + \int_0^\tau |K_2(t) - i/(\pi t)| e^{-nc^2t^2/2}\, dt + \frac{1}{\pi} \int_\tau^\infty e^{-nc^2t^2/2} \frac{dt}{t}.\end{aligned}$$

Our standardized iid laws satisfy its independence, centering and positive total-variance hypotheses. The cited convention uses a strict CDF. Continuity of Φ makes the supremum identical for closed CDFs: take one-sided limits at each evaluation point. Thus atoms require no smoothing of the original law.

For $g(u) = e^{-u^2/2}$ and envelopes $m \ge |f|$, $E_0 \ge |f - g|$, the first integrand uses $|f^n - g^n| \le \min\{E_0 \sum_{j=0}^{n-1} m^j g^{n-1-j}, m^n + g^n\}$. Every literal integer n is evaluated separately. The original 20–24 integral verifier covers $[\epsilon, \tau]$ and $[\tau, 1 - \epsilon]$ by closed cells, where $\epsilon = 1/1024$; its numerical parameter domain is $0 < c < \pi$ and $\epsilon < \tau \le 3/10 < 1 - \epsilon$. In particular $c\tau < 1$ licenses the stable low-frequency Taylor bounds. On a cell of width w and midpoint t_0, it uses either $w \sup F$ or $w\{F(t_0) + w^2 \max(0, \sup F'')/24\}$ for a fixed smooth majorant F. The latter follows by Taylor expansion and integration; a failed derivative-domain guard triggers the former bound on the entire cell. Switching branches at a sampled midpoint would not justify this rule.

For the omitted endpoint pieces, the respective upper bounds are

$$\begin{aligned}L_\epsilon &= n\left[\frac{rc^3}{6}\left(\frac{\epsilon^4}{4} + \frac{\epsilon^3}{3\pi}\right) + \frac{c^4}{8}\left(\frac{\epsilon^5}{5} + \frac{\epsilon^4}{4\pi}\right)\right], \\ N_\epsilon &= \epsilon + \pi\epsilon^2/2, \qquad H_\epsilon = \epsilon^2/2 + \pi\epsilon^3/3.\end{aligned}$$

They follow from $|K_2(t)| \le 1 + 1/(\pi t)$ near zero, cubic Taylor error for f, $e^{-u^2/2} - 1 + u^2/2 \le u^4/8$, and

$$0 \le 1/x - \cot x \le \frac{x}{3(1 - x^2/6)} \le x \quad (0 < x \le 1).$$

For this last estimate use $\sin x - x\cos x = \int_0^x t \sin t\, dt \le x^3/3$ and $\sin x \ge x - x^3/6$. Apply the same identity with $1 - t$ near one. Finally the infinite Gaussian term is at most $e^{-nc^2\tau^2/2}/(\pi nc^2\tau^2)$, by $\int_a^\infty e^{-v} v^{-1}\, dv \le e^{-a}/a$. For a strip $\rho \in [\ell, r]$, multiply the full four-term bound by $\sqrt{n}/\ell$, with outward enclosures. A complete closed strip cover and a strict rational comparison then establish the claimed CDF bound.

5.7 Sharper scalar bounds for the same integrals

For $a > 0$ write $E_1(a) = \int_a^\infty e^{-v} v^{-1}\, dv$. Integration by parts, followed by Jensen's inequality for the convex function $t \mapsto (a+t)^{-2}$ and an exponential random variable of mean one, gives

$$E_1(a) = \frac{e^{-a}}{a} - e^{-a} \int_0^\infty \frac{e^{-t}}{(a+t)^2}\, dt \le e^{-a} \left(\frac{1}{a} - \frac{1}{(a+1)^2} \right).$$

The last Gaussian term in Prawitz's inequality is $E_1(a)/(2\pi)$, where $a = nc^2\tau^2/2$. This improvement has no additional moment or support hypothesis.

Some small-index certificates require $\tau > 3/10$. Their low-frequency range is still controlled explicitly. Put $U = 6/5$. On $0 \le x \le U$, the alternating sine bounds and integration give

$$\sin x \ge x \left(1 - x^2/6 + x^4/120 - x^6/5040\right), \qquad 0 \le \sin x - x\cos x \le x^3/3 - x^5/30 + x^7/840.$$

The denominator below is at least $1 - U^2/6 - U^6/5040 > 0$, and the numerator factor is at least $1/3 - U^2/30 > 0$. Consequently

$$0 \le 1/x - \cot x \le L(x) := \frac{x(1/3 - x^2/30 + x^4/840)}{1 - x^2/6 + x^4/120 - x^6/5040}, \qquad L(0) = 0.$$

It follows that

$$|K_2(t) - i/(\pi t)| \le (1-t)\sqrt{1 + L(\pi t)^2} \quad (0 < t \le 19/50).$$

At the high endpoint put $s = 1 - t$. The identity $K_2(t) = s + i\, s(1/(\pi s) - \cot(\pi s))$ gives $|K_2(t)| \le s\sqrt{1 + L(\pi s)^2}$ for $t \ge 3/4$. The rational upper bound $\pi < 22/7$ implies $\pi(19/50) < 6/5$ and $\pi/4 < 6/5$. The numerical quadrature must therefore declare $\epsilon < \tau \le 19/50 < 1 - \epsilon$, $c < \pi$, and $c\tau \le 6/5$. It differentiates the displayed smooth rational majorant on whole cells; it does not differentiate an interval-selected minimum.

The cancellation series on this larger range also have explicit tails. Truncate the even series for cosine and $e^{-u^2/2}$ at degree twenty. For their difference, the error in each t derivative of order $j = 0, 1, 2$, with $u = ct \in [0, U]$, is at most

$$\frac{2(22)_j c^j u_{\max}^{22-j}}{2^{11} 11!}.$$

Indeed, for terms of Gaussian index $k \ge 11$, the ratios of consecutive absolute derivative terms are bounded respectively by

$$\frac{U^2}{2(k+1)}, \qquad \frac{U^2}{2k}, \qquad \frac{U^2(2k+1)}{2k(2k-1)}.$$

All decrease with k and are less than $1/8$ at $k = 11$. The cosine ratios are $U^2/\{(2k+2-j)(2k+1-j)\} < 1/8$. The first cosine tail is $2^{11}11!/22! < 1/4$ times the first Gaussian tail, for every j. Summing both geometric tails gives a factor less than $(1 + 1/4)/(1 - 1/8) = 10/7 < 2$. Uniform absolute convergence justifies these differentiations. For the removable function $2(\cos u - e^{-u^2/2})/u^3$, divide the series symbolically, so its value tail is bounded by $4u_{\max}^{19}/(2^{11}11!)$ and its value at zero is obtained directly from the series. No numerical division by a nearly zero frequency is needed.

5.8 Uniform low-frequency remainder

The next estimate requires actual fourth through sixth moments, acquired from support or separately bounded. A third-moment ceiling alone is insufficient.

Assume $n \ge N > 2$ and common bounds

$$\mu_4 := \mathbb{E}X^4 \le M \le 111/10, \quad |\gamma| \le g \le 37/20, \quad \mu_6 := \mathbb{E}X^6 \le M_6,$$

and put

$$\mathfrak{a} = (3-\mu_4)/24, \quad \mathfrak{z} = (10\gamma - \mathbb{E}X^5)/120, \quad |\mathfrak{a}| \le A, \quad |\mathfrak{z}| \le Z.$$

Here A, Z, M_6 are nonnegative common ceilings, not quantities inferred from bounded third moments. Define

$$E = M_6/720 + g^2/72 + (1/2 + g^2/144)^3/5, \qquad B = A + Z/2 + E/4,$$

$$a_0 = \frac{N-1}{2N}, \quad b_0 = \frac{45-M}{96}(1-2/N) > 0, \quad \lambda_N = \sqrt{\pi/2}\left(\frac{N}{N-1}\right)^{3/2}.$$

Proposition 5.6. *For all the laws and thresholds above,*

$$I_0(H) \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n},$$

where, for any $c_0 \ge \lambda_N$ and $c_1 \ge \lambda_N/2$,

$$\begin{aligned} L_0 &= c_0 Z + c_1 A g + 251 g^3/17280, \\ L_H &= c_0 A + 251 g^2/4800, \\ L_2 &= E/(2a_0^2) + B^2/(2b_0^3). \end{aligned}$$

Proof. Let $h(t) = \exp(-t^2/2 - i\gamma t^3/6)$. Taylor's integral remainder for e^{itX}, followed by expectation, gives its expansion through degree five with error at most $\mu_6 t^6/720$. For $w = -t^2/2 - i\gamma t^3/6$ and $0 \le t \le 1/2$,

$$|w| \le t^2(1/2 + g^2/144) < 1/6, \quad |e^w - 1 - w - w^2/2| \le |w|^3 e^{|w|}/6 \le |w|^3/5.$$

The first bound follows from $\sqrt{1/4+y} \le 1/2 + y$; for the last use $e^{1/6} \le (1-1/6)^{-1} = 6/5$. Comparing the two polynomials gives

$$f - h = -\mathfrak{a}t^4 - i\mathfrak{z}t^5 + r_6(t), \qquad |r_6(t)| \le Et^6, \qquad |f-h| \le Bt^4. \tag{21}$$

The degree-six term in $w^2/2$ costs $g^2t^6/72$; it has not been discarded.

Four integrations of $1 - \cos u \ge 0$ show $\cos u \le 1 - u^2/2 + u^4/24$ for every real u. Independence gives $\mathbb{E}(X - X')^4 = 2\mu_4 + 6$, and hence

$$|f(t)|^2 \le 1 - t^2 + (M+3)t^4/12 \le 1 - (45-M)t^2/48.$$

Thus $|f|, |h| \le e^{-(45-M)t^2/96}$ throughout the low interval. Every point on their complex joining segment obeys the same bound. Taylor's integral formula for the polynomial $v \mapsto v^n$ proves

$$f^n - h^n = n(f-h)h^{n-1} + R_2, \quad |R_2| \le \tfrac{1}{2}n(n-1)B^2t^8 e^{-(n-2)(45-M)t^2/96}.$$

The leading remainder in (21) and this power remainder cost at most L_2/n after division by t^3 and integration: use $n - 1 \ge 2a_0 n$, $(n-2)(45-M)/96 \ge b_0 n$, $n(n-1) \le n^2$, and $\int_0^\infty t^3 e^{-At^2}dt = 1/(2A^2)$, $\int_0^\infty t^5 e^{-At^2}dt = A^{-3}$.

For the remaining leading terms let $b = n - 1$ and $v = b\gamma t^3/6$. Their imaginary part after rotation by Ht is bounded by

$$ne^{-bt^2/2}\{(Z + A|H|)t^5 + bAgt^7/6\}.$$

The Gaussian moments of orders two and four, after dividing by t^3, give $n\sqrt{\pi/2}b^{-3/2}(Z + A|H| + Ag/2)$. This is at most $\lambda_N(Z + A|H| + Ag/2)/\sqrt{n}$. Finally put $u = n\gamma t^3/6$. Direct sine and cosine remainders give

$$|\Im[e^{-itH}(h^n - e_n)]| \le e^{-nt^2/2}(|u|^3/6 + |H|tu^2/2).$$

The Gaussian moments of orders six and four respectively give $\sqrt{\pi/2}g^3/(432\sqrt{n})$ times 5, and $\sqrt{\pi/2}g^2|H|/(24\sqrt{n})$. Using $\sqrt{\pi/2} < 251/200$ gives precisely the two displayed rational coefficients. Every extended integrand was nonnegative. The only index comparisons were the displayed rational inequalities; in particular the law and H can vary arbitrarily with n. □

A common Pearson envelope. For actual support $|X| \le R \le 6$ and $\mathbb{E}|X|^5 \le E_5$, the triangle LOW bound may use $M_6 = RE_5$ and $A = \max(2, M - 3)/24$. To bound its fifth coefficient, put $Q = X^2 - \gamma X - 1$. Multiplication and the first three standardized moments give

$$\mathbb{E}Q^2 = \mu_4 - 1 - \gamma^2, \qquad \mathbb{E}X^5 = \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X + 2\gamma)].$$

Reflect this auxiliary calculation so that $t = |\gamma| \ge 0$. For $r \ge \max(R, 2g)$ the two signs of $\mathbb{E}X^5 - 10\gamma$ are at most

$$P_r(t) = r(M - 1) + (10 - 2M)t - rt^2 + t^3, \qquad Q_r(t) = r(M - 1) + (2M - 10)t - rt^2 - t^3.$$

Indeed the coefficients of μ_4 before replacing it by M are $r - 2t, r + 2t \ge 0$. More precisely, for $z = (10\gamma - \mathbb{E}X^5)/120$ this proves $-Q_r(t) \le 120z \le P_r(t)$. If $M \le 5$, $P_r \ge Q_r$ and $P_r \le r(M - 1) + (10 - 2M)t - (r - g)t^2$. If $M > 5$, $P_r \le r(M - 1)$ and $Q_r \le r(M - 1) + (2M - 10)t - rt^2$. Both quadratics are maximized on $[0, g]$ at their clipped vertices. Thus for $r_0 = \max(R, 2g) > g$ the exact selected value is

$$Z_r = r_0(M - 1) + at_* - bt_*^2, \qquad t_* = \min\{g, a/(2b)\},$$
$$(a, b) = \begin{cases} (10 - 2M, r_0 - g), & M \le 5, \\ (2M - 10, r_0), & M > 5. \end{cases}$$

For the radius-six choice $P_6'' = -12 + 6t < 0$ on $[0, g]$. Consequently its selected ceiling is

$$Z_0 = \begin{cases} P_6(g), & M \le 5,\ P_6'(g) \ge 0, \\ 6(M - 1) + (10 - 2M)^2/[4(6 - g)], & M \le 5,\ P_6'(g) < 0, \\ 6(M - 1) + (2M - 10)^2/24, & M > 5. \end{cases}$$

The first branch uses monotonicity up to g; the others use the unrestricted quadratic vertices just proved. Hence $120Z = \min\{E_5 + 10g, Z_0, Z_r\}$ bounds the actual fifth coefficient. Every radius, moment and sign assumption refers to this same marginal.

5.9 The complete high-frequency half-line

A nearly lattice law can retain characteristic-function peaks away from zero. The high-frequency argument therefore uses an event involving the actual difference $X - X'$ and

partitions the oscillations around their zeros. The six contributions below cover intermediate frequencies, nonzero-cell cores, nonzero-cell wings, the small-angle part of the zero cell, the rest of the zero cell, and the signed Gaussian comparator. Each optional improvement has its own stated positivity and monotonicity conditions.

For independent copies of the same marginal suppose an actual event $\mathcal{E}$ satisfies

$$\mathbb{P}(\mathcal{E}) \geq \alpha > 0, \quad |X - X'| \geq d > 0 \text{ on } \mathcal{E}, \quad \mathbb{E}[(X - X')^2 \mid \mathcal{E}] \leq M_E.$$

Fix $b = 9/10$, $0 < r \leq 1/4$, and

$$h = 333/106 - (22/7)^3 r^2/6 > 0, \quad \kappa = \alpha(23/48)^2 d^2, \quad L(x) = \frac{x+2}{x^2+4x+2},$$

$$S_3 = \sum_{j=1}^{20} j^{-3} + 1/800, \quad S_5 = \sum_{j=1}^{20} (j-r)^{-5} + \frac{1}{4(20-r)^4}.$$

Let $c(u) = u/2 - (M+3)u^2/24$ and set

$$c_* = \min\{c(1/4), c(81/100)\}, \quad c_q = 1/8 - (M+3)/384, \quad s = [1 - (M+3)53/600]/2.$$

Proposition 5.7. *Suppose $c_* > 0$. For every $n \geq N$, $\sqrt{n} I_\infty$ is at most the sum of the following six terms:*

$$\begin{aligned}
T_1(n) &= 2\sqrt{n} e^{-nc_*}, \\
T_2(n) &= \frac{M_E}{4\pi^{3/2} h \sqrt{\alpha}} \{S_3 + 3S_5/(n\alpha h^2)\}, \\
T_3(n) &= \frac{M_E \sqrt{n}}{2\pi^2} e^{-n\alpha h^2 r^2}, \\
T_4(n) &= \frac{e^{-n\kappa b^2}}{2b^4 \kappa \sqrt{n}}, \\
T_5(n) &= \frac{\sqrt{n}}{2b^2} e^{-n\alpha(23/48)^2}, \\
T_6(n) &= \sqrt{n} e^{-n/8} \{2L(n/8) + \sqrt{M-1}/3\}.
\end{aligned}$$

If $s > 0$, T_1 may be replaced by its minimum with $2\sqrt{n} e^{-nc_q} L(ns/4)$. The fourth term may be replaced by its minimum with $\sqrt{n} e^{-n\kappa b^2} L(n\kappa b^2)/(2b^2)$. The common-floor sum at N is valid if

$$2Nc' > 1 \quad \text{for } c' \in \{c_*, \alpha h^2 r^2, \alpha(23/48)^2, 1/8\}.$$

For the optional first term also require $2Nc_q > 1$; for the optional fourth term require $N\kappa b^2 \geq 1/4$. If an optional gate fails, the simpler term remains available.

Proof. The triangle inequality separates $|f^n|$ and the entire comparator. On $[1/2, b]$, symmetrization gives $|f|^n \leq e^{-nc(t^2)}$. The function c is concave, so $c(t^2) \geq c_*$ and the integral of t^{-3} is less than two. This gives T_1. Also

$$c(u) - c_q - s(u - 1/4) = (M+3)(u - 1/4)(81/100 - u)/24 \geq 0.$$

When $s > 0$ this chord may be integrated to infinity.

For completeness, the tail calculation used here is, for $x = nab^2 > 0$,

$$\sqrt{n} \int_b^\infty e^{-nat^2} t^{-3} dt \leq \frac{\sqrt{n} e^{-x}}{2b^2} L(x). \tag{22}$$

To see where the rational function L comes from, let U have density e^{-u} on $u \ge 0$. Substitution followed by positive Fubini gives

$$\sqrt{n}\int_b^\infty e^{-nat^2}t^{-3}dt = \frac{\sqrt{n}e^{-x}}{2b^2}\mathbb{E}\frac{U}{x+U}.$$

For example, after putting $v = na(t^2-b^2)$ the remaining factor is $\int_0^\infty xe^{-v}(x+v)^{-2}dv$; integration by parts makes it $\mathbb{E}[U/(x+U)]$. For $x > 0$ put

$$J_x = \mathbb{E}\frac{1}{x+U}, \qquad v_0 = \begin{pmatrix}1\\1\end{pmatrix}, \qquad G = \mathbb{E}\left[(x+U)\begin{pmatrix}1 & U\\ U & U^2\end{pmatrix}\right] = \begin{pmatrix}x+1 & x+2\\ x+2 & 2x+6\end{pmatrix}.$$

The entries use $\mathbb{E}U = 1$, $\mathbb{E}U^2 = 2$, and $\mathbb{E}U^3 = 6$. For $q(U) = c_1 + c_2U$, weighted Cauchy–Schwarz in its squared-residual form gives

$$0 \le \mathbb{E}\frac{[1-(x+U)q(U)]^2}{x+U} = J_x - 2v_0^Tc + c^TGc.$$

The matrix G is positive definite: its leading entry is positive and $\det G = x^2+4x+2 > 0$. Choosing $c = G^{-1}v_0$ yields

$$J_x \ge v_0^TG^{-1}v_0 = \frac{x+3}{x^2+4x+2}, \qquad \mathbb{E}\frac{U}{x+U} = 1 - xJ_x \le \frac{x+2}{x^2+4x+2} = L(x).$$

This proves (22) for every $a, b, n > 0$ with the same function L. Applying it at endpoint $1/2$ and rate s proves the optional first term.

For $t \ge b$, symmetrization, the actual-event probability bound, and convex Jensen give

$$|f(t)|^n \le e^{-n\mathbb{E}\sin^2(t(X-X')/2)} \le \mathbb{E}[e^{-n\alpha\sin^2(tD/2)} \mid \mathcal{E}], \quad D = |X-X'|.$$

Positive Tonelli allows a separate partition for every $D > 0$. With $\ell = 2\pi/D$, use the zero cell $[0, \ell/2]$ and nonzero cells $\ell[j-1/2, j+1/2]$, $j \ge 1$. In a nonzero cell write $t = \ell(j+v)$. On its core $|v| \le r$, $|\sin\pi v| \ge h|v|$; on its wings $r \le |v| \le 1/2$ it is at least hr. These follow from $\sin x \ge x - x^3/6$ and monotonicity of sin on $[0, \pi/2]$, with directed bounds on π. For $|v| \le r$ Taylor's integral remainder gives

$$\{(j-v)^{-3} + (j+v)^{-3}\}/2 \le j^{-3} + 6(j-r)^{-5}v^2.$$

Extend the nonnegative Gaussian majorant over the whole real line. Its zeroth and second moments, the Jacobian $t^{-3}dt = D^2(4\pi^2)^{-1}(j+v)^{-3}dv$, and $\mathbb{E}[D^2 \mid \mathcal{E}] \le M_E$ give T_2. The two series tails are bounded by their decreasing integrals after 20. The total wing weight is at most $\int_{\ell/2}^\infty t^{-3}dt = D^2/(2\pi^2)$, giving T_3. Extending these sets below b only increases their nonnegative bounds.

In the zero cell put $u = tD/2 \le \pi/2$. Where $u \le 1/2$, $\sin u \ge 23u/24$, and $D \ge d$ yields rate κ. Extending the set to $[b, \infty)$ gives T_4 from $t^{-3} \le t/b^4$; (22) gives its optional improvement. Where $u \ge 1/2$, monotonicity gives $\sin u \ge 23/48$ and the total weight is at most $1/(2b^2)$, proving T_5. Endpoints have zero Lebesgue measure even if the conditional distance distribution has atoms.

Finally $|e_n(t)| \le e^{-nt^2/2}(1 + n|\gamma|t^3/6)$. The first part on the entire $[1/2, \infty)$ uses (22); the second uses $\int_{1/2}^\infty e^{-nt^2/2}dt \le 2e^{-n/8}/n$. Pearson's identity $\mu_4 - 1 - \gamma^2 = \mathbb{E}(X^2 - \gamma X - 1)^2 \ge 0$ gives T_6. No comparator tail is dropped.

For floor evaluation, $\sqrt{n}e^{-nc'}$ decreases for $2nc' > 1$, whereas n^{-1} and $n^{-1/2}e^{-nc'}$ decrease for every $c' > 0$. Also $L'(x) = -(x^2+4x+6)/(x^2+4x+2)^2 < 0$. Thus the optional first

term decreases under its own gate, and the comparator decreases for $n \geq 5$. The sign of the derivative of $\sqrt{x}e^{-x}L(x)$ is the sign of

$$4 - 10x - 22x^2 - 13x^3 - 2x^4,$$

which is negative at $1/4$ and strictly decreasing on $(0, \infty)$. This proves the optional zero-cell gate. Every comparison is of a common scalar majorant, not of characteristic functions for different laws. □

5.10 Uniform remainder conclusion

Under the preceding moment, actual-event and scalar gates, let K_H be any complete HIGH sum evaluated at N, including its permitted optional improvements. Then

$$|E_n| \leq \frac{2}{\pi}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\} \leq \frac{16}{25}\{L_0 + K_H + L_2/\sqrt{N} + L_H|H|\}. \tag{23}$$

The second inequality uses $\pi > 25/8$ and nonnegative coefficients. This supplies a constant Fourier budget and a raw-threshold slope. The scalar gates are imposed on the analytic majorants, not on the rounded implementation. Complete HIGH bounds for different valid windows may be minimized. Each LOW triple remains intact: a constant from one enclosure cannot be combined with another enclosure's threshold slope without a separate justification. These statements are uniform over every law and threshold satisfying the displayed hypotheses; no fixed-law limit is used.

6 The finite induction through forty-two

We prove the finite base by excluding a first violating index. Every calculation then concerns one attained positive closed-event ratio maximizer with score $C = C_n > C_{\mathrm{E}}$, and its immediate predecessor has score at most C_{E}. The computational propositions below assert inequalities on complete parameter regions; the intervening arguments prove that this same marginal lies in those regions. No bound on its number of atoms is assumed.

6.1 Quantitative starting cases

The small cases provide quantitative starting bounds for the later induction. The second and third cases use the same idea: a hypothetical violating extremizer has bounded support, and a polynomial majorizes its closed sum event on the whole support product. Independence then evaluates the polynomial using moments of the same marginal. No restriction on the number of atoms is imposed.

Theorem 6.1. *For every standardized law with finite third absolute moment,*

$$C_1 \leq \frac{2}{5}, \qquad C_2 \leq \frac{2\sqrt{2}}{7}, \qquad C_3 \leq \frac{59\sqrt{3}}{250}.$$

All three upper bounds are strictly smaller than C_{E}.

One summand. The first numerical assertion is the following checked family of 150 consecutive rational cells $[a, b]$ covering $[-2, 1]$, with two rational coefficients u, v in each cell. It proves

$$L_a - v + uy + vy^2 + \frac{2}{5}|y|^3 - \mathbf{1}_{\{y \leq b\}} \geq 0 \quad (y \in \mathbb{R}), \qquad L_a \leq \Phi(a). \tag{24}$$

Taking expectations cancels the linear and quadratic moment terms. For $t \in [a, b]$ this gives

$$F_P(t) - \Phi(t) \le F_P(b) - L_a \le \frac{2}{5}\rho.$$

For $t \le -2$, Cantelli gives $F_P(t) \le 1/(1+t^2) \le 1/5$. For $t \ge 1$, the discrepancy is at most $1 - \Phi(1) < 2/5$. Since $\rho \ge 1$, these bounds cover both exterior rays. Reflection gives the negative branch, including atoms.

A common support argument for two and three summands. Let $K_2 = 2\sqrt{2}/7$, $K_3 = 59\sqrt{3}/250$ and suppose $C_n > K_n$, where $n = 2$ or 3. Theorem 2.1 supplies a positive closed-CDF maximizing pair. Write

$$z = H/\sqrt{n}, \quad F = \mathbb{P}(S_n \le H), \quad D_0 = F - \Phi(z), \quad \lambda = D_0/\rho, \quad \beta = \mathbb{E}[X|X|].$$

Here D_0 denotes the discrepancy, not the support-gap defect used in the main argument. The contact condition is equivalently

$$n\mathbb{P}(S_{n-1} \le H - y) \le Q_n(y),$$
$$Q_n(y) = \lambda|y|^3 - \frac{z\phi(z) + 3D_0}{2}y^2 - (\sqrt{n}\phi(z) + 3\lambda\beta)y + nF + \frac{z\phi(z) + D_0}{2}, \quad (25)$$

with equality on the entire closed support. The published affine estimate

$$\frac{\sqrt{n}\,\Delta_n(P)}{\rho(P)} \le \frac{1661}{5000}\left(1 + \frac{429}{1000\rho(P)}\right) \quad (26)$$

from Shevtsova [9] rules out $\rho \ge 2$ at $n = 2$ and $\rho \ge 15/8$ at $n = 3$. Together with $|\beta| \le 1$, $\phi(z) < 2/5$ and $|z\phi(z)| < 1/4$, it gives the following lower bounds $Q_n(y) \ge P_n(|y|)$:

n	ρ ceiling	$P_n(r)$
2	2	$\frac{2}{7}r^3 - \frac{67}{56}r^2 - \frac{23}{14}r - \frac{1}{8}$
3	15/8	$\frac{8}{35}r^3 - \frac{4}{5}r^2 - \frac{8}{5}r - \frac{1}{8}$

Indeed, for $n = 2$, $\lambda > 2/7$, $D_0 < 5/7$ and $\lambda < 5/14$ suffice: $\sqrt{2} < 10/7$ gives $\sqrt{2}\phi(z) + 3\lambda < 4/7 + 15/14 = 23/14$. For $n = 3$, the corresponding bounds are $\lambda > 8/35$, $D_0 < 9/20$ and $\lambda < 2/7$. In the latter case (26) gives $C_3\rho < 77/100$, and $\sqrt{3} > 12/7$ then gives $D_0 = C_3\rho/\sqrt{3} < 9/20$. The shifted differences are

$$P_2(6+s) - 2 = \frac{2}{7}s^3 + \frac{221}{56}s^2 + \frac{104}{7}s + \frac{373}{56},$$
$$P_3(6+s) - 3 = \frac{8}{35}s^3 + \frac{116}{35}s^2 + \frac{472}{35}s + \frac{2197}{280}.$$

Every coefficient is positive. Equality in (25) therefore excludes every support point with $|y| \ge 6$, because its left side is at most n.

The two-summand majorants. Put $\gamma = \mathbb{E}X^3$ and

$$v = (1, y, y^2, y^3, |y|^3), \qquad \mu = \mathbb{E}v(X) = (1, 0, 1, \gamma, \rho).$$

For a rational symmetric coefficient array indexed by $i \le j$, let

$$M_c(x,y) = \sum_i c_{ii} v_i(x) v_i(y) + \sum_{i<j} c_{ij}\{v_i(x)v_j(y) + v_j(x)v_i(y)\},$$

and $q_c(\gamma,\rho) = \sum_i c_{ii}\mu_i^2 + 2\sum_{i<j} c_{ij}\mu_i\mu_j$. Independence gives $\mathbb{E}M_c(X_1,X_2) = q_c(\gamma,\rho)$. Lemma 2.2 puts the actual moments in

$$\mathcal{D}_2 = \{(\gamma,\rho) \in [-2,2]\times[1,2] : \rho^2(\gamma^2+4) - (\gamma^2+2)^2 \ge 0\}.$$

The 70 certificate rows use $a_j = -5/2 + j/20$, $b_j = a_j + 1/20$, $0 \le j < 70$. The checked numerical assertions, in the notation of Lemma 4.1, are

$$\begin{aligned} M_{c^{(j)}}(x,y) &\ge \mathbf{1}_{\{x+y\le b_j\}} && ((x,y)\in[-6,6]^2),\\ q_{c^{(j)}}(\gamma,\rho) &\le \Phi(a_j/\sqrt{2}) + \frac{2}{7}\rho && ((\gamma,\rho)\in\mathcal{D}_2). \end{aligned}$$

Each sign quadrant is cut by the event line and triangulated into closed triangles. Degree-six Bernstein coefficients, with exact subdivision, establish the first inequality. The second inequality is checked on moment rectangles; a rectangle is discarded only if $\rho_+^2(g_*+4) - (g_*+2)^2 < 0$, where g_* is the minimum of γ^2 on that rectangle. This is valid because the expression increases with ρ and decreases with γ^2 when $\rho \le 2$.

For a threshold in $[a_j,b_j]$, expectation of the first inequality and the second give discrepancy at most $2\rho/7$. For $H \le -5/2$, Cantelli gives $2/(2+H^2) \le 8/33 < 2/7$; for $H \ge 1$, $1-\Phi(H/\sqrt{2}) < 2/7$. These bounds contradict the chosen violating maximizer. This proves the second assertion of Theorem 6.1.

The three-summand majorants. Use the features

$$(v_0,\dots,v_6)(y) = (1, y, y^2-1, y^3, |y|^3, y^4, y^5).$$

Let $\mathcal{I}$ be the 59 nondecreasing triples from $\{0,\dots,6\}$ whose largest entry is less than 5 or which contain an entry 1 or 2. For $\alpha\in\mathcal{I}$, average the associated product over its distinct coordinate permutations, obtaining V_α. Set $M_c = \sum_\alpha c_\alpha V_\alpha$. The bounded-support argument above makes every feature integrable. Every term involving a fourth or fifth power contains a zero-mean factor X_i or X_i^2-1 in a different coordinate. Independence therefore gives the exact identity

$$\mathbb{E}M_c(X_1,X_2,X_3) = q_c(\gamma,\rho) = \sum_{\substack{\alpha\in\mathcal{I}\\ \alpha \text{ uses only } 0,3,4}} c_\alpha \gamma^{\#3(\alpha)}\rho^{\#4(\alpha)}.$$

There are ten surviving monomials and 49 zero-expectation controls. The controls improve the pointwise majorant without adding moment parameters.

The second checked numerical assertion uses 111 rational rows forming a consecutive closed cover of $[-7/2,3/2]$. They satisfy

$$\begin{aligned} M_{c^{(j)}}(x_1,x_2,x_3) &\ge \mathbf{1}_{\{x_1+x_2+x_3\le b_j\}} && ((x_1,x_2,x_3)\in[-7,7]^3),\\ q_{c^{(j)}}(\gamma,\rho) &\le \Phi(a_j/\sqrt{3}) + \frac{59}{250}\rho && ((\gamma,\rho)\in\mathcal{D}_3), \end{aligned}$$

where $\mathcal{D}_3$ replaces 2 by 15/8 in the rectangular bounds defining $\mathcal{D}_2$. Symmetry reduces the sign patterns to four ordered-coordinate orthants. A box is discarded only when it has no ordered point. On every other box, degree-five tensor Bernstein bounds establish the

pointwise inequalities. On a box crossing the event plane, the included side is checked by $M - 1 \geq s(x_1 + x_2 + x_3 - b_j)$ with $s \leq 0$, and the other side by the analogous inequality with $s \geq 0$. Existence of a slope is checked by intersecting exact rational intervals; zero plane coefficients are checked directly. This retains the closed event boundary.

As in the two-summand case, expectation bounds the discrepancy on each threshold interval. The exterior bounds are $3/(3 + H^2) \leq 12/61 < 59/250$ for $H \leq -7/2$ and $1 - \Phi(H/\sqrt{3}) < 59/250$ for $H \geq 3/2$. They contradict the violating maximizer and prove the third assertion. The first assertion was proved above. Rational square-root and Machin bounds give $K_2 < 0.405 < 0.409 < C_{\mathrm{E}}$ and $K_3 < 0.409 < C_{\mathrm{E}}$.

6.1.1 Localization at four summands

The third bound is needed quantitatively. Merely knowing $C_3 \leq C_{\mathrm{E}}$ would not give the following support reduction. Write

$$K_3 = \frac{59\sqrt{3}}{250}, \quad c_0 = \frac{256}{625}, \quad T = \frac{4097}{10000}; \qquad K_3 < c_0 < T < C_{\mathrm{E}}.$$

For any standardized law set $\beta = \mathbb{E}[X|X|]$, $u = \mathbb{E}X_+ = \mathbb{E}X_- > 0$, $v_\pm = (1 \pm \beta)/2$, and

$$a = \frac{v_-}{u}, \quad b = \frac{v_+}{u}, \quad w = \frac{4u^2}{1 - \beta^2}, \quad Q = \frac{u}{a}\delta_{-a} + (1 - w)\delta_0 + \frac{u}{b}\delta_b. \tag{27}$$

Cauchy's inequality gives $0 < w \leq 1$ and makes Q a standardized probability law. Put $r = \rho(Q) = (1 + \beta^2)/\sqrt{w(1 - \beta^2)}$.

Lemma 6.2. *The nonnegative quantity*

$$W(X) = X_+(X_+ - b)^2 + X_-(X_- - a)^2$$

has expectation $\eta = \rho(P) - r$. Thus $r \leq \rho(P)$, with equality exactly when $P = Q$. Moreover, writing $c(X) = b$ for $X > 0$ and $c(X) = a$ for $X < 0$,

$$\begin{aligned} \mathbb{E}X^3 &= b - a + \mathbb{E}[\operatorname{sgn}(X)W(X)], \\ \mathbb{E}X^4 &= a^2 - ab + b^2 + \mathbb{E}[(|X| + 2c(X))W(X)] \end{aligned} \tag{28}$$

whenever the fourth moment is finite.

Proof. On either half-line put $y = |X|$ and $c = a$ or b. The identities $y^3 = y(y-c)^2 + 2cy^2 - c^2y$ and $y^4 = (y+2c)y(y-c)^2 + 3c^2y^2 - 2c^3y$ integrate using $\mathbb{E}[y; \text{half-line}] = u$ and $\mathbb{E}[y^2; \text{half-line}] = uc$. If $\mathbb{E}W = 0$, the support is contained in $\{-a, 0, b\}$ and the first two moments fix the masses. Conversely $P = Q$ has $W = 0$. □

The three-point law is a reference; the original law remains arbitrary.

Proposition 6.3. *If $C_4 > c_0$, a positive closed-CDF maximizing pair satisfies*

$$\operatorname{supp} P \subset (-101/25, 101/25), \quad \rho(P) < 46/25, \quad 625/2116 < w \leq 1, \quad |\beta| < 321/500.$$

In particular it belongs to the closed numerical class with $|X| \leq 17/4$ and $|\beta| \leq 13/20$.

Proof. Use Theorem 2.1. The affine inequality (1.2) first gives $\rho < 979013/532500 < 46/25$, without a support assumption. Since $r \leq \rho$ and $w \leq 1$, the inequality

$$\frac{(1 + (321/500)^2)^2}{1 - (321/500)^2} > (46/25)^2$$

gives $|\beta| < 321/500$. Put $C = C_4$, $z = H/2$, $s = \sqrt{4/3}$. The contact expression, using the quantitative bound on the three-summand predecessor, obeys

$$I_{\mathrm{new}}(y) \le \mathcal{H}_4(z,y) + 4\rho(K_3 - C) + 4\rho(s-1)K_3 \\ + C\{-|y|^3 + \tfrac{3}{2}\rho y^2 + 3\beta y - \rho/2\}, \mathcal{H}_4(z,y) = \quad 8\left\{\Phi\left(\frac{z - y/2}{\sqrt{3/4}}\right) - \Phi(z)\right\} \\ + 4y\phi(z) + z\phi(z)(y^2 - 1).$$

Taylor expansion in the translation and the variance gives, uniformly in z, y,

$$|\mathcal{H}_4(z,y)| < \frac{\phi(0)}{6}|y|^3 + \frac{|y|}{5} + \frac{1}{15}.$$

Here $\sup|\phi''| = \sup|(x\phi(x))'| = \phi(0) < 2/5$ and $\sup|x(3-x^2)\phi(x)| < 3/5$ suffice. For the last bound, on $t = x^2 \le 3$ compare e^t with its cubic Taylor polynomial and use

$$\tfrac{9}{4}(1 + t + t^2/2 + t^3/6) - t(3-t)^2 = \tfrac{3}{4}\{7(t - 9/14)^2 + 3/28\} + \tfrac{5}{8}t^2(3-t) > 0.$$

For $t \ge 3$, use $|x|^3\phi(x) < 1/2$. The second derivative of $\Phi((z - y/2)/\sqrt{1-\lambda})$ has absolute value below $4/15$ on $0 \le \lambda \le 1/4$, giving the remainder $1/15$ after multiplication by eight.

Discard $4\rho(K_3 - C) < 0$ and use $4(s-1) < 5/8$. For $R = |y| \ge 4$, the remaining coefficient of C is at most $B(R) = -R^3 + 69R^2/25 + 963R/500 + 23/100$. All coefficients of $B(4+t)$ are negative, so substituting $C > c_0$ is now valid. Consequently

$$I_{\mathrm{new}}(y) < -\frac{643}{1875}R^3 + \frac{17664}{15625}R^2 + \frac{77257}{78125}R + \frac{7541}{46875}.$$

The coefficients after translation by $R = 101/25 + t$ are all negative. This excludes $|y| \ge 101/25$ from the closed support by contact. Finally $r^2 \ge 1/w$ gives $w > 625/2116$. □

6.2 The physical comparison at four summands

Write $F_{4,P}(H) = \mathbb{P}_P(S_4 \le H)$, $T = 4097/10000$, and $R_* = 17/4$. The preceding localization reduces a contrary extremizer to $|X| \le R_*$, $\rho < 46/25$, $625/2116 \le w \le 1$, $|\beta| \le 13/20$. The reference Q and excess $\eta = \rho - r$ retain the original marginal; they do not replace it by a three-point law.

One measure carries all frequencies. For $\eta > 0$ set $\nu = W\,dP/\eta$, tagged by $(\sigma, y) = (\operatorname{sgn} X, |X|)$, and write $c_+ = b, c_- = a$, $p_0 = 1 - w$, $g(q) = \varphi_Q(q)$. For $r_q(\sigma, y) = \exp(i\sigma q\,\cdot)[0, c_\sigma, c_\sigma, y]$, Hermite interpolation at $0, c, c$ gives

$$\varphi_P(q) - g(q) = \eta \int r_q\,d\nu, \qquad |r_q| \le |q|^3/6. \tag{29}$$

The linear and quadratic terms agree on each sign class; the constant terms agree after both classes and the zero atom are included. Define $k_0 = (c_\sigma^2 y)^{-1}$, $k_2 = (y - c_\sigma)^{-2}$ and $k_1 = (y - c_\sigma)^{-1}$ off the contacts, which have zero ν mass. Multiplication by $W = y(y-c)^2$ and integration give

$$\eta \int k_0\,d\nu = p_0 - P\{0\} \le p_0, \quad \eta \int_\sigma k_2\,d\nu = u - c_\sigma P\{\sigma c_\sigma\} \le u, \quad \int_\sigma k_1\,d\nu = 0.$$

Cauchy and the k_2 bound prove absolute integrability of k_1. At $\eta = 0$, $P = Q$ and no measure division is used. Also $a, b \le R_*$, since $\mathbb{E}[y^2; \sigma] \le R_*\mathbb{E}[y; \sigma]$. Thus for fixed nonnegative $\lambda_0, \lambda_\pm$ and signed $\theta_\pm$, any whole-domain bound

$$S_H \ge \sup_{\sigma,\, 0 < y \le R_*} \{A_H - \lambda_0 k_0 - \lambda_\sigma k_2 - \theta_\sigma k_1\}$$

with $A_H = \Re \sum_j 4\kappa_j(H) g(q_j)^3 r_{q_j}$ gives

$$\eta \int A_H \, d\nu \le \Gamma_H(\eta) := p_0\lambda_0 + u(\lambda_+ + \lambda_-) + S_H\eta. \tag{30}$$

The signed frequency terms are summed before taking this supremum. Neither S_H nor Γ_H is required to be nonnegative.

The nonlinear response. For $\psi = g + \delta$, the exact identity is

$$\psi^4 - g^4 = 4g^3\delta + \delta^2(\psi^2 + 2\psi g + 3g^2).$$

If $|g| \le A \le 1$, $|\psi| \le 1$, $|\delta| \le E$, valid remainder bounds are

$$E^2(p^2 + 2pA + 3A^2), \quad p = \min(1, A + E), \qquad v(4A^2 + 2 - v), \quad v = \min(E^2, 2A^2 + 1).$$

The second follows from $12\delta^2 \int_0^1 (1-t)(g+t\delta)^2 dt$ and $|g+t\delta|^2 = (1-t)|g|^2 + t|\psi|^2 - t(1-t)|\delta|^2$. The sharper disk bound used by some cells is also finite algebra. After rotating $g = a \ge 0$, put $x = \Re\delta$, $s = |\delta|^2$; the squared remainder equals

$$s^2\{36a^4 + 4a^2 s + s^2 + 8ax(6a^2 + s) + 24a^2x^2\}.$$

Convexity in x leaves collinear endpoints or $|\psi| = 1$. With $r_+ = \min(E, 1)$, $m = \min(A, 1-r_+)$, $r_- = \min(E, 1 + A)$, the collinear modulus bounds are

$$r_+^2(6m^2 + 4mr_+ + r_+^2), \quad r_-^2(6A^2 - 4Ar_- + r_-^2), \quad \begin{cases} r_-^4, & r_- \le 1, \\ r_-^2(3r_-^2 - 8r_- + 6), & r_- \ge 1. \end{cases}$$

On the unit circle maximize $Q_A(s) = s^2[18A^4 + 12A^2 + 6 - (12A^2 + 8)s + 3s^2]$ on $[(1 - A)^2, \min(E^2, (1 + A)^2)]$: only its endpoints and $s_- = (3A^2 + 2 - A\sqrt{4 - 3A^2})/2$, if inside, are needed. Convexity in a^2 reduces the remaining disk radii to $a = A$ or a collinear endpoint. The largest collinear value and square root of this circle maximum therefore bound the modulus; the minimum with either preceding complete bound is valid. On an excess cell $[L, U]$, an affine radius $e_j(\eta) \ge 0$ is used only when proved throughout the cell. Freezing $p_j = \min(1, a_j + \max(e_j(L), e_j(U)))$ gives a quadratic remainder with nonnegative leading coefficient. Freezing the entire radius instead gives a valid constant disk bound. Bounds are combined as complete alternatives, never coefficientwise.

From a finite frequency sum to every threshold. The closed-CDF majorant of Lemma 5.5, with $\widehat{K}(t) = (1 - t) + iR(t)$ and $R(t) = (1 - t)\cot(\pi t) + 1/\pi$, gives

$$F_{4,P}(H) \le J_P(H/2), \quad J_P(x) = \tfrac{1}{2} + \int_0^1 \Re\{\widehat{K}(t) e^{-6itx} \varphi_P(3t)^4\} \, dt.$$

This is a real compensated integral, including the closed boundary. Put $Z = S_4/2$, $D_x = x - Z$, $S(t) = tR(t)$. Its integrand is $(1 - t)\mathbb{E}\cos(6tD_x) + S(t)\mathbb{E}[6D_x \operatorname{sinc}(6tD_x)]$. The cotangent partial fractions give $0 \le S \le 1/3$, $|S'| < 2$, $|S''| < 5$, and therefore

$$|F_x''(t)| \le 42\mathbb{E}|D_x| + 108\mathbb{E}D_x^2 + 24\mathbb{E}|D_x|^3.$$

For the derivative constants, write $S = \pi^{-1} - 2\pi^{-1}(A + MG)$ with $A = t^2/(1+t)$, $M = t^2(1-t)$ and $G = \sum_{j \ge 2}(j^2 - t^2)^{-1}$. On $[0, 1]$, $(|A'|, |A''|, |M|, |M'|, |M''|) \le (3/4, 2, 1/6, 1, 4)$ and $(G, |G'|, |G''|) \le (3/4, 10/27, 106/135)$, by telescoping and $j^2 - t^2 \ge 3j^2/4$. For $t > 1/2$, put $v = 1 - t$ and use $v\cot(\pi v) \le 1/\pi$ to obtain $R(t) \ge 0$; below $1/2$ positivity is immediate. These

estimates imply the stated bounds; the sinc integral representation justifies differentiation by the third moment.

From (28) choose whole-cell bounds

$$\kappa_* \ge \min(\rho, |b-a|+\eta), \quad M_{4,*} \ge \min(R_*\rho, a^2-ab+b^2+(R_*+2\max(a,b))\eta).$$

Then $D_2 = x^2+1$, $D_4 = x^4+6x^2+2|x|\kappa_*+(M_{4,*}+9)/4$ bound $\mathbb{E}D_x^2, \mathbb{E}D_x^4$, and Cauchy bounds the odd absolute moments. For $N = 256$, $t_j = (j+1/2)/N$, $q_j = 3t_j$, $\kappa_j(H) = \widehat{K}(t_j)e^{-iq_jH}/N$, set

$$C_H(Q) = 1 + 2\Re\sum_j \kappa_j(H)g(q_j)^4 - 2\Phi(H/2),$$

$$Q_H = \min\left\{\frac{687}{8N^2}, \frac{42\sqrt{D_2}+108D_2+24\sqrt{D_2D_4}}{12N^2}\right\}.$$

This midpoint reserve bounds the original physical integral once. The constant alternative uses $|x| \le 2$, $D_2 \le 5$, $D_4 < 54$. To bridge adjacent raw thresholds $H_\ell = -4+\ell/8$, $0 \le \ell \le 57$, use $P_j = \min(1, a_j + e_j + 3/(2N))$, where $a_j \ge |g(q_j)|$ and $e_j \ge |\varphi_P(q_j) - g(q_j)|$ on the excess cell. The increment follows from $\mathbb{E}|X| \le 1$. If d_j is the smaller of the exact integrals over the frequency cell of $t^2(1-t)+t/3$ and $t^2(1-t)+t(1-t)/2$, then $|J_P''| \le B = 36\sum_j d_j P_j^4$. Indeed both polynomials bound $t^2|\widehat{K}(t)|$; for the second, retaining the first cotangent pole gives $S(t) < (1-t)/2$. The single threshold-interpolation reserve is $E_H = (B+1/4)/(16\cdot 8^2)$, since $\sup|\Phi''| < 1/4$. The radii $\eta q_j^3/6$, $\max(a,b)^2 p_0 q_j^2$, and $1+a_j$ are all valid; the second uses $yf[0,c,c,y] = f[c,c,y] - f[0,c,c]$, so $|r_q| \le q^2/y$, and then the k_0 budget. Thus no contact division occurs.

The finite inequalities. Let M_H bound $C_H(Q) - Tr$ on a geometry rectangle, and let $R_2(\eta)$ bound the weighted nonlinear response on $[L,U]$. The assembled bound is

$$2\{F_{4,P}(H) - \Phi(H/2)\} - T\rho \le P_H(\eta) := M_H + Q_H + E_H + 2\Gamma_H(\eta) + 2R_2(\eta) - T\eta. \quad (31)$$

It is convex quadratic in η, so its two nonpositive endpoints suffice. The optional score bound checks both denominator faces $46/25$ and $r_L + \eta$. For the latter the derivative numerator has derivative $P_H''(\eta)(r_L+\eta) \ge 0$, hence no interior maximum. All functions defining M_H, S_H, R_2 are explicit finite expressions. Their whole-cell enclosures have the following licenses. For the reference, Taylor's theorem uses $\partial_i C_H = 8\Re\sum_j \kappa_j g^3 \partial_i g$ and $|\partial_{ik} C_H| \le 8\sum_j |\kappa_j|(3D_iD_k + D_{ik})$. A supporting plane bounds r below: for $f(\beta) = (1+\beta^2)/\sqrt{1-\beta^2}$, $3ff'' - (f')^2 = (9+9\beta^2+15\beta^4-\beta^6)/(1-\beta^2)^3 > 0$ proves convexity of $r = w^{-1/2}f(\beta)$. The response is evaluated without singular divided quotients by

$$r_q = \sum_{m=0}^{60} (i\sigma)^{m+3} b_m h_m + E_q, \quad |E_q| \le (q^3/6)(qR_*)^{61}/61!,$$

where $b_0 = q^3/6$, $b_m = b_{m-1}qR_*/(m+3)$, $h_0 = 1$, $h_m = (y/R_*)h_{m-1} + (m+1)(c/R_*)^m$. Hermite–Genocchi proves this expansion and $|\partial_y r_q| \le q^4/24$, $|\partial_c r_q| \le q^4/12$. The 273 nearest-node cells $y_k = k/64$, clipped to $[0, R_*]$, therefore need radius $q^4/(48\cdot 64)$. The geometry correction uses $|g^3 - g_0^3| \le |g-g_0|(|g|^2 + |g||g_0| + |g_0|^2)$. The nonnegative prices are rational data. Each fixed choice is bounded on every tagged cell; $k_0 \ge 1/(c_U^2 y_U)$, while the other cost is $-\lambda_\sigma z^2 - \theta_\sigma z$, $z = 1/(y-c)$. Its endpoints and admissible vertex, on both reciprocal rays if the cell meets a contact, bound it. Require $\theta_\sigma = 0$ when $\lambda_\sigma = 0$. These rules prove the priced supremum over the whole tagged domain. Factoring the difference of fourth powers

also gives $|\psi^4 - g^4| \le E(p^3 + p^2A + pA^2 + A^3)$, $p = \min(1, A + E)$. This absolute bound, or the full linear-plus-remainder expression, may be used as a complete alternative.

For large excess, condition on j nonnegative summands, assigning zero to that class. With $p = P(X \ge 0)$,

$$\mu_j = \frac{u(j - 4p)}{2p(1-p)}, \quad V_j = \frac{1}{4}\left[j\left(\frac{v_+}{p} - \frac{u^2}{p^2}\right) + (4-j)\left(\frac{v_-}{1-p} - \frac{u^2}{(1-p)^2}\right)\right].$$

Cantelli bounds each conditional CDF by 1 if $x \ge \mu_j$ and by $V_j/(V_j + (\mu_j - x)^2)$ otherwise. The closed sign-mass interval is $w(1-\beta)/2 \le p \le 1 - w(1+\beta)/2$. On each threshold cell use the upper endpoint in Cantelli and the lower endpoint in Φ; degree-four Bernstein bounds retain the joint binomial mixture. A sign-mass cell supplies numerator $d_i \ge 0$ and correlated lower $r_i \le r$, giving score $d_i/(r_i + \eta)$. With $r_B \le r$ on the parent, set

$$c_C^0 = \max\{0, \max_i(d_i/T - r_i), 2/(5T) - r_B\}, \qquad c_C = 10^{-4}\lceil 10^4 c_C^0 \rceil + 10^{-4},$$
$$c_S = \max(0, 979013/533000 - r_B), \qquad c_B = \min(c_C, c_S).$$

Cantelli gives the first cutoff; the affine bound gives the second. The two threshold rays $H \le -4$ and $H \ge 25/8$ have score at most $2/5$ by Cantelli and $\Phi(25/16) > 4/5$. Thus $[c_B, \infty)$ is covered, and it meets the low-excess interval at its included endpoint.

Proposition 6.4 (Four-summand numerical assertion)**.** *Divide* $[625/2116, 1] \times [0, 13/20]$ *into* 20×13 *closed rectangles. A rectangle with* $\min r^2 \ge (46/25)^2$ *is excluded. Every other parent, or all four of its coordinate-bisected children, passes the preceding large-excess bound and, on sixteen equal closed subintervals of* $[0, c_B]$*, the low-excess inequalities at all 58 thresholds, their joins, both tails and both reflected orientations. All 260 dispositions and all five parent/child request slots per parent are verified by exact acceptance. Consequently* $C_4 \le T < C_{\mathrm{E}}$.

Derivation from the numerical assertion. Every localized contrary law belongs to a closed parent. Its reference and common defect measure satisfy the derived inequalities on that parent or one of its four children. The midpoint and threshold reserves cover all central thresholds, and the two tails cover the rest. Every $\eta \ge 0$ belongs to the low or large-excess range, including $\eta = 0$ and the common endpoint. Hence its score is at most T, contradicting $C_4 > T$. Reflection accounts for the other discrepancy sign. □

6.3 One support criterion for the remaining finite indices

The support calculations throughout the finite range use one elementary criterion. Its inputs are a lower bound for the attained score, upper bounds for moments of the same marginal, and a predecessor allowance. The Gaussian calculation keeps the actual integer n throughout.

Lemma 6.5 (Direct Gaussian support criterion)**.** *Let* $n \ge 2$*, and let* (P, H) *be an attained positive closed-event ratio maximizer with score* C*. Write* $\rho = \mathbb{E}|X|^3$*,* $\beta = \mathbb{E}[X|X|]$ *and* $A = \sqrt{n-1}\Delta_{n-1}(P)/\rho$*. Suppose known numbers* c, r, b_-, b_+, S, e, q *satisfy*

$$0 < c \le C, \quad 1 \le \rho \le r, \quad -1 < b_- \le \beta \le b_+ < 1, \quad q \ge 0,$$

$$\sqrt{n/(n-1)} \le S, \qquad n(\sqrt{n/(n-1)} - 1) - 1/2 \le e, \qquad n(A - C)_+ \le q.$$

Fix $R \ge 1$ *and a support sign* $\eta \in \{-1, 1\}$*, and put* $B_+ = b_+$*,* $B_- = -b_-$*. Define*

$$Q_\eta(v) = -v^3 + \tfrac{3}{2}rv^2 + 3B_\eta v + re, \qquad L_\eta(v) = \tfrac{2}{5}(v^2/2 + 1/2 + e) + cQ'_\eta(v).$$

Suppose that

$$Q_\eta(R), Q'_\eta(R), Q''_\eta(R), L_\eta(R), L'_\eta(R) < 0, \qquad 1/5 - 3c < 0. \tag{32}$$

Define the actual-index Gaussian expression

$$G_n(H,t) = n\sqrt{n}\{\Phi((H-t)/\sqrt{n-1}) - \Phi(H/\sqrt{n})\} + nt\phi(H/\sqrt{n}) + \tfrac{H}{2}(t^2-1)\phi(H/\sqrt{n}),$$

and suppose U_η satisfies

$$\sup_{H\in\mathbb{R}} \eta G_n(H,R) \le U_\eta, \qquad U_\eta + cQ_\eta(R) + rSq < 0.$$

Then $\operatorname{supp} P$ *contains no point in the ray* $\{\eta v : v \ge R\}$*. The two signs can be checked separately, with different signed-moment bounds.*

Proof. Contact gives

$$0 \le G_n(H,t) + C\{-|t|^3 + \tfrac{3}{2}\rho t^2 + 3\beta t + \rho e_n\} + \rho s_n n(A-C),$$

where $s_n = \sqrt{n/(n-1)}$ and $e_n = n(s_n-1) - 1/2$. The last term is at most rSq. At $t = \eta v$ with $v \ge R$, the bracket is at most $Q_\eta(v)$. Since $Q'''_\eta = -6$, the first three gates make this polynomial strictly negative on the full ray. Thus multiplication by $C \ge c$ gives an upper bound $cQ_\eta(v)$.

The derivative is $n\phi(z) + Ht\phi(z) - ns_n\phi(s_n y)$, where $z = H/\sqrt{n}$ and $y = (H-t)/\sqrt{n}$. Taylor's theorem and $\|\phi''\|_\infty = \phi(0)$ bound $|n\phi(y) - n\phi(z) - Ht\phi(z)|$ by $\phi(0)t^2/2$. The identity $\partial_s[s\phi(sy)] = -\phi''(sy)$ bounds the remaining scaling error by $n\phi(0)(s_n-1)$. This proves, for every $n \ge 2$, $|\partial_t G_n(H,t)| \le \phi(0)(t^2/2 + 1/2 + e)$. Therefore the derivative in v of $G_n(H,\eta v) + cQ_\eta(v) + rSq$ is at most $L_\eta(v)$. Its second derivative as a polynomial bound is $L''_\eta = 2/5 - 6c < 0$; the last two endpoint gates imply $L_\eta(v) < 0$ on $[R,\infty)$. Finally $G_n(H,-R) = -G_n(-H,R)$, so the assumed U_η bounds $G_n(H,\eta R)$ uniformly in H. The strict endpoint inequality and monotonicity contradict contact on every point of the specified support ray. □

A finite certificate for the whole threshold line. Let $z = H/\sqrt{n}$ and $a = (H-R)/\sqrt{n-1}$. Direct differentiation gives

$$\partial_H G_n(H,R) = ns_n\phi(a) + \left[-n - RH + \tfrac{1}{2}(R^2-1)(1-H^2/n)\right]\phi(z).$$

Choose a positive rational H_0 such that $H_0 \ge 5\sqrt{n}$ and $H_0 - R \ge 5\sqrt{n-1}$. An outward enclosure of the square roots makes this choice exact. On the whole real line,

$$|\partial_H^2 G_n(H,R)| \le M_{n,R} := \frac{\sqrt{n}}{4}\left(\frac{n}{n-1} + 1\right) + \frac{2}{5}R + \frac{2|R^2-1|}{5\sqrt{n}}.$$

On a closed threshold cell with midpoint h and half-width w, let $[g_-, g_+]$ and $[d_-, d_+]$ enclose $G_n(h,R)$ and $\partial_H G_n(h,R)$. Taylor's theorem then gives the two upper bounds

$$g_+ + \max(|d_-|,|d_+|)w + M_{n,R}w^2/2, \qquad -g_- + \max(|d_-|,|d_+|)w + M_{n,R}w^2/2.$$

They bound $G_n(H,R)$ and $-G_n(H,R)$ respectively throughout the cell. The certificate includes every consecutive cell from $-H_0$ to H_0, including their endpoints.

On the two exterior rays, both distribution-function arguments lie in the same normal tail, so

$$|G_n(H,R)| \le n\sqrt{n}\,\tau + \left(nR + \tfrac{1}{2}H_0|R^2-1|\right)\phi(H_0/\sqrt{n}), \quad |H| \ge H_0,$$

where $\tau = 3\cdot 10^{-7} > \Phi(-5)$ is sufficient. Take the largest positive-cell bound and the exterior bound for U_+, and the largest negative-cell bound and the exterior bound for U_-. This proves a statement about the entire threshold line, not just about its grid points.

Proposition 6.6 (Actual-index support rows)**.** *Under $C_{n-1} \le C_{\mathrm{E}} < C_n$, each row below bounds the support of the same positive closed maximizer by $(-R, R)$. Repeated index ranges mean one check at every actual integer, not a transfer between integers.*

n	*third-moment interval*	R
5	$[1, 11/10], [11/10, 23/20], [23/20, 237/200], [237/200, 6/5]$	$11/5, 47/20, 49/20, 99/40$
5	$[6/5, 5/4], [5/4, 7/5], [7/5, 3/2]$	$13/5, 117/40, 25/8$
5	$[3/2, 8/5], [8/5, 17/10], [17/10, 7361/4000]$	$133/40, 7/2, 15/4$
6–14	$[1, 11/10], [11/10, 23/20]$	$23/10, 12/5$
6–9	$[23/20, 237/200], [23/20, 6/5]$	$5/2, 13/5$
6–9	$[6/5, 7/5], [7/5, 3/2], [3/2, 8/5]$	$3, 16/5, 17/5$
6–9	$[8/5, 17/10], [17/10, 7361/4000]$	$18/5, 19/5$
10–14	$[23/20, 71/50]$	3

The last five lines comprise all 26 low-moment and 25 complementary actual-index rows. Their endpoint tests are instances of the same criterion, including both support rays and all real thresholds.

Mathematical meaning and finite inequalities. Use $c = 819/2000 < C_{\mathrm{E}}$, $q = 0$, and $e \le 91/1000$. For each upper moment endpoint r, the signed-moment bound supplies $|\beta| \le b$, where b is the outward rational root of $(1+b^2)^2 \ge r^2(1-b^2)$ recorded in the support certificate. The certificate verifies the five strict polynomial gates and the strict Gaussian endpoint test. For $n = 5$ use $H_0 = 16$; for $6 \le n \le 9$ use $H_0 = 35/2$, enlarged to 18 for the complementary rows at $n = 9$; for $10 \le n \le 14$ use $H_0 = 22$. All threshold meshes have width $1/10$. The value at each endpoint, enlarged by $M_{n,R}/800$, bounds both signs on the intervening cell: the interpolation error for $|f''| \le M$ is $Mh^2/8$, by comparison with $\pm M(x-a)(b-x)/2$. The displayed exterior-ray argument applies because $H_0^2 \ge 25n$ and $(H_0 - R)^2 \ge 25(n-1)$. All numerical comparisons are strict rational inequalities with the enclosures of Section 4.2. Thus the preceding support criterion applies before any of these radii enters a Fourier estimate. Additional actual-index rows at 15–39, and the separate two-endpoint support test at 40–42, use the same contact license; their complete input domains are part of the numerical propositions below. □

6.4 Finite-frequency refinements

At a small index, replacing a finite frequency integral by an integral over the whole half-line can lose much of the available margin. The following estimates retain the finite interval and the actual integer power. They use the moments of the same supported marginal throughout.

6.4.1 A finite-interval compensated estimate

Let $n > 2$ be an integer, $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $|X| \le R \le 6$, $\gamma = \mathbb{E}X^3$, $\mu_j = \mathbb{E}X^j$, and $f(t) = \mathbb{E}e^{itX}$. Assume $\mu_4 \le M \le 111/10$ and $|\gamma| \le g \le 37/20$. The simultaneous coefficient and moment bounds in the signed moment atlas of Section 7.2 give

$$|a| \le A_*, \quad |z| \le Z, \quad \mu_6 \le M_6, \qquad a = (3 - \mu_4)/24, \quad z = (10\gamma - \mu_5)/120.$$

Define

$$E = \frac{M_6}{720} + \frac{g^2}{72} + \frac{(1/2 + g^2/144)^3}{5}, \qquad B = A_* + Z/2 + E/4,$$

$$a_n = (n-1)/2, \qquad b_n = (n-2)(45 - M)/96, \qquad I_j(A) = \int_0^{1/2} t^j e^{-At^2}\, dt.$$

All these quantities are nonnegative, and the two damping rates are positive. Put

$$L_0 = \sqrt{n}\left\{nZI_2(a_n) + \frac{n(n-1)A_*g}{6}I_4(a_n) + \frac{n^3g^3}{1296}I_6(n/2)\right\},$$
$$L_H = \sqrt{n}\left\{nA_*I_2(a_n) + \frac{n^2g^2}{72}I_4(n/2)\right\},$$
$$L_2 = n\left\{nEI_3(a_n) + \frac{n(n-1)B^2}{2}I_5(b_n)\right\}.$$

For every real raw threshold H, the complete low-frequency estimate is

$$\int_0^{1/2} \frac{\left|\Im\left[e^{-itH}\{f(t)^n - e^{-nt^2/2}(1 - in\gamma t^3/6)\}\right]\right|}{t^3}\,dt \le \frac{L_0 + L_H|H|}{\sqrt{n}} + \frac{L_2}{n}. \tag{33}$$

To prove this finite-interval version, use the Taylor decomposition (21) with $h(t) = e^{-t^2/2 - i\gamma t^3/6}$. The same polynomial-power remainder gives $|f^n - h^n - n(f-h)h^{n-1}| \le n(n-1)B^2t^8e^{-b_nt^2}/2$. Retain the integration interval $[0, 1/2]$ in that proof. The leading rotated imaginary terms are bounded by $ne^{-a_nt^2}[(Z + A_*|H|)t^5 + (n-1)A_*gt^7/6]$; the cubic Gaussian phase remainder contributes $e^{-nt^2/2}(|n\gamma t^3/6|^3/6 + |H|t(n\gamma t^3/6)^2/2)$. Division by t^3 and integration give exactly L_0, L_H, L_2 above. The damping and all Taylor hypotheses are those of the same proof; no half-line extension or asymptotic replacement is made.

For the small indices the quantities $I_j(A)$ are enclosed directly. The degree-20 and degree-21 Taylor polynomials of e^{-x} are, respectively, upper and lower bounds on all $x \ge 0$, by the sign of the integral remainder. Their integrated values are the rational sums

$$\sum_{k=0}^{m} \frac{(-A)^k}{k!(j+2k+1)2^{j+2k+1}}, \qquad m = 20, 21.$$

The selected routines require $0 \le A \le 8$ and $0 \le j \le 16$; the rates above meet those gates for $5 \le n \le 9$. Every coefficient multiplying an integral is nonnegative. When two complete triples (L_0, L_H, L_2) are available, either may be selected by a rule independent of H. Selecting different coefficients from different triples is not justified by this argument.

6.4.2 Containing rectangles and exact finite powers

The supported fourth-moment bound also improves the real coordinate. If $\rho \in [\rho_l, \rho_u]$, $|X| \le R$, and $\mu_4 \le M_4$, then

$$\Re f(t) \le 1 - t^2/2 + M_4t^4/24.$$

If $R^2t^2 \le 30$, a second lower endpoint is

$$\Re f(t) \ge 1 - t^2/2 + \rho_l^2t^4(1 - R^2t^2/30)/24.$$

These follow from the global fourth-degree upper and sixth-degree lower Taylor polynomials of cosine, $\mu_6 \le R^2\mu_4$, and $\mu_4 \ge \rho^2$ by Cauchy's inequality. The nonnegative coefficient gate is needed before substituting ρ_l^2. Intersect these real intervals with the anchored bounds of Lemma 5.3.

Lemma 6.7 (Two corners of a complex-power rectangle)**.** *Let $n \ge 2$ be an integer, $a \ge 0$, $0 < x_0 \le x_1$ and $Y \ge 0$. If*

$$(n-2)Y \le 3x_0/2, \qquad (2n-1)x_0^n \ge (n-1)a,$$

then for $x \in [x_0, x_1]$ and $|y| \le Y$,

$$|(x+iy)^n - a|^2 \le \max\{|(x_0 + iY)^n - a|^2, |(x_1 + iY)^n - a|^2\}.$$

Proof. Write $z = x + iy$, $r = |z|$ and $F(x, y) = |z^n - a|^2$. Since $|\arg z| \le Y/x_0$, the first gate and $3/2 < \pi/2$ imply $\Re z^{n-2} \ge 0$. Direct differentiation gives

$$F_{yy} = 2nr^{2n-4}\{r^2 + 2(n-1)y^2\} + 2an(n-1)\Re z^{n-2} \ge 0,$$
$$F_{xx} = 2nr^{2n-4}\{r^2 + 2(n-1)x^2\} - 2an(n-1)\Re z^{n-2}.$$

Divide the latter by $2nr^{n-2} > 0$. Its first term is $r^{n-2}((2n-1)x^2 + y^2) \ge (2n-1)x_0^n$; the subtracted term is at most $a(n-1)$. Thus F is convex in each coordinate. Apply convexity first in y and then in x, and use $F(x, Y) = F(x, -Y)$. □

For Fourier smoothing take $a = e^{-nt^2/2}$ and any proved containing rectangle for $f(t)$. Both gates must hold throughout the frequency cell. If either fails, the geometric-sum estimate remains available. The maximum of the two pointwise corner bounds is enclosed before integration; the maximum of their separate integrals would not suffice. Complex powers are finite polynomial recurrences, so zero or equal bases create no division. A fourth-Taylor endpoint is kept fixed on each integration cell when differentiating its smooth majorant.

Cancellation can be retained when computing the real endpoints. For example, on $0 \le u \le 1$,

$$1 - u^2/2 + M_4u^4/24 - e^{-u^2/2} = \frac{M_4 - 3}{24}u^4 + \sum_{k=3}^{10} \frac{(-1)^{k+1}u^{2k}}{2^k k!} + R(u).$$

For derivative order $j = 0, 1, 2$ with $u = ct$, the absolute remainder is at most $2(22)_j u_{\max}^{22-j} c^j/(2^{11}11!)$; consecutive absolute tail terms have ratio below $1/2$ on this domain. Alternatively, at higher frequencies enclose each explicit smooth residual $d(u) = \cos u - e^{-u^2/2}$ or $d(u) + (\rho_u - 1)u^3/6$ by its midpoint value plus the half-width times an enclosure of its derivative on the entire cell. Intersecting two such valid enclosures preserves their common function. These operations differentiate scalar envelopes, not the law's CDF.

6.4.3 A sine improvement on the acquired 26–30 region

Let the positive and negative support magnitudes lie in $[a_+, R_+]$ and $[a_-, R_-]$, respectively. Assume

$$0 < A \le a_+^2 + a_-^2, \qquad B \ge R_+^2 + R_-^2, \qquad M \ge \max(R_+, R_-).$$

Put $R(u) = 1 - Au^2(1 - Bu^2/42)/20$. For $u \ge 0$, $Mu \le \pi$ and $Bu^2 \le 42$,

$$|\Im f(u)|^2 \le (\rho^2 - \zeta^2)u^6 R(u)^2/36. \tag{34}$$

In particular $R(u) \ge 1 - 21A/(40B) \ge 19/40$. Outside these frequency gates the ordinary sine-circle inequality remains available.

Here is a proof retaining the same marginal. Write $\operatorname{sinc} t = \int_0^1 \cos(vt)\,dv$. For $u\sqrt{z} \le \pi$,

$$0 \le -\frac{d}{dz}\operatorname{sinc}(u\sqrt{z}) = \frac{u^2}{2}\int_0^1 v^2 \operatorname{sinc}(uv\sqrt{z})\,dv \le \frac{u^2}{6} - \frac{u^4 z}{60} + \frac{u^6 z^2}{1680}.$$

The upper bound follows by integrating the global polynomial $\operatorname{sinc} t \le 1 - t^2/6 + t^4/120$; nonnegativity follows from $0 \le uv\sqrt{z} \le \pi$. Integrating between a^2 and b^2 gives

$$|\operatorname{sinc}(au) - \operatorname{sinc}(bu)| \le \frac{|a^2 - b^2|u^2}{6}\left(1 - \frac{(a^2 + b^2)u^2}{20} + \frac{(a^4 + a^2b^2 + b^4)u^4}{840}\right).$$

The last bracket is at most $R(u)$, since $a^4 + a^2b^2 + b^4 \le B(a^2 + b^2)$ and $1 - Bu^2/42 \ge 0$. At $a = b$ use the undivided inequality.

For the centered two-point law at $a, -b$, put

$$e_{a,b} = \frac{2a^2b^2}{a+b}, \quad r_{a,b} = \frac{ab(a^2+b^2)}{a+b}, \quad h_{a,b} = \frac{abu}{a+b}\{\operatorname{sinc}(au) - \operatorname{sinc}(bu)\}.$$

The identity $(2ab)^2 + (a^2-b^2)^2 = (a^2+b^2)^2$ gives $\|(e_{a,b}, 6h_{a,b}/(u^3R(u)))\|_2 \le r_{a,b}$. Integrate against the exact centered two-point mixture from the ordinary sine-circle proof. The three integrals are ζ, ρ and $\Im f(u)$, respectively. The triangle inequality proves (34). The case $u = 0$ is immediate.

For each of the 55 acquired rows ($26 \le n \le 30$, $32 \le j \le 42$), retain its closed probability interval $[p_l, p_u]$, an actual-noise bound $s \le s_u$, and an upper bound $d > 0$ on the inward sum $a_0 + a_1$. The interface requires $d < p_l$, $d \le 1 - 2p_u$ and

$$\frac{p_u^2 + (1-p_u-d)^2}{p_u(1-p_u) + s_u} \ge \frac{17}{10}, \qquad R_-^2 + R_+^2 \le \frac{1037}{100}, \qquad \max(R_-, R_+) \le \frac{13}{5}. \tag{35}$$

Indeed the minimum of $(p-a_0)^2 + (1-p-a_1)^2$ under $a_0, a_1 \ge 0$, $a_0 + a_1 \le d \le 1 - 2p$ is $p^2 + (1-p-d)^2$: use the full inward sum and differentiate in a_0, obtaining $2(1-2p-d)+4a_0 \ge 0$. This numerator decreases in p on the retained interval, while $p(1-p) + s$ increases. Thus (35) gives the required standardized inner support sum. It licenses (34) with $A = 17/10$, $B = 1037/100$, $M = 13/5$ on every row.

A direct resource-to-sine map at indices 26–30. The three common constants in (35) follow directly from the acquired floor-25 resource. Write its original simultaneous intercepts as d_0, L, and its prices as b, g, h, so that

$$d_{\text{floor}} - b = gd_0 = hL > 0, \qquad b \ge 0.$$

These identities are checked exactly against the exported residual; L is the original joint intercept, even when another bound further clips the variance. The weighted resource implication gives

$$\frac{n}{25}(b + g\delta) + h\lambda_n \le d_{\text{floor}}, \qquad \frac{n}{25}\frac{\delta}{d_0} + \frac{\lambda_n}{L} \le 1.$$

The second inequality follows by subtracting $nb/25$, dividing by $d_{\text{floor}} - b$, and using $n \ge 25$. Therefore

$$\delta \le \delta_u := \min\left\{\frac{25d_0}{n}, \frac{25d_{\text{floor}}/n - b}{g}\right\}.$$

The bounded conditional supports give $s \le \mathcal{B}\delta$, where $\mathcal{B} = \max(O_0, O_1)\max(p_u, 1-p_l)$ and $\lambda_n = ns$. Substitution in the joint plane yields

$$\lambda_n \le \left\{L^{-1} + (25\mathcal{B}d_0)^{-1}\right\}^{-1} =: \lambda_u.$$

This step uses one joint plane before optimizing. For all 55 rows, the choices $d = \delta_u$ and $s_u = \lambda_u/n$ satisfy the two preceding gates and

$$\frac{p_u^2 + (1-p_u-\delta_u)^2}{p_u(1-p_u) + \lambda_u/n} > \frac{17}{10}.$$

The smallest left side is greater than $1716/1000$ and occurs at $n = 26$ on P_{38}. The same rows satisfy $R_-^2 + R_+^2 \le 1037/100$ and $\max(R_-, R_+) \le 13/5$. The actual-index support proofs and the full 100-panel resource cover are checked separately. Thus the finite-sine majorant applies on every surviving panel. Its integral uses only these common constants and the rational moment and frequency intervals below.

Here η, ζ denote the symmetrization quantities of Lemma 5.4, not the four-summand excess. Their bounds follow directly from that lemma. Since $(13/5)^2 < (11/10)^2 + 8$, the general bounded-support inequality for η gives $\eta \le b(\rho)$ throughout $[11/10, 6/5]$, where $b(r) = (\sqrt{r^2+8} - r)/2$. Lemma 5.4 supplies $\zeta \ge b(\rho)$ and the supported bound for η. Since b decreases and $r + b(r)$ increases, on a strip $[r_l, r_u]$ we may use simultaneously

$$\rho + \eta \le r_u + b(r_u), \qquad \rho^2 - \zeta^2 \le r_u^2 - b(r_u)^2.$$

Rational brackets z_l, z_u for $b(r_u)$ are checked by $z_l^2 + r_u z_l \le 2 \le z_u^2 + r_u z_u$ with $0 < z_l \le z_u$. The integrator uses $r_u + z_u$ and $r_u^2 - z_l^2$ in their indicated upper-bound directions. The value z_u is not separately asserted to bound η at the lower end of the strip.

The selected 23 closed strips cover $[11/10, 6/5]$. All five actual indices are evaluated on each strip, giving 115 complete four-term Fourier inequalities. The optional fourth-Taylor real endpoint and Lemma 6.7 are used only when their whole-cell gates hold. The remaining cells use a previously valid geometric-sum or direct interval bound.

The implementation sometimes integrates the maximum of the two corner majorants more accurately. If f_0, f_1 are twice continuously differentiable on $[t_0 - h, t_0 + h]$, enclose their midpoint values above by a_i, their midpoint derivatives by $[d_i^-, d_i^+]$, and both second derivatives above by $C \ge 0$. Taylor's theorem bounds the integral of their pointwise maximum by

$$\int_0^h \max_i (a_i + d_i^+ x)\, dx + \int_0^h \max_i (a_i - d_i^- x)\, dx + \frac{Ch^3}{3}.$$

Each affine maximum has at most one crossing, found exactly. This bounds the maximum before integration and assumes nothing about how often the original smooth curves cross.

6.5 A common count comparison

The finite argument follows the number of positive summands. Conditional on that number, the sum is a binomial lattice point plus a centered residual. The residual can move mass across a threshold in both directions. Keeping its outgoing and incoming contributions together is essential near a binomial jump. Everything in this section concerns one common marginal; no finite-atom assumption is made.

6.5.1 Exact conditioning and the common moments

Use the sign normalization (5), writing $h = 1/\ell = pq/m$ and $v = pq$. Thus $hX = B - p + U$, $h^2 = v + s$, $\lambda = ns$ and $w_0 = \lambda(1-\alpha)/(nq)$, $w_1 = \lambda\alpha/(np)$. At $s = 0$ both conditional residuals vanish and any $\alpha \in [0,1]$ may be used; zero atoms belong to $B = 0$ as before.

Write W_j for a sum of $n - j$ independent copies of $U \mid B = 0$ and j independent copies of $U \mid B = 1$, all independent. For $b_j = \binom{n}{j} p^j q^{n-j}$ and $z = np + hH = k + u$,

$$\mathbb{P}(S_n \le H) = \sum_{j=0}^{n} b_j \mathbb{P}(W_j \le k + u - j), \tag{36}$$

$$\mathbb{P}(S_n \le H) \le F_{\mathrm{Bin}(n,p)}(k) - b_k \mathbb{P}(W_k > u) + \sum_{j>k} b_j \mathbb{P}(W_j \le u - (j-k)). \tag{37}$$

The second line discards nonnegative outgoing probabilities at $j < k$. The outgoing event is strict and the incoming event is closed. In particular, an atom exactly on the threshold is retained on the CDF side. Also

$$V_j := \mathbb{E}W_j^2 = (n-j)w_0 + jw_1 = \lambda\left(1 + \frac{(j - np)(\alpha - p)}{npq}\right).$$

Suppose support acquisition gives $U_0 \in [-O_0, a_0]$ and $U_1 \in [-a_1, O_1]$, with nonnegative endpoints. For a centered $T \in [-L, R]$ of variance w,

$$w \le LR, \quad w^2/L - Lw \le \mathbb{E}T^3 \le Rw - w^2/R, \quad \mathbb{E}T^4 \le (R-L)\mathbb{E}T^3 + LRw. \tag{38}$$

The first inequality integrates $(T+L)(R-T) \ge 0$. The two cubic bounds integrate $(T+L)(T-w/L)^2 \ge 0$ and $(R-T)(T+w/R)^2 \ge 0$. Multiplying the first polynomial by T^2 gives the fourth-moment bound. If $L = 0$ or $R = 0$, centering forces $T = 0$; the divisions are replaced by this deterministic case.

The common cubic identities (6), in these coordinates, read $\rho h^3 = v(1-2v) + 3v(w_0 + w_1) - q\mu_0 + p\mu_1$, $\gamma h^3 = v(1-2p) + 3v(w_1 - w_0) + q\mu_0 + p\mu_1$, and $\beta h^2 = v(1-2p) + s(2\alpha - 1)$. The same pair (μ_0, μ_1) is retained in every probability bound below.

6.5.2 Outgoing and incoming probability bounds

A useful outgoing minorant is

$$G_a(x) = x(x+a)^2(2a-x), \qquad a > 0.$$

It is nonpositive outside $[0, 2a]$, and its maximum is Ka^4, where $K = 9/4 + 3\sqrt{3}/2$. Consequently

$$\mathbb{P}(W_k > u) \ge \frac{\mathbb{E}G_a(W_k - u)}{Ka^4}.$$

This is a global polynomial inequality, including $W_k = u$. Its expectation uses

$$\mathbb{E}W_k^3 = (n-k)\mu_0 + k\mu_1,$$

$$\mathbb{E}W_k^4 = (n-k)\mathbb{E}U_0^4 + k\mathbb{E}U_1^4 + 3\{V_k^2 - (n-k)w_0^2 - kw_1^2\}.$$

Thus (38) makes the outgoing bound affine in the two common cubic moments. Negative lower bounds remain valid; they are not mistaken for negative probabilities.

For an incoming count $j > k$ and phase $u \le 1/2$, put $d = j - k - u > 0$. Cantelli gives

$$\mathbb{P}(W_j \le -d) \le \frac{V_j}{V_j + d^2}.$$

For completeness, for a centered Y of variance V and $a > 0$, $\mathbb{P}(Y \ge d) \le \mathbb{E}(Y+a)^2/(d+a)^2 = (V+a^2)/(d+a)^2$. Choosing $a = V/d$ proves the formula; $V = 0$ is deterministic. Exponential bounds can be sharper. If $\mathbb{E}Y = 0$, $Y \le B$, $B > 0$, and $\mathbb{E}Y^2 \le w$, then for $t \ge 0$,

$$\mathbb{E}e^{tY} \le M(w; B, t) := \frac{B^2 e^{-tw/B} + we^{tB}}{B^2 + w}. \tag{39}$$

Indeed, the quadratic interpolating e^{tx} at $-w/B$ with multiplicity two and at B majorizes it on $x \le B$: the Hermite remainder has sign $(x + w/B)^2(x - B) \le 0$. Its quadratic coefficient is nonnegative. The centered two-point law on $\{-w/B, B\}$ with variance w evaluates its expectation exactly. This proves the formula also with an upper variance bound. At zero variance use $Y = 0$. Differentiation shows that M is increasing and concave in w: with $x = t(B^2 + w)/B$,

$$M''(w) = \frac{B^2 e^{-tw/B}}{(B^2+w)^3}(x^2 + 2x + 2 - 2e^x) \le 0.$$

Also

$$M'(w) = \frac{B^2}{(B^2+w)^2}\{e^{tB} - e^{-tw/B}(1+x)\} \ge 0,$$

since $e^x \geq 1 + x$. Its tangents therefore give affine upper bounds throughout the variance half-line. Applying (39) to $-U_0, -U_1$ and multiplying the corresponding factors bounds $\mathbb{E}e^{-tW_j}$. Chernoff then gives $e^{-td}\mathbb{E}e^{-tW_j}$ as an incoming upper bound.

One can also retain the common cubic moments. On $[A, B]$, $A < 0 < B$, the function

$$r_t(x) = \frac{e^{tx} - 1 - tx}{x^2} = t^2 \int_0^1 (1 - r)e^{trx}\, dr$$

is increasing and convex. Its chord gives $e^{tx} \leq 1 + tx + c_2x^2 + c_3x^3$, with $c_2, c_3 \geq 0$. Applied to $-U_i$, choose $C_2 \geq c_2$ and $0 \leq C_{3,-} \leq c_3 \leq C_{3,+}$, and put

$$Q_i(\mu_i) = \max\{0, 1 + C_2w_{i,+} - C_{3,-}\mu_i, 1 + C_2w_{i,+} - C_{3,+}\mu_i\}.$$

This is a nonnegative convex upper bound for its MGF. The maximum with zero also makes it nonnegative at infeasible vertices of a relaxed moment polygon. For fixed positive pivots P_0, P_1, weighted AM–GM yields

$$Q_0^{n-j}Q_1^j \leq P_0^{n-j}P_1^j \left(\frac{n-j}{n}\frac{Q_0}{P_0} + \frac{j}{n}\frac{Q_1}{P_1}\right)^n.$$

The right side is convex in (μ_0, μ_1). A fixed exponential rate and its phase chord produce a convex common-moment bound on an entire closed phase interval. Choose a strategy on that whole interval and moment polygon before comparing different strategies; a pointwise minimum of convex functions need not be convex.

6.5.3 Normalization and a finite closed cover

Let $\mathcal{M}$ be a compact polygon containing the actual common cubic moments, obtained from proved moment and defect restrictions. Suppose F is convex on $\mathcal{M}$ and L is affine and strictly positive there. Then

$$\frac{F(x)}{L(x)} \leq \max_{v \text{ vertex of } \mathcal{M}} \frac{F(v)}{L(v)}. \tag{40}$$

To see this, write $x = \sum_v \theta_v v$. Convexity bounds the numerator by $\sum_v \theta_v F(v)$; affinity identifies the denominator with $\sum_v \theta_v L(v)$. The ratio is therefore bounded by the largest vertex ratio. Segments and single points are included. If the lower bound for ρ is $\max\{r_0, L\}$, split $\mathcal{M}$ along $L = r_0$ and retain both closed pieces.

Proposition 6.8 (Whole-cell count certificate)**.** *Fix a closed box of acquired parameters, a count k, and a closed phase interval. Let B_k bound the binomial CDF above, b_k^- the own mass below, and b_j^+ the incoming masses above. Let L_k be a proved outgoing lower bound, $A_j \geq 0$ proved incoming upper bounds, and N a lower bound for $\Phi((k + u - np)/(\sqrt{n}h))$. Suppose the combination*

$$B_k - b_k^- L_k + \sum_{j>k} b_j^+ A_j - N$$

is bounded on the whole phase interval by convex Bernstein control functions $F_i(\mu_0, \mu_1)$. Let $r > 0$ be the affine lower bound for ρ on each closed normalization piece and let $n_+ \geq \sqrt{n}$. Then the positive normalized CDF error is at most

$$n_+ \max_{i,\, v \text{ vertex}} \frac{\max\{0, F_i(v)\}}{r(v)}.$$

Proof. Insert the probability bounds into (37). For the outgoing term, $b_k\mathbb{P}(W_k > u) \geq b_k^-\mathbb{P}(W_k > u) \geq b_k^- L_k$, even if $L_k < 0$. The nonnegativity of the incoming bounds justifies the upper mass substitutions. The Bernstein basis is nonnegative and sums to one. Take the positive part and apply (40); the positive part of a convex function is convex. Finally enlarge $\sqrt{n}$ to n_+. □

The acquired p, h, H bounds must first put every actual threshold into one of the original roots $k = 0, \dots, n$, $u \in [-1/2, 1/2]$. If they do not, prove the remaining phase tails separately. A tied nearest count belongs to both closed roots. Apply Proposition 4.3 only after every leaf is either a proved necessary-condition contradiction or satisfies the whole-cell score bound. The finite tree then covers infinitely many parameters. Its leaves never assert that every feasible moment tuple is an iid law; they need only contain all actual laws.

6.5.4 Necessary filters for the same count domain

In the following formulas write $s_i = w_i$, $x = \mu_0$, $y = \mu_1$. The preceding cubic identities and support-moment inequalities give all three quantities for the same law; they may not be chosen separately. These identities restrict the single pair (x, y) to a rectangle and linear strips. For example, replacing p by the midpoint p_m in $-qx + py$ incurs error $(p - p_m)(x + y)$. On a closed cubic rectangle its absolute value is at most the probability half-width times $\max(|x_l + y_l|, |x_u + y_u|)$. In the signed-cubic identity the corresponding factor is $y - x$, bounded by $\max(|y_l - x_u|, |y_u - x_l|)$. All other products use directed interval endpoints. Thus these strips contain every actual common pair.

6.5.5 The physical defect as a further half-plane

Let $\sigma = \sqrt{10}$ and retain the physical gap defect $D = (3 + \sigma)\rho - \gamma - 3J$. The endpoints of the central gap are $A = (p - a_0)/h$, $B = (q - a_1)/h$. The outer-gap inequality gives

$$D \geq \sigma\rho - \gamma - 6mAB + 6B(\mathbb{E}X_+^2 - mB) + 6A(\mathbb{E}X_-^2 - mA).$$

Substituting the count coordinates and the two cubic identities yields

$$\begin{aligned} Dh^3 \geq\ & B_\sigma(p) + 6vF(a_0, a_1) \\ & + qs_0\{3p(\sigma + 3) - 6a_0\} + ps_1\{3q(\sigma + 1) - 6a_1\} \\ & - (\sigma + 1)qx + (\sigma - 1)py, \\ B_\sigma(p) =\ & v\{(\sigma - 1) + 2p - (2\sigma + 6)v\}, \\ F(a_0, a_1) =\ & a_0 + a_1 - a_0^2 - a_1^2 - a_0a_1. \end{aligned} \tag{41}$$

The polynomial B_σ is the Bernoulli defect square. Replacing σ by a lower enclosure decreases the complete right side, because its coefficient is exactly $\rho h^3 \geq 0$. After that replacement, signed interval products and the explicit nonnegative-coefficient gates license the termwise lower bounds. If such a gate fails, this additional half-plane is omitted.

Substitution of (38) into (41) gives the stronger scalar inequality

$$\begin{aligned} Dh^3 \geq\ & B_\sigma + 6vF(a_0, a_1) \\ & + q\left\{[3p(\sigma + 3) - (\sigma + 7)a_0]s_0 + (\sigma + 1)s_0^2/a_0\right\} \\ & + p\left\{[3q(\sigma + 1) - (\sigma + 5)a_1]s_1 + (\sigma - 1)s_1^2/a_1\right\}. \end{aligned} \tag{42}$$

The terms with $a_i = 0$ are treated by $s_i = 0$, not by division. Also $s_i \leq O_i a_i$, by averaging the nonnegative quadratic $(U_i + O_i)(a_i - U_i)$, with reflection for class one.

On a rectangle clipped by $a_0 + a_1 \leq d < 1$, the function F is nonnegative and concave: its Hessian is $\left(\begin{smallmatrix} -2 & -1 \\ -1 & -2 \end{smallmatrix}\right)$. Its minimum therefore occurs at a vertex of the clipped polygon. There is no requirement $d < 1/2$ for this vertex argument. The older coordinate-monotonicity shortcut requires separate gates. For example, a lower bound for the a_0 derivative in (42) is

$$6v_l(1 - 2d) - q_u(\sigma_u + 7)(s_0)_u - q_u(\sigma_u + 1)O_0^2.$$

The reflected expression uses $\sigma + 5$ and $\sigma - 1$. Positivity of these gates implies $d < 1/2$ and licenses that shortcut only on its smaller domain. A bisection discards an interval only when its entire defect lower bound exceeds the already acquired upper bound.

6.5.6 Second moments and necessary noise floors

The exact second-moment relations are

$$\beta = \frac{v(1-2p) + s(2\alpha - 1)}{v + s}, \qquad m = \frac{v}{\sqrt{v+s}}. \tag{43}$$

The second expression increases with v and decreases with s. For $s > 0$, the first can be solved for α; the case $s = 0$ is tested directly. If an acquired upper bound β_u is below $t_l = 1 - 2p_u$, then

$$v_l(t_l - \beta_u) \le s(\beta_u + 1 - 2\alpha_l).$$

A nonpositive right coefficient excludes the box; a positive one gives a lower bound for s. If an acquired lower bound β_l exceeds $t_u = 1 - 2p_l$, use instead $v_l(\beta_l - t_u) \le s(2\alpha_u - 1 - \beta_l)$. These tests retain the same α as the conditional probabilities.

There is a further floor from the absolute third moment. The elementary bounds $-x \le O_0 s_0$, $y \le O_1 s_1$ in (6) give

$$\rho_l (v+s)^{3/2} \le v(1-2v) + Ms,$$

where M bounds $(3p + O_0)(1-\alpha) + (3q + O_1)\alpha$ at all four p, α corners. Convexity of $(v+s)^{3/2}$ implies

$$A(v) := \rho_l v^{3/2} - v + 2v^2 \le [M - \tfrac{3}{2}\rho_l \sqrt{v}]s.$$

If $A(v_l) > 0$, then $\rho_l \sqrt{v_l} > 1 - 2v_l$, so $A'(v_l) > 1/2 + v_l > 0$; moreover $A'' > 0$. Hence a positive lower bound for $A(v_l)$ and an upper bound for the right coefficient give a necessary noise floor on the whole box. A nonpositive upper coefficient proves infeasibility. Directed square roots preserve both conclusions.

Finally, the common inward bound $a_0 + a_1 \le d$, $d > 0$, together with (38), implies

$$\rho (v+s)^{3/2} \ge v(1-2v) + B_* s + H_* s^2/d, \tag{44}$$

where

$$B_* = 3p(1-\alpha) + 3q\alpha - d\max(\alpha, 1-\alpha), \qquad H_* = \left(\frac{1-\alpha}{\sqrt{q}} + \frac{\alpha}{\sqrt{p}} \right)^2.$$

For the quadratic term use

$$\frac{(1-\alpha)^2}{q a_0} + \frac{\alpha^2}{p a_1} \ge \frac{((1-\alpha)/\sqrt{q} + \alpha/\sqrt{p})^2}{a_0 + a_1}.$$

The zero-variance restrictions cover vanishing endpoints by a limit. The minimum of B_* is at the four p, α corners, because it is affine in p and concave in α. On a p interval $[p_l, p_u]$ a positive lower envelope for $\sqrt{H_*}$ is $(1-\alpha)/\sqrt{q_u} + \alpha/\sqrt{p_u}$, where $q_u = 1 - p_l$. This affine function is minimized at an α endpoint. The global upper Taylor bound

$$(v+s)^{3/2} \le v^{3/2} + \tfrac{3}{2}\sqrt{v}\, s + \tfrac{3}{8} s^2/\sqrt{v}$$

turns (44), with $\rho \le \rho_u$, into a necessary quadratic inequality in s. Its exact minimum on the closed variance interval is tested at both endpoints and any interior vertex. Strict positivity excludes the whole box. The implementation defers $d = 0$ to its deterministic branch.

6.5.7 An additional outgoing event and full phase coverage

For centered $W \in [-a_1, O_1]$ and $-a_1 \le b < O_1$,

$$(W + a_1)(W - b) \le \mathbf{1}_{\{W>b\}}(O_1 + a_1)(O_1 - b).$$

Thus $\mathbb{P}(W > b)$ is at least the nonnegative part of $(s_1 - a_1 b)/((O_1 + a_1)(O_1 - b))$. For $b < -a_1$ the strict event is certain; for $b \ge O_1$ use zero. If any of the k class-one variables exceeds $b = u + B + (k-1)a_1$, their sum exceeds $u + B$. Independence gives the lower probability $1 - (1 - r)^k$, where r is the preceding bound. The remaining centered sum, of variance V_0 and upper support $(n-k)a_0$, is at least $-B$ with probability at least

$$\max\left\{\frac{B^2}{V_0 + B^2}, \frac{B}{B + (n-k)a_0}\right\} \quad (B > 0).$$

These follow from Cantelli and Markov, respectively. At $B = 0$ use one if $V_0 = 0$ and zero otherwise. Multiplication of the two lower bounds is valid because the two conditional groups are independent. This supplies the optional single-event outgoing witness.

The acquired lower bound $m_l > 0$ also gives

$$\lambda = n(v^2/m^2 - v) \le n(v^2/m_l^2 - v), \qquad h \le v_u/m_l.$$

The quadratic on the right is convex, so its maximum on the v interval occurs at an endpoint. If $-H_- \le H \le H_+$, the gates

$$np_l - H_- h_u > -1/2, \qquad np_u + H_+ h_u < n + 1/2$$

ensure that all nearest counts belong to $\{0, \dots, n\}$. The original joint budget $d/d_0 + \lambda/\Lambda \le 1$ retains its own intercept Λ when a separate sharper noise cap clips the count domain. Replacing that intercept by the sharper cap would assert an additional joint inequality and is not licensed here.

6.6 The signed fifth-index corner

The last small moment rectangle can be treated directly by conditioning on signs. This avoids contact and Fourier estimates in that rectangle. Let $\beta = \mathbb{E}[X|X|]$, $p = \mathbb{P}(X > 0)$, $q = 1 - p$, $m = \mathbb{E}X_+ = \mathbb{E}X_-$ and $a_\pm = (1 \pm \beta)/2$. Zero atoms belong to the nonpositive class. Signwise Cauchy gives

$$m \ge \frac{1 + \beta^2}{2\rho}, \qquad \frac{m^2}{a_+} \le p \le 1 - \frac{m^2}{a_-}, \qquad m^2 \le pq.$$

Set $h = pq/m$, $Y = p + hX$, $B = \mathbf{1}_{\{X>0\}}$, and $U = Y - B$. Both conditional means of U vanish, with variances

$$v_0 = p^2(qa_-/m^2 - 1), \qquad v_1 = q^2(pa_+/m^2 - 1).$$

Given $\sum_{i=1}^5 B_i = j$, the noises are independent with variance $V_j = (5-j)v_0 + jv_1$. Write $b_j = \binom{5}{j} p^j q^{5-j}$ and define

$$T(V, d) = \begin{cases} 1, & d \le 0, \\ V/(V + d^2), & d > 0. \end{cases}$$

Conditional Cantelli and the exact normal coordinate yield

$$\frac{\sqrt{5}}{\rho}\{\mathbb{P}(S_5 \le t) - \Phi(t/\sqrt{5})\} \le \frac{\sqrt{5}}{\rho}\left\{\sum_{j=0}^{5} b_j T(V_j, j - z) - \Phi\left(\frac{(z - 5p)m}{\sqrt{5}pq}\right)\right\}, \qquad z = 5p + ht. \tag{45}$$

The value $T(0,0)=1$ is essential: a deterministic sum at the threshold belongs to the closed event.

Proposition 6.9 (The positive corner)**.** *For every standardized iid marginal satisfying*

$$\frac{477}{400}\le\rho\le\frac{6}{5},\qquad \frac{17}{50}\le\beta\le\frac{35139006743001}{10^{14}},$$

the positive closed-CDF score in (45) *is strictly below* $4097/10000<C_{\mathrm{E}}$ *at every real threshold.*

Proof. Here are the enclosures defining the finite proof. On a closed β slab $[b_l,b_u]$ put $m_l=(1+b_l^2)/(2(6/5))$. The moment inequalities enclose p in $[2m_l^2/(1+b_u),\,1-2m_l^2/(1-b_l)]$. On a closed p,z box $[l,u]\times[a,b]\subset(0,1/2)\times[0,5]$ use

$$v_0\le\max\{0,u^2((1-l)(1-b_l)/(2m_l^2)-1)\},$$
$$v_1\le\max\{0,(1-l)^2(u(1+b_u)/(2m_l^2)-1)\}.$$

Each binomial mass is maximized at the projection of $j/5$ onto $[l,u]$, and each Cantelli factor is maximized at $z=b$ and the variance upper bound. Clip the sum of the six upper bounds at one. If $a-5u\ge0$, the normal argument is at least $(a-5u)m_l/(\sqrt{5}\,u(1-u))$. Otherwise it is at least $(a-5u)/\sqrt{5u(1-u)}$: use $m\le\sqrt{pq}$ on the negative part and the fact that $(z-5p)/\sqrt{pq}$ decreases in p for $z\ge0$, $p<1/2$. Its derivative has numerator $-[5p+z(1-2p)]/2$. Outward square roots and a rational Gaussian lower enclosure therefore bound the complete box. Take the nonnegative part of the CDF difference before multiplying by $\sqrt{5}/\rho_l$.

The twelve closed β slabs have endpoints

$$17/50,\quad 341/1000,\dots,351/1000,\quad 35139006743001/10^{14}.$$

Together with the five intervals $z\in[k,k+1]$, $0\le k<5$, these give sixty roots. The supplied binary subdivisions have 1,027 leaves. Exact evaluation on every closed leaf gives a maximum upper bound

$$204848338979/500000000000<4097/10000.$$

Both omitted rays have direct bounds. Use the full rectangle's m_l and enclosing $[p_l,p_u]$, put $v_u=p_u(1-p_u)$ and $s_u=v_u^2/m_l^2-v_u$. On the feasible domain $m^2\le v$, v^2/m^2-v increases with v and decreases with m. Thus $V_j\le\max\{j/p_l,(5-j)/(1-p_u)\}s_u$. For $z\le0$, discard the nonnegative Gaussian CDF and evaluate the Cantelli upper sum at zero, using binomial upper masses $\binom{5}{j}p_u^j(1-p_l)^{5-j}$. The resulting score is below $331/1000$. For $z\ge5$, the CDF is at most one and normal Cantelli gives the score upper bound

$$\frac{\sqrt{5}}{\rho_l}\frac{1}{1+5m_l^2/p_u^2}<187/1000.$$

These exact rational comparisons and the sixty roots cover all real z. The proposition concerns the stated positive branch; it is not silently extended to the opposite β rectangle by reflection. □

6.7 The numerical regions and the induction

The preceding lemmas reduce each finite index to a closed cover of moment and probability parameters. Table 1 shows how the methods meet. All displayed intervals are closed; adjoining rows overlap at their endpoints. The target of each Fourier region is $T=4097/10000<C_{\mathrm{E}}$. The affine bound closes the indicated upper tail; at $\rho=7361/4000$ it is already at most $819/2000<T$.

Table 1: The complete third-moment cover for indices five through thirty-nine. The count column includes necessary-condition exclusions and, where specified, the signed corner or finite-sine comparison.

n	Count region	Fourier complement and required support	Affine tail
5	$[1, 6/5]$, with the positive corner below	$[6/5, 7361/4000]$; the six support rows of Proposition 6.6	$\rho \geq 7361/4000$
6–9	$[1, 237/200]$	$[237/200, 6/5]$, radius $13/5$; then the five support rows through $7361/4000$	$\rho \geq 7361/4000$
10–14	$[1, 23/20]$	$[23/20, 71/50]$, radius 3; $[71/50, 15/8]$, unrestricted	$\rho \geq 7361/4000$
15–19	$[1, 23/20]$	$[23/20, 33/25]$, radius 3; $[33/25, 19/10]$, unrestricted	$\rho \geq 19/10$
20–24	$[1, 9/8]$	$[9/8, 6/5]$, radius 3; $[6/5, 19/10]$, unrestricted	$\rho \geq 19/10$
25–34	$[1, 6/5]$	$[6/5, 19/10]$, unrestricted	$\rho \geq 19/10$
35–39	$[1, 6/5]$	$[6/5, 19/10]$, unrestricted	$\rho \geq 19/10$

The count regions split at $11/10$; at indices five through nine they also split at $23/20$. At five the exact signed cover is as follows. Put $b_2 = 84729828287999/(250000000000000)$ and $b_3 = 35139006743001/10^{14}$. The signed-moment inequality gives $|\beta| \leq b_2$ up to $\rho = 237/200$ and $|\beta| \leq b_3$ up to $\rho = 6/5$: at each endpoint the rational test is $(1 + b_i^2)^2 \geq \rho_i^2(1 - b_i^2)$, with the difference increasing in $b_i \geq 0$. The count cover contains every feasible signed moment for $1 \leq \rho \leq 477/400$, and every such moment with $\beta \leq 17/50$ for $477/400 \leq \rho \leq 6/5$. The remaining closed rectangle is exactly Proposition 6.9; its positive sign is retained. The splits at $237/200$, $467/400$, $477/400$, $1/5$ and $3/10$ are refinements of this union, not additional assumptions.

What the numerical assertion checks. The finite acquisition theorem 7.8 transports one actual marginal through every update. Support is proved before higher moments are used, and an initial raw remainder before defect feedback. Its simultaneous state includes the score, support, moments, defect and signed threshold. A failed analytic gate gives no exclusion. Each state must instead reach a necessary-condition contradiction, a complete Fourier bound, or a complete count forest.

Write $P_j = [j/100, (j + 1)/100]$, $0 \leq j < 100$. Every probability panel is closed. When several moment cells export to it, upper bounds are maximized and lower bounds minimized over *all* those cells. Thus seams retain the source law. A count forest covers every original root $k = 0, \ldots, n$, including all signed phase intervals. Missing panels are justified exclusions; refinements must cover their entire parent. The exact partitions, score/HIGH caps, refinement schedules and forest counts are certificate data. Their logical content is the complete-domain assertion below, not the number of successful files.

The following application-specific prerequisites are retained. At five the first own-score bound is support-free; later Fourier inputs use Proposition 6.6, including every narrower signed profile. For six through nine, the wider Fourier domain $[23/20, 6/5]$ uses radius $13/5$, while its low-band resource uses radius $5/2$. At fifteen through nineteen, support radius three on $\rho \leq 33/25$ precedes the own-score and HIGH bounds; at twenty, radius three on $\rho \leq 6/5$ precedes them. At twenty-one through twenty-four the own-score bounds are unrestricted. The twenty through twenty-four initializers require no earlier numerical seed, and each reused coarse count contract at twenty is admitted only after full simultaneous containment has been established. A cellwise Fourier bound evaluates the complete compensated expression, not

a presumed scalar HIGH ceiling. Each unrestricted region includes the complete auxiliary-moment domain.

At 26–30 and 32–34, respectively, the acquired floors are $N = 25$ and $N = 31$. Transfer retains $nD \le Nd$ as well as the raw remainder bound. All binomial probabilities, normal arguments and support tests still use the actual integer n. Local refinements of a parent replace only the part they cover. At 26–30 the surviving second-band panels $P_{32}, \dots, P_{42}$ are closed by the finite-sine inequality; the unrestricted Fourier complement makes a third count band at twenty-five unnecessary. All omitted count panels, including the two sign-mass exclusions at twenty and the weighted-budget exclusions at 26–30, have their own necessary-condition checks.

Here the five weighted exclusions have a direct explanation. For $(n, j) = (30, 52)$ and for $27 \le n \le 30$, $j = 53$, the acquired pairs (d, Θ) in $D \le 25d/n$, $\theta \le \Theta$ are respectively $(3/10, 1/500)$ and $(5/16, 1/2500)$. Put $s = 316227766/10^8 < \sqrt{10}$ and $l = j/100$. Since $p \ge l > p_*$, $\sqrt{p(1-p)} \le 1/2$ and $(1-\theta)^{3/2} \ge (1-\Theta)^2$, the joint budget gives

$$D \ge (12 + 4s)(1-\Theta)^2(l - (4-s)/2)^2 > \begin{cases} 251/1000, & j = 52, \\ 38/125, & j = 53. \end{cases}$$

These contradict $D \le 1/4$ at $(30, 52)$ and $D \le 125/432 < 38/125$ at $j = 53$, $n \ge 27$. The separate exclusions of P_{36} and P_{34} at twenty use the sign-mass inequality on the respective bands, including endpoints.

The prerequisite at thirty-nine. Sharpness through 38 and the recurrence give

$$39(C_{39} - C_{\mathrm{E}}) < 55887/190000 < 3/10, \qquad C_{39} < .4226,$$

with both deleted scores below .4253. Before the sharp-39 acquisition, apply the separate comparison with actual $n = 39$, $K = 741/4000$, predecessor allowance .33 and excess allowance $1/2$. Under $C_{39} \ge C_{\mathrm{E}} + K/39$, its support input is $(N, L, K, q, R) = (39, 39, 741/4000, .33, 3.84)$, and the positive charge $K\rho_l$ remains in the resource. Its full acquisition excludes the probability complement of $[.302, .458]$. Nearest count covers all phases for $0 < \lambda \le .08$; on $[.08, .10]$ use nearest count for $|u| \ge .19$ and whole count for $|u| \le .19$; on $[.10, .12]$ use nearest count for $|u| \ge .27$ and whole count for $|u| \le .27$; above .12 whole count covers every phase. Each band is clipped to its own noise ceiling and includes its seams. Zero noise is Bernoulli. The signed allowance tests of Section 7.7 exclude the hypothesized score. Hence $C_{39} < .41449$, since $C_{\mathrm{E}} < .40974$ and $K/39 = .00475$.

At 35–38, the entry bound $C_n < 41/100 + 4177/(13824n)$ supplies the own-score caps; at 39 use the bound just proved. The initial support radii are $39/10$ at 35 and $19/5$ at 36–39. The independent support bound $96/25 < 39/10$ licenses the first, and all five initial and resource calculations use the actual index. Any local support refinement is proved on its whole source interval before export. Thus sharpness at 39 is never used to establish its own entry cap.

Proposition 6.10 (Complete numerical finite base through thirty-nine)**.** *The support, Fourier, initialization, resource and count inequalities in the preceding closed cover hold on their complete domains, including the separate positive-39 prerequisite. Every exported row implies its entire original count contract. The necessary exclusions and retained forests cover every original moment and probability panel.*

This assertion combines acquisition for one marginal, directed containment of every scalar and signed-phase condition, count acceptance at the actual integer, and the complete closed union. In particular, if $q_i(H)$ are the acquired phase polynomials and $\widetilde{q}_j(H)$ the contract

polynomials, the join proves, for each i, some j with $q_i \le \widetilde{q}_j$ throughout that signed interval. Endpoints and the possible interior minimum of the quadratic difference suffice. Hence $\max_i q_i \le \max_j \widetilde{q}_j$. Selecting one row retains the whole contract's actual index and moment domain.

Derivation of $C_n \le C_{\mathrm{E}}$ for $5 \le n \le 39$. Choose the least failing index and an attained positive closed maximizer. Its predecessor is sharp, so contact loss is zero and every larger retained allowance is valid. The table covers every $\rho \ge 1$. Fourier and signed-corner regions have score below $T < C_{\mathrm{E}}$. Otherwise acquisition retains the same law in an exported panel. It cannot satisfy a necessary-condition exclusion; the directed join therefore puts it in a complete count root, whose bound is below C_{E}. At 39 the positive comparison is applied first. Every case contradicts the assumed failure. □

6.8 The actual indices forty, forty-one and forty-two

A first failure here has sharp predecessors through two deletions. The recurrence gives

$$0 < n(C_n - C_{\mathrm{E}}) < 1433/4875 < 3/10, \qquad C_n < 167/400 < .4226.$$

Thus the deleted bounds are below .4253, and the weaker allowances $n(C_n - C_{\mathrm{E}}) \le .54$ and contact loss .19 are valid. The floor-40 argument preserves $\mathcal{R}_n \le r$ and $nD \le 40d$; its support test uses both endpoints 40 and 42 and proves radius 96/25 before higher moments are used. Complete acquisition leaves left rows covering $[.282, .38]$ and central rows covering $[.38, .476]$; their complements are excluded. Confinement starts with widths $.55, .575, .6$ on the left and .55 centrally, and cubic constants 30 and 24, with every width and positive-slope gate checked.

The central phase cover uses nearest count for $0 < \lambda \le .08$, then nearest count for $|u| \ge .19$ and whole count for $|u| \le .19$ on $[.08, .10]$, and nearest count for $|u| \ge .27$ and whole count for $|u| \le .27$ on $[.10, .12]$; whole count applies above .12. The left cover uses nearest count through .0925, then nearest count for $|u| \ge .16$ and whole count for $|u| \le .16$ on $[.0925, .10]$, and nearest count for $|u| \ge .25$ and whole count for $|u| \le .25$ on $[.10, .12]$; then nearest count for $u \le -.27$ and whole count for $u \ge -.27$ on $[.12, .135]$; above .135 use whole count. Each closed band is clipped to its acquired noise ceiling. Zero noise is Bernoulli.

These three indices admit a positive finite-binomial reserve. For $[a, b] \subset (0, 1/2)$ put $F_j(p) = \Pr(\mathrm{Bin}(n, p) \le j)$ and $z_j(p) = (j - np)/\sqrt{np(1-p)}$. Both decrease with p, while $\sqrt{np(1-p)}/(1 - 2p(1-p))$ increases. Therefore both sides of every jump are bounded by

$$\frac{\sqrt{nb(1-b)}}{1 - 2b(1-b)} \max\{F_j(a) - \Phi(z_j(b)), \Phi(z_j(a)) - F_{j-1}(b), 0\}, \qquad F_{-1} = 0. \tag{46}$$

Between jumps the CDF is constant, so $j = 0, \dots, n$ suffices. Exact endpoint probabilities and outward Gaussian bounds on a complete closed subdivision of $[.28, .48]$ give a positive uniform lower bound ϵ_p for $C_{\mathrm{E}} - C^{\mathrm{bin}}_{n,p}$ on each parent, at the actual $n = 40, 41, 42$.

Retain both signs in the finite corrections: $I/n \le \max(I/40, I/42)$ and $\sqrt{n}\pi_k J \le \max(b_0 J, J)$ when $b_0 \le \sqrt{n}\pi_k \le 1$. If S bounds $\rho_B(C - C_B)/\lambda$ on a positive-noise band $[a, b]$, it suffices that $S < \rho_{B,l}\epsilon_p/b$. For $S < 0$ this is immediate; otherwise use $\lambda \le b$ and $\rho_B \ge \rho_{B,l}$. The scalar bounds are convex in positive noise, being affine terms plus nonnegative reciprocals of positive affine terms; their two endpoint tests cover the band. No assertion for all $n \ge 40$ follows from this finite calculation.

Proposition 6.11 (Numerical floor-40 assertion)**.** *The preceding acquisition, two-endpoint support, actual-index binomial reserve and complete signed local covers satisfy their strict tests at $n = 40, 41, 42$. Each local row consumes its acquired simultaneous profile. Consequently $C_{40}, C_{41}, C_{42} \le C_{\mathrm{E}}$.*

Proof. The base through 39 and a least failure supply all entry premises. Acquisition and confinement preserve its witness. The local comparison with the signed finite corrections excludes every positive-noise case; the Bernoulli theorem excludes zero noise. □

Theorem 6.12 (The finite base)**.** *For every integer* $1 \le n \le 42$ *and every standardized iid marginal with finite third absolute moment,* $\sqrt{n}\,\Delta_n(P) \le C_{\mathrm{E}}\rho(P)$.

Proof. The starting bounds and the four-summand comparison establish indices one through four. Proposition 6.10 and its same-law induction establish five through thirty-nine; Proposition 6.11 supplies the last three indices. □

7 Acquisition and comparison near a two-point law

The defect identity has two uses. Contact and a complete Fourier bound first confine one extremizing marginal to a neighborhood of a two-point law. We then compare the actual sum with that law's own binomial count. The second step needs no extremality beyond the simultaneous hypotheses already obtained in the first. All bounds below include zero excess $e = C_n - C_{\mathrm{E}} = 0$; positive conditional noise and zero noise are treated separately at the end.

Write $\sigma = \sqrt{10}$, $m = \mathbb{E}X_+ = \mathbb{E}X_-$, $\rho = \mathbb{E}|X|^3$, $\gamma = \mathbb{E}X^3$, $\beta = \mathbb{E}[X|X|]$, $\mu_j = \mathbb{E}X^j$ and $M_j = \mathbb{E}|X|^j$ (also $\eta_5 = M_5$). In the central gap we use $(-A, B)$ for the endpoints, equivalently $A = A_-$ and $B = A_+$ in Section 2. The quantities $J, J_1, J_2, D, L, K_1, K_q$ and $\mathcal{R}$ are those of Theorem 2.5; $U = J_1 + (\mu_4 - 3)/6$ below is a coefficient, distinct from the conditional noise introduced later. The letter B denotes the positive endpoint until Bernoulli coordinates are introduced; there its meaning is explicitly reset to a label.

7.1 Support from the contact inequality

The scale correction. Write $\phi_0 = \phi(0)$. For $n > 1$ put $s_n = \sqrt{n/(n-1)}$ and $e_n = n(s_n - 1) - 1/2$.

Lemma 7.1. *For every integer* $n \ge 36$*, one has* $0 < e_n < .384/n$.

Proof. Put $x = 1/n$ and $c = .384 = 48/125$. Direct expansion gives

$$(1-x)(1 + x/2 + cx^2)^2 - 1 = x^2\{(2c - 3/4) - (c + 1/4)x + (c^2 - c)x^2 - c^2x^3\}.$$

The bracket is decreasing for $x \ge 0$, since $0 < c < 1$. At $x = 1/36$ its exact rational value is $823/4050000 > 0$. Squaring positive quantities gives $s_n < 1 + x/2 + cx^2$ for $0 < x \le 1/36$. Similarly $(1-x)(1+x/2)^2 - 1 = -3x^2/4 - x^3/4 < 0$, which gives the lower bound. □

A signed Gaussian envelope. Define, for every real H, t and $n > 1$,

$$\mathcal{H}_n(H,t) = n^{3/2}\{\Phi((H-t)/\sqrt{n-1}) - \Phi(H/\sqrt{n})\} + nt\phi(H/\sqrt{n}) + \tfrac{H}{2}(t^2-1)\phi(H/\sqrt{n}).$$

Write $v = n - 1$, $s = \sqrt{n/v}$, $u = |t|$, and

$$A = \frac{u^3 - 3u}{6}, \quad B = \frac{u^4 - 6u^2 + 3}{24\sqrt{v}}, \quad D = \frac{u^5 - 15u}{120v}.$$

Lemma 7.2 (Fifth-order identity with a uniform finite remainder)**.** *If* $u > \sqrt{3}$*, then, with* $z = H/\sqrt{v}$*,*

$$\frac{\mathcal{H}_n(H, \pm u)}{s^3} = \pm A(1 - z^2)\phi(z) + B\phi^{(3)}(z) \mp D\phi^{(4)}(z) + R,$$

where the sign of the remainder need not be fixed and

$$|R| \le E_v^{(5)}(u) := \frac{51u^6}{14400v^{3/2}} + \frac{51(3u^2-2)}{960v^{3/2}} + \frac{u}{8v^2}.$$

This statement is uniform over every real H.

Proof. Let $F_w(x) = \Phi(x/\sqrt{w})$, with primes denoting x derivatives. The heat identity $\partial_w F_w = F_w''/2$ and ordinary Taylor's theorem give, with unlabelled derivatives at (v, H),

$$\begin{aligned}
F_{v+1} &= F_v + F_v''/2 + F_v^{(4)}/8 + R_0, & |R_0| &\le \sup_{w\in[v,v+1]} \|F_w^{(6)}\|_\infty/48,\\
F_{v+1}' &= F_v' + F_v'''/2 + F_v^{(5)}/8 + R_1, & |R_1| &\le \sup_{w\in[v,v+1]} \|F_w^{(7)}\|_\infty/48,\\
F_{v+1}'' &= F_v'' + F_v^{(4)}/2 + R_2, & |R_2| &\le \sup_{w\in[v,v+1]} \|F_w^{(6)}\|_\infty/8.
\end{aligned}$$

Expand $F_v(H-t)$ through degree five, with remainder at most $u^6\|F_v^{(6)}\|_\infty/720$. In

$$\frac{\mathcal{H}_n}{n^{3/2}} = F_v(H-t) - F_{v+1}(H) + tF_{v+1}'(H) - \frac{t^2-1}{2}F_{v+1}''(H),$$

the first three derivative coefficients cancel. The next three are

$$\left(\frac{t}{2} - \frac{t^3}{6}\right)F_v''' + \frac{t^4-6t^2+3}{24}F_v^{(4)} + \left(\frac{t}{8} - \frac{t^5}{120}\right)F_v^{(5)}.$$

The remaining absolute value is bounded by

$$\left\{\frac{u^6}{720} + \frac{1}{48} + \frac{u^2-1}{16}\right\}\sup_{w\in[v,v+1]}\|F_w^{(6)}\|_\infty + \frac{u}{48}\sup_{w\in[v,v+1]}\|F_w^{(7)}\|_\infty.$$

Fourier inversion of the Gaussian, differentiated under its absolutely integrable Fourier transform, gives

$$\|\phi^{(k)}\|_\infty \le \frac{1}{\pi}\int_0^\infty y^k e^{-y^2/2}\,dy.$$

Integration by parts yields $\|\phi^{(5)}\|_\infty \le 8/\pi < 2.55$ and $\|\phi^{(6)}\|_\infty \le 15\phi_0 < 6$. Since $F_w^{(j)} = w^{-j/2}\phi^{(j-1)}$, the displayed remainder becomes $s^3E_v^{(5)}(u)$ after multiplication by $n^{3/2}$. The three retained derivative coefficients give the identity. □

Lemma 7.3 (A lower tangent bound for the fourth Gaussian derivative)**.** *For every real z,*

$$\phi^{(4)}(z) \ge \phi_0(3 - 15z^2/2).$$

Proof. The same Gaussian Fourier integral gives $\phi^{(4)}(z) = \pi^{-1}\int_0^\infty y^4 e^{-y^2/2}\cos(yz)\,dy$. Use the globally valid inequality $\cos w \ge 1 - w^2/2$ inside this integral. Its fourth and sixth Gaussian moments are $3\phi_0$ and $15\phi_0$, respectively. □

Lemma 7.4 (Uniform signed fifth-order envelope)**.** *Suppose $u > \sqrt{3}$ and $A, B, D \ge 0$. If $W = A - 15D/2 > 0$, put*

$$T = A - 3D + \frac{9B^2}{4W}.$$

If $T \ge A/2 + 2B + 3D$, then, for every real H and either sign,

$$\mathcal{H}_n(H, \pm u) \le s^3\{\phi_0 T + E_v^{(5)}(u)\}.$$

Proof. For positive $t = u$ and $|z| \le 1$, divide the retained terms by ϕ_0 and use the preceding tangent bound to obtain

$$A - 3D - (A - 15D/2)z^2 + 3B|z| \le A - 3D + 9B^2/(4W).$$

Here $(1 - z^2)\phi(z) \le (1 - z^2)\phi_0$ and $|\phi^{(3)}(z)| \le 3|z|\phi_0$ on this interval. For $|z| \ge 1$ the first term is nonpositive, and the Fourier bounds $\|\phi^{(3)}\|_\infty < 2\phi_0$ and $\|\phi^{(4)}\|_\infty \le 3\phi_0$ give $2B + 3D$. For negative $t = -u$, the first term has global upper bound $2e^{-3/2}A\phi_0 < A\phi_0/2$, and the other terms give $(2B+3D)\phi_0$. The hypothesis on T covers both exterior cases. The inequality $2e^{-3/2} < 1/2$ follows from four positive terms of the exponential series at $3/2$. Finally add the uniform remainder. □

Lemma 7.5 (A global Gaussian remainder bound)**.** *For every integer $n \ge 2$ and all real H, t,*

$$|\mathcal{H}_n(H,t)| < \frac{|t|^3}{15} + \frac{|t|}{5} + \frac{1}{4} < \frac{|t|^3}{15} + \frac{|t|}{5} + \frac{1}{2}.$$

Proof. Write $F_w(x) = \Phi(x/\sqrt{w})$, $v = n - 1$ and $y = H - t$. The identity defining $\mathcal{H}_n$ is

$$n^{-3/2}\mathcal{H}_n(H,t) = F_v(y) - F_n(H) + tF_n'(H) - \frac{t^2 - 1}{2}F_n''(H).$$

Taylor expansion of $F_n(y)$ at H through degree two has remainder at most $\phi(0)|t|^3/(6n^{3/2})$, since $\|\phi''\|_\infty = \phi(0)$. After this expansion the remaining correction is

$$\frac{1}{2}\{F_n''(H) - F_n''(y)\} + \frac{1}{2}\int_v^n \{F_n''(y) - F_w''(y)\}\, dw.$$

Here we used $\partial_w F_w = F_w''/2$. The first term is at most $\phi(0)|t|/(2n^{3/2})$ in absolute value. Applying the heat identity once more bounds the second term by

$$\frac{1}{4}\int_v^n \int_w^n |F_r^{(4)}(y)|\, dr\, dw \le \frac{\|\phi'''\|_\infty}{8v^2} \le \frac{1}{4\pi v^2}.$$

The last inequality follows by differentiating the absolutely convergent Gaussian Fourier integral, which gives $\|\phi'''\|_\infty \le 2/\pi$. Multiply by $n^{3/2}$ and use $\phi(0) < 2/5$. The function $x^{3/2}/(x-1)^2$ decreases for $x > 1$, because its logarithmic derivative is $3/(2x) - 2/(x-1) = -(x+3)/(2x(x-1)) < 0$. Hence, for $n \ge 2$,

$$\frac{n^{3/2}}{4\pi(n-1)^2} \le \frac{\sqrt{2}}{2\pi} < \frac{1}{4},$$

using $\sqrt{2} < 3/2$ and $\pi > 3$. This proves the stronger bound and therefore the exact ray majorant used by the original checkers. □

Finite bands and the exact support polynomial. For integers $36 \le N \le L$, use the simultaneous endpoint bounds

$$B_+ = \frac{u^4 - 6u^2 + 3}{24\sqrt{N-1}}, \quad D_+ = \frac{u^5 - 15u}{120(N-1)}, \quad D_- = \frac{u^5 - 15u}{120(L-1)}.$$

For an infinite index range, set $D_- = 0$ instead. Put

$$W_* = A - 15D_+/2, \qquad T_* = A - 3D_- + \frac{9B_+^2}{4W_*}.$$

The proof of Lemma 7.4 applies uniformly on the whole band if $W_* > 0$ and $T_* \geq A/2 + 2B_+ + 3D_+$. In particular the favorable term is frozen at the upper index endpoint, and the adverse terms at the lower one. No monotonicity of a signed coefficient is assumed.

Let (P, H) be an attained positive closed maximizing pair, standardized to mean zero and variance one. Write $C = C_n \geq C_E$, $\rho = \mathbb{E}|X|^3$, $\beta = \mathbb{E}[X|X|]$, and $d = n(C_{n-1} - C_n)_+$. The finite contact/predecessor bound is

$$\rho I(t) \leq \mathcal{H}_n(H, t) + C\{-|t|^3 + 3\rho t^2/2 + 3\beta t + \rho e_n\} + \rho s_n d,$$

with $I = 0$ on the entire closed support. Suppose also, for a finite band, that $C_n - C_E \geq K/n$, where $K > 0$. The finite moment bounds of [9, p. 125], namely $C\rho \leq .3031(\rho + .646)$ and $C\rho \leq .3322(\rho + .429)$, give

$$c = .4095 + K/L < C_n, \qquad \rho < r = \min\left\{\frac{.3031(.646)}{c - .3031}, \frac{.3322(.429)}{c - .3322}\right\}.$$

For the unrestricted nonnegative case put $K = 0$, $c = .4095$. The sign-moment inequality $\rho \geq (1 + \beta^2)/\sqrt{1 - \beta^2}$ supplies $|\beta| < b$ whenever $(1 + b^2)^2 > r^2(1 - b^2)$ and $0 < b < 1$. All these statements concern the same current law. The lower endpoint cannot replace L in K/L.

When a preceding resource enclosure supplies sharper actual bounds $\rho \leq r$ and $|\beta| \leq b$, these may instead be used in the same polynomials. Their validity is the only extra premise.

Use rational $\ell^2 \leq N - 1$ and $S^2 > N/(N - 1)$. For $u = |t|$ define

$$Q(u) = -u^3 + 3ru^2/2 + 3bu + .384r/N.$$

First check $Q < 0$, before replacing C by its lower bound c. Replace $1/\sqrt{N - 1}$ by $1/\ell$ in the adverse remainder and in the term $2B_+$ of the domination gate. Write E_* for that rational remainder. Strict negative contact for every n in the band follows from the polynomial inequality

$$\frac{2}{5}S^3\left\{(A - 3D_-)W_* + \frac{(u^4 - 6u^2 + 3)^2}{256(N - 1)}\right\} + S^3E_*W_* + \{cQ + rSq\}W_* < 0.$$

The second polynomial gate is $W_* > 0$. The third is

$$\{A/2 - 3D_- - 3D_+ - 2B_+\}W_* + \frac{(u^4 - 6u^2 + 3)^2}{256(N - 1)} > 0.$$

This is the cleared-denominator version of the domination hypothesis. The bound on T_* is positive, so the replacement $\phi_0 < .4$ and $s_n < S$ both have their correct direction.

The exact kernel tests these polynomials on all adjacent closed intervals of length .05 from R to 4.5. Positive Bernstein coefficients prove their signs on every point of each cell, including endpoints. On the rays $u \geq 4.5$ use Lemma 7.5, which gives the same bound $|\mathcal{H}_n(H, t)| \leq u^3/15 + u/5 + 1/2$. The resulting cubic and its first three derivatives are strictly negative at 4.5, as is the coefficient Q before replacing C by c. This excludes the whole rays. Thus, if $d \leq q$, actual support is in $(-R, R)$, including exclusion of arbitrarily small exterior atoms and exterior accumulation points.

The support rows used in this proof. For the all-index local argument, take $K = 0$, $R = 3.84$, $q = .19$ on

$$[45, 49],\ [50, 59],\ [60, 89],\ [90, 134],\ [135, 199],\ [200, \infty).$$

Only indices at least 47 are used. The last row uses $D_- = 0$; there is no upper index restriction. The exact checker reconstructs all closed u cells and the two exterior rays. Missing an index band or the infinite row is an acceptance failure.

One bounded-threshold series. Put $c_k = (-1)^k/[2^k k!(2k+1)]$, and let $(x)_j$ denote the rising factorial. Define the following rational polynomials:

$$P_j(H,t) = \sum_{k=0}^{j+1} \frac{c_k(k+1/2)_{j+1-k}}{(j+1-k)!}(H-t)^{2k+1} - c_{j+1}H^{2j+3} + \frac{(-1)^{j+1}tH^{2j+2}}{2^{j+1}(j+1)!} + \frac{(-1)^j H^{2j+1}(t^2-1)}{2^{j+1}j!}.$$

In particular,

$$P_0 = (t^3-3t)/6, \qquad P_1 = \frac{3t-t^3}{4}H^2 + \frac{t^4-6t^2+3}{8}H + \frac{-t^5+10t^3-15t}{40}.$$

Fix $h > 0$, $E(x) = (193/71)^{\lceil x \rceil}$, and

$$B_r = \frac{7(h+r)}{6}E((h+r)^2/6) + \left(h + r + \frac{h(r^2+1)}{8}\right)E(h^2/8).$$

Lemma 7.6. *For every $n \geq 4$, $|H| \leq h$ and $|t| \leq r$,*

$$\left| \mathcal{H}_n/\phi(0) - \sum_{j=0}^{5} P_j/n^j \right| \leq 4^7 B_r/n^6.$$

Proof. Put $z = 1/n$. Integration of the Gaussian exponential series gives $\mathcal{H}_n/\phi(0) = S(z)/z$, where

$$S(z) = \sum_{k\geq 0} c_k\{(H-t)^{2k+1}(1-z)^{-k-1/2} - H^{2k+1}\}z^k + te^{-H^2z/2} + \tfrac{1}{2}zH(t^2-1)e^{-H^2z/2}.$$

Its constant coefficient vanishes, and expansion of the binomial factor gives $[z^{j+1}]S = P_j$. At $z_0 = 1/4$, the sum obtained by replacing all coefficients by their absolute-value majorants is at most B_r: use $(1-z_0)^{-1/2} < 7/6$, $1/(2k+1) \leq 1$, and $e < 193/71$. The latter bound follows, for example, by summing the exponential series through degree six and bounding the rest by a geometric series with ratio $1/8$. All products and rearrangements are therefore absolutely convergent. The discarded terms have degrees at least seven in S, so their sum at $0 \leq z \leq z_0$ is at most $(z/z_0)^7 B_r$. Divide by z to get the result. This proves a finite uniform bound, including every threshold in the stated interval and every $n \geq 4$. □

A common contact test for radius and strip refinement. Let $N \geq 36$, $E, q \geq 0$ and $C_0 > 0$. Suppose $n \geq N$, $0 \leq C - C_E \leq E/n$, $\delta_n := n(C_{n-1} - C_n)_+ \leq q$, $C \leq C_0$ and $|X| \leq R_0$. A previously acquired row gives

$$\rho \leq r, \quad 0 \leq b_l \leq \beta \leq b_u < 1, \quad |H| \leq h.$$

Define $Q(t) = -|t|^3 + 3\rho t^2/2 + 3\beta t$ and $L_P(t) = C_E Q(t) + \phi(0)(t^3-3t)/6$. For $t > 0$, upper polynomials for $L_P(t), Q(t)$ are

$$(1/15 - .4095)t^3 + 1.5(.40974)rt^2 + (3(.40974)b_u - .3988/2)t, \quad -t^3 + 1.5rt^2 + 3b_u t.$$

For $t < 0$, write $u = |t|$ and use

$$(-.3988/6 - .4095)u^3 + 1.5(.40974)ru^2 + (.2 - 3(.4095)b_l)u, \quad -u^3 + 1.5ru^2 - 3b_l u.$$

These coefficient choices are valid because $b_l \ge 0$ is checked first. On any closed cell of one sign, let L_J be an upper bound for the first polynomial and let $Q_J \ge 0$ bound the positive part of the second. For $j = 1, 2, 3$, take m_j to be the positive part of the largest tensor Bernstein coefficient of P_j on $[-h, h] \times J$. For $j = 4, 5$, the absolute monomial bound $\sum_{a,b} |[H^a t^b] P_j| h^a R_J^b$ suffices, where $R_J = \max_{t \in J} |t|$. The support contact is then bounded above by

$$L_J + \frac{EQ_J}{N} + .4 \left(\sum_{j=1}^{5} \frac{m_j}{N^j} + \frac{4^7 B_{R_J}}{N^6} \right) + r \left(qS + \frac{.384 C_0}{N} \right), \qquad S > \sqrt{N/(N-1)}. \tag{47}$$

Indeed substitute the finite series in contact, split $C = C_E + (C - C_E)$, and use the scale estimate $e_n < .384/n$. Every error retained in (47) is nonnegative and decreases with n. A strictly negative rational result therefore excludes the entire cell for every integer $n \ge N$, including atoms and accumulation points. The uniform-tail specialization is $(N, E, q, C_0) = (47, .7, .19, .425)$; its score cap follows from $C < .40974 + .7/47 < .425$. The finite bands use their own acquired E, q, C_0 at the corresponding floor, with exactly the same formula.

For a coordinate strip, conditional centering gives

$$X = \ell(Y - p), \quad \ell = \sqrt{(1-\theta)/(p(1-p))}, \quad \beta = (1-\theta)(1-2p) + \kappa, \quad |\kappa| \le \theta.$$

If $p \in [p_l, p_u] \subset (0, 1/2)$ and $\theta \le \Theta$, put $\ell_l \le \sqrt{(1-\Theta)/(p_u(1-p_u))}$. The forbidden strip $Y \notin (-.55, 1.55)$ maps into

$$X \le -\ell_l(p_l + .55), \qquad X \ge \ell_l(1.55 - p_u).$$

Apply (47) between these thresholds and $\pm R_0$. An interval beyond the known radius is empty; an equal endpoint is a singleton that is still checked. If only an upper β bound is given, use $b_l = (1-\Theta)(1-2p_u) - \Theta$, after checking its nonnegativity.

The selected consequences used in the resource schedule are:

input rows	prior radius	conclusion	closed cells
10 central preparation rows	3.84	$\lvert X\rvert < 2.5$	1080
45 central rows	2.5	$-.55 < Y < 1.55$	1431
38 left rows	2.7	$-.55 < Y < 1.55$	3073

The respective largest contact bounds are below $-.084$, $-.097$ and $-.01658$. These implications acquire no row by themselves: each is used only after its input row is proved by the preceding resource step. The left radii are applications of the all-threshold support polynomial, with $(N, L, K, q) = (47, \infty, 0, .19)$ and the actual moment caps

$$(r, b, R) = (1.390, .430, 3), \qquad (1.2563, .3926, 2.7).$$

Thus the same heat proof supplies the initial radius, both left contractions, and the five finite-base rows. The bounded-threshold series supplies the central contraction and both coordinate strips.

7.2 Acquiring simultaneous resources for one marginal

For a ratio maximizer $C = C_n \ge C_{\mathrm{E}}$, put $e = C - C_{\mathrm{E}}$ and $w = H^2/(2n)$. The core identity is

$$ne\rho + nC_{\mathrm{E}}\rho(1 - e^{-w}) + n\phi(0)e^{-w}D/6 + n^{3/2}L = \mathcal{R}.$$

The first complete raw enclosure must give $\mathcal{R} < r < .4N$, for the actual $n \ge N$. Since $C\rho \ge C_{\mathrm{E}} > .4$, this licenses Corollary 2.6, including

$$ne\rho + n\phi(0)D/6 + n^{3/2}L \le \mathcal{R}, \quad e^{-w} \ge 1 - 5r/(2n), \quad nD \le 15000r/997.$$

The proof uses $e \ge 0$ and never divides by e. If a previous pass gives $nD \le Nd$, subtracting the favorable phase term gives the sharper shared budget

$$ne\rho + \kappa nD + n^{3/2}L \le \mathcal{B}(H), \qquad \kappa = 997/15000, \tag{48}$$

by increasing each raw quadratic curvature by

$$q_{\mathrm{extra}}(\rho) = \tfrac{1}{2}(1 - 5r/(2N))(.4095\rho - d/15) > 0. \tag{49}$$

Indeed $1 - e^{-w} \ge we^{-w}$, $D \le d$, $\phi(0) < .4$ and $C_{\mathrm{E}} > .4095$. The positivity gate is checked before this use. The field called `Qbudget` already contains this addition. For a cell with $D \ge d_l \ge 0$ and $\rho \ge \rho_l > 0$, the same identity also gives the score update

$$C \le .40974 + \{r/N - .3988(1 - 5r/(2N))d_l/6\}/\rho_l.$$

A negative numerator excludes the cell by $e \ge 0$. Otherwise the numerator increases with $1/n$, proving this bound at every $n \ge N$.

One positive-measure rule. Suppose zero lies in the gap $(-A, B)$, $A, B > 0$. Let

$$p = \mathbb{P}(X > 0), \quad q = 1 - p, \quad m = \mathbb{E}X_+ = \mathbb{E}X_-, \quad V = \mathbb{E}[X \mid \mathbf{1}_{X>0}],$$
$$W = X - V, \quad \theta_\pm = \mathbb{E}[W^2; \pm X > 0], \quad \theta = \theta_+ + \theta_-.$$

Then $m^2 = (1 - \theta)pq$, and the conditional means have magnitudes $\bar{A} = m/q$, $\bar{B} = m/p$. Set $\Delta_A = \bar{A} - A \ge 0$, $\Delta_B = \bar{B} - B \ge 0$. We also use the unnormalised endpoint displacements

$$M_+ = \mathbb{E}[(X - B)\mathbf{1}_{X>0}] = p\Delta_B, \qquad M_- = \mathbb{E}[(-X - A)\mathbf{1}_{X<0}] = q\Delta_A.$$

Lemma 7.7 (Positive-measure pricing)**.** *For $x = |X|$, $a = |V|$, and a function h constant on each sign class,*

$$\mathbb{E}[h(|X|^j - |V|^j)] = \mathbb{E}\left[hW^2 \sum_{k=0}^{j-2} (k+1)a^k x^{j-2-k}\right], \quad j \ge 2. \tag{50}$$

Also, for $d\mu_0 = (|X| - 1)^2(|X| + 1/2)\, dP$,

$$\begin{aligned}
\mu_0(\mathbb{R}) &= \rho - 1, \\
M_4 - 1 &= \int \frac{(x+1)^2}{x + 1/2}\, d\mu_0, \\
M_5 - 1 &= \int \frac{x^3 + 2x^2 + 3x + 3/2}{x + 1/2}\, d\mu_0, \\
K_1 + 3\rho\beta &= \int \frac{\operatorname{sgn}(X)x(x + 2 - 3\rho/2)}{x + 1/2}\, d\mu_0.
\end{aligned}$$

For any nonnegative resources z_i satisfying $\sum a_i z_i \le r$, $a_i > 0$, a signed linear cost obeys

$$\sum b_i z_i \le r \max\{0, b_1/a_1, \dots, b_k/a_k\}. \tag{51}$$

Proof. Divide $x^j - a^j - ja^{j-1}(x-a)$ by $(x-a)^2$; the linear term has zero conditional mean. The radial identities follow by multiplying out and using $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$. Finally each b_i is at most a_i times the displayed maximum, and all z_i are nonnegative. □

This one rule accounts for the moment, signed Taylor, endpoint movement, outer-gap, and confinement-error estimates. In particular, no two costs may each spend the full residual and then be added.

The gap identity and simultaneous defect budget are (9) and (11), with $(A_-, A_+) = (A, B)$ and $(\Delta_-, \Delta_+) = (\Delta_A, \Delta_B)$. The outer-gap estimate (8) bounds the same $T_\pm, R_\pm, J_o$. One resource pays for all their charges together.

A state, its update, and its exported resource. A state for a closed probability interval $I = [l, u]$ records

$$|X| \le R, \quad C \in [c_0, c_1], \quad \mathcal{R} \le r, \quad nD \le Nd, \quad \theta \le \Theta < 1, \quad A \ge A_l > 0, \quad B \ge B_l > 0,$$

together with simultaneous signed-moment and coefficient intervals. The distinction between $nD \le Nd$ and $D \le d$ is essential. Let $s_+ = \max_I p(1-p)$, $s_- = \min_I p(1-p)$ and

$$b_I = \frac{(6+2\sigma)(1-\Theta)^{3/2}}{\sqrt{s_+}} \operatorname{dist}(p_*, I)^2, \qquad r_I = d - b_I,$$
$$K_+ = 3(\sigma+1)B_l, \quad K_- = 3(\sigma+3)A_l, \quad K = \min(K_+, K_-), \quad m_l = \sqrt{(1-\Theta)s_-}.$$

If $r_I < 0$, exclude the entire closed cell. Otherwise (11) gives, simultaneously,

$$6m(A_l + B_l)(\Delta_A + \Delta_B) + K_+\theta_+ + K_-\theta_- \le r_I. \tag{52}$$

Thus the actual law satisfies the intersected bounds

$$\begin{aligned} \theta &\le \min\{\Theta, r_I/K\}, \\ A &\ge \max\left\{A_l, \sqrt{(1-\Theta)l/(1-l)} - \frac{r_I}{6m_l(A_l+B_l)}\right\}, \\ B &\ge \max\left\{B_l, \sqrt{(1-\Theta)(1-u)/u} - \frac{r_I}{6m_l(A_l+B_l)}\right\}. \end{aligned} \tag{53}$$

Each right side is evaluated from established bounds; sequential improvements may then be used by the next expression. A nonpositive endpoint estimate fails this update, rather than excluding a law. Initialization uses the same formulas with the valid lower sum $A + B \ge \sqrt{2-d}$.

In Bernoulli coordinates set $Y = B_0 + U$, $\mathbb{E}[U \mid B_0] = 0$, $B_0 \sim \text{Bernoulli}(p)$, $X = \ell(Y - p)$ and $d_Y = (\Delta_A + \Delta_B)/(\bar{A} + \bar{B})$. Then $\ell^2 = (1-\theta)/(pq)$ and $\lambda = n\mathbb{E}U^2 = npq\theta/(1-\theta)$. The *weighted* versions of (52) prove

$$n\theta \le \frac{Nr_I}{K}, \quad nd_Y \le \frac{Nr_I}{6(1-\Theta)(A_l+B_l)}, \quad \lambda \le \frac{Ns_+r_I}{(1-\Theta)K}. \tag{54}$$

Indeed $nD \le Nd$ whereas $nb_I \ge Nb_I$. These exports are derived from the weighted resource; multiplying an unweighted variance ceiling by N would not prove them.

For $j = 3, 4, 5$ set

$$L_3(v) = R + 2v, \quad L_4(v) = R^2 + 2Rv + 3v^2, \quad L_5(v) = R^3 + 2R^2v + 3Rv^2 + 4v^3.$$

Put $v_+ = \sqrt{(1-l)/l}$, $v_- = \sqrt{u/(1-u)}$ and $k_j = [1-(1-\Theta)^{j/2}]/\Theta$, with $k_j = j/2$ at $\Theta = 0$. For $u < 1/2$, valid Bernoulli moment ceilings are, with $s = l(1-l)$,

$$B_3^* = (1-2s)/\sqrt{s}, \quad B_4^* = 1/s - 3, \quad B_5^* = (1-4s+2s^2)/s^{3/2}.$$

They decrease towards $p = 1/2$. Convexity places $(1-\theta)^{j/2}$ below its endpoint chord on $[0, \Theta]$; positive-measure pricing gives

$$M_j \le B_j^* + r_I \max\left\{0, \frac{L_j(v_+) - k_j B_j^*}{K_+}, \frac{L_j(v_-) - k_j B_j^*}{K_-}\right\}. \tag{55}$$

The same rule, using $T_+ \le 2mv_+\Delta_B + L_3(v_+)\theta_+$ and its reflection, bounds

$$W_o := (\sigma-1)T_+ + (\sigma+1)T_- \le r_I \max\left\{\begin{array}{c} \dfrac{(\sigma-1)v_+}{3(A_l+B_l)}, \dfrac{(\sigma+1)v_-}{3(A_l+B_l)}, \\ \dfrac{(\sigma-1)L_3(v_+)}{K_+}, \dfrac{(\sigma+1)L_3(v_-)}{K_-} \end{array}\right\}. \tag{56}$$

Intersect every new bound with the previous one. The versions which merge θ_+ and θ_- are obtained by replacing both costs by their maximum and both prices by their minimum; they need no new proof.

The enclosure theorem. On each cell, the formula atlas bounds the four coefficients in (12). The result, for each actual sign $s = \operatorname{sgn} H$, is a finite family of quadratics

$$\mathcal{R} \le \max_j (A_j + F_{j,s}|H| - Q_j^{\mathrm{raw}} H^2), \qquad Q_j^{\mathrm{raw}} > 0. \tag{57}$$

The maxima retain all score–moment corners. In the initial signed cover, intervals contain both signs of β, γ; no orientation is discarded. In the probability cover, the central integral bound is a maximum of affine functions of ρ, hence convex. The other ρ-dependence is affine or the maximum of convex quadratics. For fixed H, the score dependence is convex. Endpoints are therefore taken *before* completing the square.

Theorem 7.8 (Finite resource enclosure)**.** *Consider a finite ordered schedule of the following operations:*

1. *Split a cell into an explicitly covered family of closed cells.*
2. *Intersect its enclosures with an inequality proved in the atlas.*
3. *Exclude a cell by an empty necessary interval or a strictly negative nonnegative resource.*
4. *Apply a separately proved support or confinement implication, checking all its hypotheses against this cell's already acquired data.*
5. *Export a raw, weighted, or signed phase bound by* (57)*,* (48)*, and* (54)*.*

If exact outward arithmetic checks every domain, corner and cover, every output contains all attaining laws satisfying the input. The conclusion holds for all integers $n \ge N$ wherever the formula's monotonicity gates are checked. It includes zero conditional variance, singleton intersections and arbitrarily small atoms.

Proof. Induct over the schedule. A split preserves the union, intersections use necessary inequalities, and an excluded cell contains no feasible law. Each callback consumes only earlier hypotheses for that same law. For a scalar raw output, use $fh - qh^2 \le (f_+)^2/(4q)$ for $h \ge 0$, $q > 0$; using f^2 is a valid weaker bound. For a resource output first add (49) to each curvature. A negative maximum excludes the cell; a nonnegative one may be divided by κN to give its next defect parameter. The phase bound follows from the positive root of $A + fh - qh^2 \ge 0$, replacing A by A_+ if necessary. A strict output is exported only when the certificate retains a strict upper gap, such as its strict Fourier or target enclosure; closed-domain coverage does not change strictness. All constants are frozen before evaluating corners. Thus no iteration step relies on an unproved fixed point. The stated index uniformity is that of the displayed finite formulas, not a fixed-law limit. □

For the Fourier cancellation estimate only, reflect (X, H) if needed: $I_0(-X, -H) = I_0(X, H)$. Both odd coefficients change sign, and $\int e^{-x^2/2}|zx^2 - cx^4|\,dx$ is unchanged by $(z, c) \mapsto (-z, -c)$. Consequently a lower endpoint used there bounds $|\gamma|$, not necessarily the original signed γ. The actual contact, gap and phase data keep their original orientation. A useful additional signed lower bound is

$$\gamma \ge (1-\Theta)^{3/2}\frac{1-2u}{\sqrt{u(1-u)}} - \left(R + 2\sqrt{\frac{u}{1-u}}\right)\Theta. \tag{58}$$

To prove it expand $\mathbb{E}X^3 = \mathbb{E}V^3 + \mathbb{E}[W^2(X+2V)]$; the positive residual is nonnegative and the negative residual has the displayed lower bound. The binary term decreases in both $p < 1/2$ and θ. Thus $\max(0, \text{any signed lower bound})$ is a valid lower bound for $|\gamma|$ inside this auxiliary Fourier estimate.

Signed moments with a common first moment. Put $a_\pm = (1 \pm \beta)/2$, $b_\pm = (\rho \pm \gamma)/2$. Cauchy gives $m^2 \le a_+a_-$ and $b_\pm m \ge a_\pm^2$. Consequently

$$\frac{(1+\beta)^{3/2}}{\sqrt{1-\beta}} - \rho \le \gamma \le \rho - \frac{(1-\beta)^{3/2}}{\sqrt{1+\beta}}, \qquad m \ge \max\left\{\frac{1+\beta^2}{2\rho}, \frac{a_+^2}{b_+}, \frac{a_-^2}{b_-}\right\}.$$

These statements also hold with a zero atom. Projection of $X|X|$ onto the span of X and $|X|$ gives

$$\mu_4 \ge \rho^2 + \frac{(\gamma - \rho\beta)^2}{1-\beta^2}.$$

For completeness, this follows by minimizing the nonnegative expectation $E(X|X| - uX - v|X|)^2$ over u, v; its Gram matrix has diagonal entries one and off-diagonal entry β, and is positive definite.

The signwise moment matrices below, valid also for different radii, provide the bounded-support fourth- and fifth-moment contractions. A second useful upper bound for m follows by integrating $(x-a)^2(R-x) \ge 0$ for $x = |X| \in [0, R]$. For every fixed $a > 0$,

$$2m \le \frac{R - \rho + 2a + Ra^2}{a(2R+a)}.$$

The cover uses only the eleven rational choices $a = 2/10, \dots, 12/10$.

Write $\delta = \gamma - \rho\beta$. Cauchy and a square completion on each sign give the signed constant bound

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \le \frac{\rho^3}{4} - \frac{(\rho+\gamma)(\delta_+)^2}{8(1+\beta)^2} - \frac{(\rho-\gamma)((-\delta)_+)^2}{8(1-\beta)^2}.$$

Indeed on either sign, write $a = \mathbb{E}X_s^2$, $b = \mathbb{E}X_s^3$, $c = \mathbb{E}X_s^4$, $f = \mathbb{E}X_s^5$. Cauchy gives $c \geq b^2/a$, $f \geq c^2/b$, and $\rho c/2 - f/4 \leq \rho^2 b/4 - (c - \rho b)^2/(4b)$. Distance to the half-line $c \geq b^2/a$ proves the stated penalty on summing both signs. Independently, $\mu_4^2 \leq \rho\eta_5$ and the projection bound above give

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\eta_5 \leq \frac{\rho^3}{4} - \frac{(\gamma - \rho\beta)^4}{4\rho(1-\beta^2)^2}.$$

The larger of the two nonnegative deductions is retained.

An actual central gap without assuming $D < 2$. If zero belongs to the support, the signwise gap calculation gives $J \leq \rho$. Thus $D \geq \sqrt{10}\rho - \gamma \geq (\sqrt{10} - 1)\rho$. Whenever $D < (\sqrt{10} - 1)\rho$, there is therefore a central gap $(-A, B)$, $A, B > 0$, with no zero atom. Put $p = P(X > 0)$, $q = 1 - p$, $T_+ = b_+ - mB^2$ and $T_- = b_- - mA^2$. The exact gap decomposition is

$$D = mQ + (\sigma - 1)T_+ + (\sigma + 1)T_- + 3(T - J_o), \qquad Q = (\sqrt{\sigma + 1}A - \sqrt{\sigma - 1}B)^2, \quad \sigma = \sqrt{10}.$$

All terms are nonnegative. Hence $T \leq D/(\sigma - 1)$ and

$$(A + B)^2 \geq A^2 + B^2 \geq \frac{\rho - D/(\sigma - 1)}{m}.$$

Moreover $A \leq a_-/m$, $B \leq a_+/m$ and $J \leq \rho + 2m$. The last bound gives the compulsory defect

$$D \geq \sigma\rho - \gamma - 6m.$$

The inequalities $T - J_o \geq 2BR_+ + 2AR_-$ give the useful additional endpoint constraint

$$D \geq (\sigma - 1)b_+ + 6a_+B - (\sigma + 5)mB^2.$$

If $D < (\sigma - 1)b_+$, this downward quadratic forces B to be at least its positive root. The reflected constraint has a_-, b_-, A and the coefficients $\sigma + 1, \sigma + 7$. In a closed moment box, positive coefficients are rounded down, negative coefficients up, and the lower square-root endpoint is used in the resulting root bound.

The actual opposite-sign event has probability $2pq$. The first two moment constraints give lower bounds for both p and q. For independent copies one has exactly

$$E[(X - X')^2 \mid XX' < 0] = \frac{1 + \beta(1 - 2p) + 4m^2}{2pq} = \frac{1 + 4m^2 + \beta}{2p} + \frac{1 + 4m^2 - \beta}{2q}.$$

For an upper bound, use the upper m endpoint and both endpoints of p and β. The expression is convex in p and affine in β. This improves the former $2/P(XX' < 0)$ bound inside the complete HIGH estimate. Both copies still have the same actual marginal.

Preserving central and outer integral costs. Let

$$P_1(A, B) = B^3/3 + AB^2/2 - A^3/6, \quad P_2(A, B) = B^4/4 + AB^3/3 - A^4/12, \quad G = \rho P_1 - P_2.$$

Then $J_{1,c} = mP_1$, $J_{2,c} = mP_2$. Since $\partial_B G = B(A + B)(\rho - B)$,

$$G(A, B; \rho) \leq G(A, \rho; \rho) = \rho^4/12 + \rho^3 A/6 - \rho A^3/6 + A^4/12.$$

The last expression increases with A because its derivative is $(\rho - A)^2(\rho + 2A)/6$, and is convex in ρ because its second ρ derivative is $\rho^2 + \rho A$. Its closed-rectangle maximum is therefore bounded using the upper A endpoint and both ρ endpoints.

Positive outer gaps satisfy $(t-a)(\rho-t) \le (\rho-B)_+^2/4$. On negative outer gaps the positive part is bounded by $(R-\rho)^2/4$ after extracting $|t|$. Integration gives

$$3(\rho J_1 - J_2)_o \le \frac{3}{8}\{(\rho - B)_+^2 T_+ + (R-\rho)^2 T_-\}.$$

The available weighted outer budget is $D - mQ$, not the entire D. The cover intersects this budget with the individual bounds $T_+ \le b_{+,u} - m_l B_l^2$ and $T_- \le b_{-,u} - m_l A_l^2$.

For the signed phase term $U = J_1 + (\mu_4 - 3)/6$, retain

$$-RT_-/2 \le J_{1,o} \le RT_+/2.$$

The central minimum of P_1 occurs at B_l and an A endpoint; its maximum occurs at B_u and A clipped to the interval at $A = B_u$. A nonnegative polynomial bound is multiplied by m_u for an upper bound and m_l for a lower bound; a negative one uses the opposite endpoint. For example, the binary endpoints $A = 3/2$, $B = 2/3$ make this maximum negative; the sign switch is necessary even when $D < 2$. The fourth-moment sign intervals also sharpen $K_1 = E[X|X|^3] - 3\beta - 3\rho\gamma/2$. The complete K_1, U, ϕ rectangles are used in the signed expression $|CK_1 + \phi U|$, without replacing it by two separate absolute values.

A complete coarse bound before a central gap is licensed. Always $-\mu_4/6 \le J_1 \le \mu_4/2$ and $-3J_2 - \eta_5/4 \le 0$. On a gap of length $h = b - a$, these follow from

$$\int_a^b (t-a)t\,dt + \tfrac{1}{2}\int_a^b t^2 dt = hb^2/2, \quad \tfrac{3}{2}\int_a^b t^2 dt - \int_a^b (t-a)t\,dt = h(2a+b)^2/6,$$

and, on a negative gap $(-v, -u)$, from $\frac{1}{3}\int_u^v x^3 dx - \int_u^v (v-x)x^2 dx = u^3(v-u)/3$. For $1 \le r \le \rho$ and actual support radius $R \le 6$, set

$$\Gamma(R, r) \ge \max_{r \le v \le R} \left\{ -\frac{v^2}{4} + \frac{(R+r)v}{3} - \frac{Rr}{2} + \frac{r^3(2R-r)}{12v^2} \right\}_+ .$$

Then

$$3(\rho J_1 - J_2) \le \frac{3}{2}\Gamma(R, r)(\rho - \gamma) + \frac{3}{8}\rho^2 J.$$

On negative gaps extract $x = |t|$ and discard the nonpositive part $x < \rho$. Replacing ρ by r and the outer endpoint by R gives $\int_r^v (R-x)x(x-r)\,dx$. Dividing this integral by v^2 gives exactly the displayed expression. Signwise Tonelli now charges $\Gamma \mathbb{E} X_-^3 = \Gamma(\rho - \gamma)/2$. On positive gaps $(t-a)(\rho - t) \le \rho^2/4$, giving $\rho^2 J/8$. The implementation uses 64 closed v-intervals: on each, maximize the concave quadratic at its clipped vertex $2(R+r)/3$ and the positive reciprocal term at the left endpoint. This proves its interval bound. The alternative bound $\Gamma \le 5/2$ follows at $r = 1, R = 6$ by $3v^4 - 28v^3 + 66v^2 - 11 > 0$ on $[1, 6]$: the endpoint values are $30, 205$, and the interior critical values are $(85 \mp 35\sqrt{5})/2 > 0$. These monotone enclosing parameters cover all $r \ge 1, R \le 6$.

Moment and gap coefficients for a whole cell. The following coefficient rules are collected here to make the finite calculation transparent. Every rule is an inequality for an actual law; there is no extremal-law assumption inside a cell. Write $v = 1 - \theta$. For

$p \in [l,u] \subset (0,1)$ set

$$s_0 = \min\{l(1-l), u(1-u)\}, \quad v_* = \max\{\sqrt{u/(1-u)}, \sqrt{(1-l)/l}\},$$
$$r_B = (1-2s_0)/\sqrt{s_0}, \quad g_B = \max_{t=l,u} |1-2t|/\sqrt{t(1-t)}, \quad b_B = \max_{t=l,u} |1-2t|,$$
$$L_3 = R + 2v_*, \quad L_4 = R^2 + 2Rv_* + 3v_*^2,$$
$$L_5 = R^3 + 2R^2 v_* + 3Rv_*^2 + 4v_*^3, \quad e_B = r_B \frac{1 - 4s_0 + 2s_0^2}{s_0(1-2s_0)}.$$

The polynomials $T_3 = x + 2a$, $T_4 = x^2 + 2ax + 3a^2$ and $T_5 = x^3 + 2ax^2 + 3a^2 x + 4a^3$ are nonnegative. Their maxima at $x = R$, $a = v_*$, in Lemma 7.7, give, if $\theta \le \Theta < 1$,

$$\begin{aligned} \rho &\le r_B + (L_3 - r_B)\Theta, & |\gamma| &\le g_B + (L_3 - g_B)\Theta, \\ |\beta| &\le b_B + (1 - b_B)\Theta, & \mu_4 &\le (1-\Theta)^2(1/s_0 - 3) + L_4\Theta, \\ \eta_5 &\le (1-\Theta)^2 e_B + L_5\Theta. \end{aligned} \tag{59}$$

Replacing θ by Θ uses the explicitly checked inequalities $L_3 > r_B, g_B$, $L_4 > 2(1/s_0 - 3)$ and $L_5 > 2e_B$. The exact binary factors are $v^{3/2}, v, v^2, v^{5/2}$; the upper bounds use $v^{3/2} \le v$ and $v^{5/2} \le v^2$. Thus cells crossing $1/2$ and zero conditional variances are included. The same cubic expansion, in the later domain $p \in [.37, .47]$, $A \ge .75$, $B \ge 1.05$, gives

$$\rho \ge v^{3/2}\rho_B(p) + 3\min(A,B)\theta \ge \rho_B(p) + .45\theta, \qquad \rho_B(p) = \frac{1-2pq}{\sqrt{pq}}. \tag{60}$$

Indeed $\rho_B < 1.11$ there and $v^{3/2} \ge 1 - 3\theta/2$. Also $\beta \ge 1 - 2p - 2(1-p)\theta$.

For the signed coefficient $K_1 = \mathbb{E}[X|X|^3] - 3\beta - 3\rho\gamma/2$, define

$$K(p,v) = (1-2p)\{(3v^3/2 - v^2)/(pq) - 3v^3 + v^2 + 3v\}.$$

Direct substitution gives $K_1(V) = -K(p,v)$, with the coefficient 3 unchanged. For $v \ge 2/3$, K decreases in p: both coefficients in braces are nonnegative and $(1-2p)/(pq)$ decreases. Its derivative in v has the sign of $1 - 2p$, since the derivative of the braces is at least $9v^2 - 6v + 3 > 0$. Hence both variance endpoints are retained. The error is

$$|K_1(X) - K_1(V)| \le \Theta\{L_4 + 3 + \tfrac{3}{2}L_3(R_\rho + g_B)\} \tag{61}$$

when $\rho \le R_\rho$. This follows from the conditional expansions and $\rho\gamma - \rho_V\gamma_V = \rho(\gamma - \gamma_V) + \gamma_V(\rho - \rho_V)$.

Here are the remaining signed charges. They also explain the two alternative bounds retained in the initial, coarse moment cover:

$$-\mu_4/6 \le J_1 \le \mu_4/2, \qquad -3J_2 - \eta_5/4 \le 0, \tag{62}$$
$$\rho J_1 - J_2 \le \tfrac{5}{4}(\rho - \gamma) + \tfrac{1}{8}\rho^2(\rho + 1), \tag{63}$$
$$-75D/104 \le J_{1,o} \le 3D/2. \tag{64}$$

The first two lines follow from the gap integral identities above, using $\Gamma \le 5/2$ and $J \le \rho + 1$. For the outer J_1 bounds, $J_{1,o} \ge -3T_-$, $J_{1,o} \le 3J_o \le 3(T_+ + T_-)$ and (9) give the stated constants.

For the previously defined P_1, P_2, $J_{j,c} = mP_j(A,B)$. Both polynomials increase in B; their A derivative changes from positive to negative at $A = B$. A rectangle minimum is therefore at B_l and an A endpoint. Multiply a negative minimum by an upper bound for m,

and a nonnegative minimum by a lower bound. For the upper central charge, maximizing at $(\bar{a}, \bar{b})$ if $p \le 1/2$ gives $J_{1,c} \le v^2(2-3p)/(6pq)$, a decreasing function of p on that interval. If $p \ge 1/2$, maximize first at $A = B$ to get $J_{1,c} \le (2/3)v^2q^2/p \le 1/3$. Thus every cell may use

$$J_{1,c} \le \max\{1/3, (2-3l)/(6l(1-l))\}. \tag{65}$$

For $|X| \le R$, $R \ge 1$, define

$$k_4(R) = \frac{(R+1)^2}{R+1/2}, \qquad k_5(R) = \frac{R^3+2R^2+3R+3/2}{R+1/2}.$$

Then

$$\mu_4 \le \min\{R\rho, 1+k_4(R)(\rho-1)\}, \quad \eta_5 \le \min\{R\mu_4, 1+k_5(R)(\rho-1)\}, \quad \mu_6 \le R\eta_5. \tag{66}$$

The first two bounds are the supported moment estimates proved in Section 5.3; the last is pointwise. The following coarse cancellation estimates retain the radius-six majorants. The last identity in (7.7) gives

$$|K_1 + 3\rho\beta| \le (96-18\rho)(\rho-1)/13. \tag{67}$$

Indeed $u(u+2-3r/2)/(u+1/2)$ is convex on $[0,6]$, its maximum is $(96-18r)/13$, and its negative part is at most $31/40$, smaller than that maximum for $1 \le r \le 37/20$. The following bounds are used only for $|X| \le 6$ and $1 \le \rho \le 37/20 < 2$, with the signed moment hypotheses below. The other bounds needed on this domain are

$$\begin{aligned}
&|K_1| \le 6\rho - 3\rho^2/2 + 3|\beta|, \\
&K_1 \le M + 3|\beta| - 3\rho g_-/2, && -K_1 \le 3\rho + 3|\beta| + (3\rho/2-3)g_-, \\
&-K_q \le (\rho/2-3/8)M - \rho/2 + 1/8 + 3|\beta|g/2, && -K_q \le 8\rho^3/27 + (3|\beta|/2-1/2)\rho,
\end{aligned} \tag{68}$$

where $\mu_4 \le M$, $|\gamma| \le g$, $\gamma \ge g_-$, and $K_q = \eta_5/4 - \rho\mu_4/2 - 3\beta\gamma/2 + \rho/2$. For the first two lines use $|\mathbb{E}[X|X|^3] - 3\gamma| \le 3\rho$, and note the negative coefficients of γ. For the last line use $\eta_5 \ge \mu_4^{3/2} \ge 3\mu_4/2 - 1/2$, or maximize $\rho y^2/2 - y^3/4$ over $y \ge 0$.

Compatible weaker resource constants. The implementation also permits the weaker binary charge $b_I = (616/25)(1-\Theta)^2 \operatorname{dist}(p_*, I)^2$. It follows from $2(6+2\sigma) > 616/25$ and $(1-\theta)^{3/2} \ge (1-\theta)^2$. With $K = \min(K_+, K_-)$ and $v_* = \max(v_+, v_-)$, the merged charge (52) gives

$$T_+ + T_- \le T_* = (d - b_I)\max\{v_*/[3(A_l+B_l)], (R+2v_*)/K\}, \qquad d_o = \min\{d, (\sigma+1)T_*\}.$$

Thus $W_o \le d_o$ and, for $R \le 6$,

$$-75d_o/104 \le J_{1,o} \le 3d_o/2, \qquad 3(\rho J_{1,o} - J_{2,o}) \le \frac{3}{8}\max\{\rho^2/(\sigma-1), (R-\rho)^2/(\sigma+1)\}d_o.$$

These are precisely the merged-cost versions of the sign-specific resource rule; they do not introduce another available budget.

Conditional lower moments improve two signed charges. Let $s_u = \max_{p\in[l,u]} pq$ and $\theta \le \Theta$. The unsigned cases of (50) give

$$\mu_4 \ge M_l := \max\{1, (1-\Theta)^2(1/s_u - 3)\}, \tag{69}$$

$$\eta_5 \ge E_l := (1-\Theta)^{5/2}\frac{1 - 4s_u + 2s_u^2}{s_u^{3/2}}. \tag{70}$$

Both binary moment functions decrease in $s \in (0, 1/4]$. For the fifth this follows because the derivative has the sign of $s^2 + 2s - 3/2 < 0$; the fourth is immediate. The fourth-moment lower bound sharpens the lower endpoint of $J_1 + (\mu_4 - 3)/6$; its explicit use appears in the cell map below.

The constant signed charge is

$$-K_q = \frac{\rho\mu_4}{2} - \frac{\eta_5}{4} + \frac{3}{2}\beta\gamma - \frac{\rho}{2}.$$

The old coherent estimate used $\eta_5 \ge 3\mu_4/2 - 1/2$ and $\mu_4 \le M$. A second estimate uses (70). Their difference is independent of ρ. Consequently the improved bound is exactly the old one minus

$$\frac{1}{4}\max\{0, E_l - 3M/2 + 1/2\}. \tag{71}$$

This is a constant shift on the cell and does not spoil convexity. It does not assert the generally invalid implication $\eta_5 \ge 3M/2 - 1/2$ from an upper bound on μ_4.

Signed conditional Taylor cancellation. Let $K_1 = \mathbb{E}[X|X|^3] - 3\beta - \frac{3}{2}\rho\gamma$. On a sign class put $x = |X|$, $a = |V|$, $W = X - V$ and $\varepsilon = 1$ on the positive class, -1 on the negative class. In the actual central-gap domain there is no atom at zero. With $w_3 = x + 2a$, $w_4 = x^2 + 2ax + 3a^2$ and $\gamma_V = \mathbb{E}V^3$, exact conditional expansion gives

$$K_1(X) - K_1(V) = \mathbb{E}\Big[W^2\left\{\varepsilon(w_4 - 3 - \tfrac{3}{2}\rho w_3) - \tfrac{3}{2}\gamma_V w_3\right\}\Big]. \tag{72}$$

Apply (50) to $j = 2, 3, 4$, with $h = 1$ or $h = \varepsilon$, and use $\rho\gamma - \rho_V\gamma_V = \rho(\gamma - \gamma_V) + \gamma_V(\rho - \rho_V)$.

The positive coefficient in (72) is $F(x, a; t) = x^2 + 2ax + 3a^2 - 3 - \frac{3}{2}t(x + 2a)$ with $t = \rho + \gamma_V$; the negative coefficient is $-F$ with $t = \rho - \gamma_V$. Their ranges are bounded on complete rectangles of actual conditional means and support magnitudes. The positive-support magnitude is at most R. The same formula permits separate actual bounds on the two signed support magnitudes. On the late domain $p < 1/2$, bounds for γ_V are

$$\sqrt{(1-\Theta)^3(1-2u)^2/[u(1-u)]} \le \gamma_V \le (1-2l)/\sqrt{l(1-l)}.$$

The positive mean lies in $[\sqrt{(1-\Theta)(1-u)/u}, \sqrt{(1-l)/l}]$, and the negative magnitude mean lies in $[\sqrt{(1-\Theta)l/(1-l)}, \sqrt{u/(1-u)}]$. Use the lower bound $\rho \ge \rho_B(u)$ only after checking $3\min(A_l, B_l) \ge (3/2)\sup_I \rho_B$. Indeed conditional expansion and $(1-\theta)^{3/2} \ge 1 - 3\theta/2$ then imply $\rho \ge (1-\theta)^{3/2}\rho_B(p) + 3\min(A, B)\theta \ge \rho_B(p)$. Otherwise retain the universal lower bound one.

Since $x + 2a > 0$, F decreases with t. For fixed t its Hessian is $\left(\begin{smallmatrix} 2 & 2 \\ 2 & 6 \end{smallmatrix}\right) > 0$. Its rectangle maximum is at a corner. Its minimum is among the four corners, the minimizer on each edge, and the stationary point $(x, a) = (3t/8, 3t/8)$ if that lies in the rectangle. On fixed-x edges use $a = t/2 - x/3$, clipped to the edge; on fixed-a edges use $x = 3t/4 - a$, also clipped. This is a complete finite extremum list. Multiplication by $0 \le \theta \le \Theta$ uses $\Theta\min(0, L)$ and $\Theta\max(0, U)$ for the lower and upper coefficient bounds. It intersects this new interval with the retained absolute-error interval; neither need be discarded.

The complete central rectangle. The central part of $\rho J_1 - J_2$ is exactly $mG(A, B; \rho)$, where $G = \rho P_1 - P_2$. Let a proved probability cell give $A \in [A_l, A_u]$, $B \in [B_l, B_u]$, $m \in [m_l, m_u]$. Here

$$A_u = \sqrt{u/(1-u)}, \quad B_u = \sqrt{(1-l)/l}, \quad m_l = \sqrt{(1-\Theta)s_0}, \quad m_u = \sqrt{\max_{p\in[l,u]} pq}.$$

Lemma 7.9 (Complete central rectangle)**.** *Write* $G(A_l + (A_u - A_l)x, B_l + (B_u - B_l)y; \rho) = \sum t_{kh} x^k y^h$. *Define the 25 rational coefficients*

$$b_{ij}(\rho) = \sum_{\substack{k\le i\\ h\le j}} t_{kh} \frac{\binom{i}{k}}{\binom{4}{k}} \frac{\binom{j}{h}}{\binom{4}{h}}, \qquad 0 \le i, j \le 4,$$

and

$$\mathcal{G}(\rho) = \max_{i,j} \{m_l b_{ij}(\rho),\, m_u b_{ij}(\rho)\}.$$

Then $\rho J_{1,c} - J_{2,c} \le \mathcal{G}(\rho)$. *Moreover* $\mathcal{G}$ *is convex in* ρ.

Proof. Apply Lemma 4.2 separately to the two coordinates after the affine substitution. The Bernstein basis is nonnegative and sums to one on the closed square, so G is a convex combination of its 25 coefficients. For each coefficient retain both m endpoints, or equivalently use m_u when it is nonnegative and m_l otherwise. This gives a uniform bound on the entire three-dimensional rectangle, including boundaries. Each endpoint-multiplied coefficient is affine in ρ; their maximum is convex. No interior stationary point is omitted. □

One interval-feedback theorem. The signed identity from the preceding section is

$$\mathcal{R} = C\{3\beta J/2 + 3\rho J_1 - 3J_2 - K_q\} + H(CK_1 + \varphi U) - H^2(C\rho + \gamma\varphi/2) + E_n, \tag{73}$$

where $U = J_1 + (\mu_4 - 3)/6$, $\varphi = \phi(H/\sqrt{n})$. For any one coherent low-frequency triple and complete high-frequency value K_H at floor N,

$$|E_n| \le K_F + s_F|H|, \quad K_F = c_\pi(L_0 + K_H + L_2/\sqrt{N}), \quad s_F = c_\pi L_H, \quad c_\pi \ge 2/\pi. \tag{74}$$

The implementation uses 16/25 before the signed phase and 212/333 afterward; both are valid since $\pi > 333/106$.

Lemma 7.10 (Continuous cells and feedback)**.** *Suppose bounds for one actual cell give*

$$r_0 \le \rho \le r_1, \quad 0 < c_0 \le C \le c_1, \quad |\beta| \le b, \quad |\gamma| \le g, \quad \mu_4 \le M, \quad D \le d,$$

$K_1 \in [k_l, k_u]$, $U \in [u_l, u_u]$, *and* $0 \le \varphi_- \le \varphi \le 2/5$. *Freeze the shared outer charge* d_o, *lower fifth moment* E_l, *upper* $s_u = \max_{p\in[l,u]} p(1-p)$, *and central envelope* $\mathcal{G}$ *from the preceding bounds. Define*

$$\begin{aligned} A(r) &= \tfrac{3}{2}b(r + 2\sqrt{s_u}) + 3\mathcal{G}(r) + \tfrac{3}{8}\max\{r^2/(\sigma-1), (R-r)^2/(\sigma+1)\}d_o \\ &\quad + (r/2 - 3/8)M - r/2 + 1/8 + \tfrac{3}{2}bg - \tfrac{1}{4}\max\{0, E_l - 3M/2 + 1/2\}, \\ B(c) &= \max_{k\in\{k_l,k_u\},\, u\in\{u_l,u_u\},\, \psi\in\{\varphi_-, 2/5\}} |ck + \psi u|, \\ Q(c,r) &= cr + \tfrac{1}{2}\varphi_-(3.16r - 3 - d) - \tfrac{1}{2}(2/5 - \varphi_-)(d - .16)_+. \end{aligned}$$

If all four corner values of Q *are positive, then, for every real* H,

$$\begin{aligned} \mathcal{R} &\le \max_{(c,r)\in\{c_0,c_1\}\times\{r_0,r_1\}} \{K_F + cA(r) + (B(c) + s_F)|H| - Q(c,r)H^2\} \\ &\le \max_{c,r\ \text{corners}} \{K_F + cA(r) + (B(c) + s_F)^2/(4Q(c,r))\}. \end{aligned} \tag{75}$$

For the following feedback, assume in addition that the cell contains a ratio extremizer with $e = C_n - C_{\mathrm{E}} \geq 0$, $n \geq N$, *and that* $0 < T < .4N$ *is a strict upper bound for the final maximum in* (75). *Then* $T < nC_n\rho$, *since* $C_n\rho \geq C_{\mathrm{E}} > .4$, *and the weight-removal step in Corollary 2.6 gives*

$$D < \frac{15000T}{997N}, \quad \varphi > \frac{997}{2500}\left(1 - \frac{5T}{2N}\right), \quad C < C_{\mathrm{E}} + V/N \quad \text{if also } \mathcal{R}/\rho < V. \tag{76}$$

Only positive density floors are used. These statements remain valid at $H = 0$, *on closed cell edges, and for zero conditional variance.*

Proof. The constant coefficient follows from the shared outer budget, Lemma 7.9, and (68), using

$$J \leq \rho + 2ABm \leq \rho + 2(1-\theta)^{3/2}\sqrt{pq} \leq \rho + 2\sqrt{s_u}.$$

This follows from $A \leq \bar{a}$, $B \leq \bar{b}$, $m^2 = (1-\theta)pq$; it is used only in the charge $3|\beta|J/2$. Since $\gamma \geq 3.16\rho - 3 - d$, its negative part is at most $(d - .16)_+$. Using the upper density only for that negative part gives $C\rho + \gamma\varphi/2 \geq Q$. Absolute value is separately convex in each coordinate of $ck + \psi u$, proving the formula for B. For fixed r and $|H|$, the resulting upper bound is convex in c. Take those endpoints *before* completing the square. At either endpoint $c > 0$, it is convex in r, since $\mathcal{G}$ and the maximum of the two outer quadratics are convex, every other term of A is affine, and Q is affine in r. This proves the first line of (75); $fh - qh^2 \leq f^2/(4q)$ proves the second. Bilinearity of Q proves its positivity throughout the rectangle from its corners.

□

Signed phase curvature in the resource schedule. The auxiliary Fourier orientation does not change the signed third moment in the actual transfer identity. The paired phase estimate must therefore retain its sign. On a fixed resource cell put

$$L = \frac{997}{2500}\left(1 - \frac{5r}{2N}\right), \qquad g_0(\rho) = 3.16\rho - 3 - d, \qquad U = .4,$$

where r, d are the preceding raw and defect upper bounds, and the checker requires $0 \leq L \leq U$. The actual density lies in $[L, U]$ and the actual signed moment satisfies $\gamma \geq g_0(\rho)$. Consequently

$$C\rho + \tfrac{1}{2}\gamma\phi \geq q_{\mathrm{s}}(C, \rho) := C\rho + \tfrac{1}{2}\min\{Lg_0(\rho), Ug_0(\rho)\}. \tag{77}$$

This follows by first multiplying the lower bound on γ by the nonnegative actual density, and then minimizing over its two endpoints. The simpler expression $C\rho + Lg_0(\rho)/2$ is licensed if $g_0 \geq 0$, or if nonnegativity of the actual γ has been separately proved. It is not licensed merely by reflecting the auxiliary Fourier integral.

The corrected price preserves the four-corner reduction. For fixed C, q_{s} is concave in ρ, being affine plus the minimum of two affine functions. For fixed ρ it is affine in C. In the phase envelope $A + f|H| - q_{\mathrm{s}}H^2$, the remaining terms are separately convex in C, ρ, by the moment and coefficient bounds already proved. Thus the entire envelope is separately convex, and two successive one-dimensional endpoint bounds place its maximum at the four corners. The same argument applies after adding the affine conserved-resource curvature e_ρ and subtracting a barrier price $K\rho$. If the four corrected divisors are positive, concavity makes them positive throughout the rectangle, and square completion gives the valid raw and resource bounds

$$\max_{C,\rho \text{ corners}}\left(A + \frac{f^2}{4q_{\mathrm{s}}}\right), \qquad \max_{C,\rho \text{ corners}}\left(A + \frac{f^2}{4(q_{\mathrm{s}} + e_\rho)} - K\rho\right).$$

Any threshold-root export likewise uses the corrected corner curvature.

For the later acquired domains, the conditional cubic bound (58) applies in the actual contact orientation. It requires the same marginal's probability interval, support radius and variance ceiling, all established before the later paired calls. The exact domain check covers 302 input rows: 45 central-input rows, 45 left-input rows, 45 first-confinement feedback rows, 45 Sperner feedback rows, 84 signed-resource rows, and 38 final left-radius rows. With the common already acquired radius $R = 3.84$, each right side is greater than .096. Thus all later paired calls have an actual nonnegative signed third moment, independently of any auxiliary Fourier orientation.

There is also a convenient cell-constant correction. Since $\rho \geq 1$, $(-g_0(\rho))_+ \leq (d - .16)_+$, so the weaker price

$$C\rho + \tfrac{1}{2}Lg_0(\rho) - \tfrac{1}{2}(U - L)(d - .16)_+$$

is valid. This is the correction selected for the additional index–46 calculation. Both forms keep the raw remainder separate from the barrier-subtracted resource budget.

The broad preliminary and complementary resource passes are replayed with (77); their subsequent consumers retain the same fixed target tables only when the complete corrected replay verifies every exported resource. A diagnostic equality of two corner maxima alone does not replace this check, since threshold bounds also consume the phase coefficients. Later uses of the simpler formula require their own explicit nonnegative signed-moment or small-defect license.

Using a floor calculation at a later index. A floor calculation encloses one fixed attaining ratio marginal with $C \geq C_{\mathrm{E}}$; it does not assert monotonicity of C_n. For actual $n \geq N$, every saved pass preserves the two simultaneous statements

$$\mathcal{R}_n \leq r, \qquad nD \leq Nd. \tag{78}$$

In particular $D \leq d$ may be used in index-free moment and gap inequalities. The stronger weighted statement is retained whenever a cost is proportional to the sample size.

Here is why each operation in the selected schedules preserves this invariant. Keep the actual marginal's moments, support, signed phase and admissible frequency windows fixed while increasing n. All numerical gates mentioned below are checked before the operation.

Fourier bounds. Positive sums and products of nonnegative decreasing factors remain decreasing. For the complete HIGH bound of Proposition 5.7, these factors are

$$\sqrt{n}e^{-cn}\ (2Nc > 1), \quad n^{-1}, \quad e^{-knb^2}/\sqrt{n}, \quad r(x) = \frac{x+2}{x^2+4x+2}, \quad x \text{ increasing in } n.$$

Indeed $r' = -(x^2+4x+6)/(x^2+4x+2)^2 < 0$. The comparator $\sqrt{n}e^{-n/8}\{2r(n/8) + g_c/3\}$ decreases for $N \geq 5$. The improved first piece has the same factors with its separately checked positive rate and slope. For the optional zero piece, the derivative of $\sqrt{x}e^{-x}r(x)$ has the sign of

$$4 - 10x - 22x^2 - 13x^3 - 2x^4.$$

This polynomial decreases for $x > 0$ and equals $-11/128$ at $x = 1/4$, so the native gate $Nkb^2 \geq 1/4$ suffices. Every use retains the event/window conditions $0 < a \leq 1$, $m_e > 0$, $0 < d_0 \leq 12$, $1 \leq M \leq 111/10$, $0 < w \leq 1/4$ and positive sine slope. Taking the minimum of complete admissible windows is valid even when the minimizing window changes with n.

For LOW, the nonconstant factors are

$$(n/(n-1))^{3/2}, \quad (n/(n-1))^{3/2}/(n-1), \quad a_n^{-j}, \quad b_n^{-j}, \quad n^{-1/2},$$

where $a_n = (n-1)/(2n)$, $b_n = (45-M)(1-2/n)/96$ and $j > 0$. They decrease on the stated positive domains. This covers the full error $e/(2a_n^2) + B^2/(2b_n^3)$ and $L_2/\sqrt{n}$; the paired moment-rectangle hypotheses still have to hold.

Density and score updates. Under the raw sign and positivity license $r < .4N$, the invariant (78) gives

$$\phi(H/\sqrt{n}) \geq \phi_{0,l}(1 - 5r/(2n)).$$

Its lower bound increases with n, whereas $r/n - \phi_{0,l}(1-5r/(2n))D_l/6$ decreases for $r, D_l \geq 0$. Both may therefore be evaluated at N. The general curvature correction has derivative $(g_0 + t)/2 \geq 0$ with respect to this density floor, where $g_0 = (79/25)\rho - 3 - d$, $t = (d - 4/25)_+$: if $d \leq 4/25$, use $g_0 \geq 4/25 - d$; otherwise $g_0 + t = (79/25)(\rho - 1)$. The signed curvature

$$C\rho + \tfrac{1}{2}\min\{\phi_l g_0, (2/5)g_0\}$$

also increases with ϕ_l when $g_0 \geq 0$ and is constant when $g_0 < 0$. Thus both signs are retained. Positive added curvatures and all divisors are checked before square completion.

For $C \geq C_{\mathrm{E}}$ and $C_{\mathrm{E}} < 40974/10^5$, the compulsory charge is

$$n(C - C_{\mathrm{E}})\rho \geq N \max\{0, C - 40974/10^5\}\rho_l.$$

Split the score cover at $40974/10^5$; the charge is affine on each side. The coherent corner calculation then exports $nD \leq B/\kappa$, hence $d = B/(\kappa N)$, for $\kappa = 997/15000$. A negative budget excludes the cell before division. Positive K/n calculations separately retain their native $K\rho$ charge; two lower bounds on the same excess are not added. Closed covers, upward rounding and intersection with an earlier bound now preserve (78) by finite induction, without a fixed-point or limiting argument.

Weighted physical resources. If an acquired inequality is

$$D \geq b + gd_{\mathrm{in}} + (N/n)h_N\lambda_n, \qquad \lambda_n = ns, \quad b, g, h_N \geq 0,$$

then the invariant gives $(n/N)(b + gd_{\mathrm{in}}) + h_N\lambda_n \leq d$, hence $b + gd_{\mathrm{in}} + h_N\lambda_n \leq d$. This retains the original joint intercepts. The actual n is still used in every binomial probability. The transfers used here are $N = 25$ to $26 \leq n \leq 30$ and $N = 31$ to $32 \leq n \leq 34$; each actual index has its own verified support calculation. Index-free support refinements use only already acquired data. For example $6mAB \geq P > 0$ gives $(A+B)^2 \geq 2P/(3m_u)$ by $4AB \leq (A+B)^2$; it does not establish an initial central gap.

Phase and variance exports from the same budget. Suppose one signed phase family gives, for $h = |H| \geq 0$,

$$0 \leq D \leq c_n \max_j (A_j + F_j h - Q_j h^2), \qquad c_n > 0, \quad Q_j > 0.$$

Some corner is nonnegative at the actual h. For that corner $\Delta_j = F_j^2 + 4Q_jA_j \geq 0$ and $h \leq (F_j + \sqrt{\Delta_j})/(2Q_j)$. If $u_j \geq 0$ is a rational upper enclosure of $\sqrt{\Delta_j}$, the complete family therefore gives

$$h \leq \max_j r_j, \qquad r_j = \begin{cases} 0, & \Delta_j < 0, \\ \max\{0, (F_j + u_j)/(2Q_j)\}, & \Delta_j \geq 0. \end{cases}$$

This includes double roots and $h = 0$. Keep separate maxima for the two phase signs and intersect each with its previous cap. At floor31 the normalization is $15000/(997 \cdot 31)$; its actual-index version $c_n = 15000/(997n) > 0$ leaves this argument unchanged. Positivity of every Q_j is required.

Finally suppose bounded conditional supports imply $s \le B\delta$. Writing O_0, O_1 for their outward offsets, take $B = \max(O_0, O_1)\max(p_u, 1-p_l) > 0$. Combining this with the original joint plane $\delta/d_0 + \lambda_n/L \le 1$ and $\lambda_n = ns$ gives

$$\lambda_n \le \{L^{-1} + (nBd_0)^{-1}\}^{-1},$$
$$\theta_n = \frac{\lambda_n}{npq + \lambda_n} \le \frac{1}{1 + nv_l/L + v_l/(Bd_0)} \le \frac{1}{1 + Nv_l/L + v_l/(Bd_0)},$$

where $v_l = \min_{p\in[p_l,p_u]} pq > 0$ and $d_0, L > 0$. Upward rounding and intersection with the previous variance bound preserve this conclusion. A downward support-scale enclosure enlarges O_0, O_1, so also weakens the bound safely. The refinement concerns physical θ_n; it does not identify λ_n with a floor's diagnostic λ_N or replace the plane's intercept L. All 193 saved first-band refinements at floor 25 are checked at each actual index 26–30.

Here is the version of the moment matrices that permits different positive and negative support radii. On either sign write $Y = |X|$, $0 \le Y \le r$, and let p_s, m, a, b, c_4, c_5 be its unnormalized moments of orders zero through five. Positivity of the weights $r - Y$, $Y(r-Y)$, and $Y^2(r-Y)$ gives

$$\begin{pmatrix} rp_s - m & rm - a \\ rm - a & ra - b \end{pmatrix}, \quad \begin{pmatrix} rm - a & ra - b \\ ra - b & rb - c_4 \end{pmatrix}, \quad \begin{pmatrix} ra - b & rb - c_4 \\ rb - c_4 & rc_4 - c_5 \end{pmatrix} \succeq 0.$$

For positive denominators their determinant inequalities imply

$$p_s \ge m/r + (rm - a)^2/[r(ra - b)],$$
$$c_4 \le rb - (ra - b)^2/(rm - a),$$
$$c_5 \le rc_4 - (rb - c_4)^2/(ra - b).$$

A zero diagonal entry forces the corresponding off-diagonal entry to vanish. One may check that condition or discard the nonnegative correction, but one cannot divide by zero. In interval form a correction is bounded below by the square of a positive-part lower numerator divided by an upper positive denominator. The sign radius r is used only for its own class. Summing the first inequality over the two signs and using $p_+ + p_- \le 1$ can improve the common upper mean.

One further bound used at small indices strengthens the signed constant coefficient. Set $\delta = \gamma - \rho\beta$ and $Y = |X|$. Then

$$V := \mathbb{E}\{Y(Y^2 - \rho Y)^2\} = \mathbb{E}|X|^5 - 2\rho\mu_4 + \rho^3, \qquad \delta = \mathbb{E}\{Y \operatorname{sgn}(X)(Y^2 - \rho Y)\}.$$

Here $m > 0$ by centering and unit variance. Cauchy's inequality for the nonnegative measure $Y\,dP$ gives $\delta^2 \le 2mV$. Consequently

$$\frac{\rho}{2}\mu_4 - \frac{1}{4}\mathbb{E}|X|^5 \le \frac{\rho^3}{4} - \frac{(\gamma - \rho\beta)^2}{8m}. \tag{79}$$

This deduction can be combined with the two other deductions in Section 7.2 by taking their maximum. Their sum would spend the same nonnegative quantity more than once.

A universal lower bound for each sign probability. Six probability panels at index six are excluded before the count calculation. The exclusion is the following elementary moment inequality; it does not require support, a central gap, or absence of a zero atom.

Lemma 7.11 (Sign mass and third moment)**.** *If $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, and $\rho = \mathbb{E}|X|^3 < \infty$, then*

$$\rho^2\mathbb{P}(X > 0), \quad \rho^2\mathbb{P}(X < 0) \;\ge\; \frac{8(7\sqrt{7} - 17)}{27} \;>\; \frac{11261}{25000}.$$

Proof. Write $p = \mathbb{P}(X > 0)$, $a = \mathbb{E}X_+^2$, and $m = \mathbb{E}X_+ = \mathbb{E}X_- > 0$. Centering and unit variance imply $0 < a < 1$. Cauchy's inequality, first with factors $X_+^{1/2}$ and $X_+^{3/2}$ and then with their negative-sign counterparts, gives

$$a^2 \le m\mathbb{E}X_+^3, \qquad (1-a)^2 \le m\mathbb{E}X_-^3.$$

Also $m^2 \le pa$. Consequently

$$p\rho^2 \ge f(a) := \frac{(a^2 + (1-a)^2)^2}{a}.$$

The derivative has the sign of $6a^2 - 2a - 1$, because

$$f'(a) = \frac{(2a^2 - 2a + 1)(6a^2 - 2a - 1)}{a^2}.$$

Thus the unique minimum on $(0,1)$ is at $a = (1+\sqrt{7})/6$, where $f(a) = 8(7\sqrt{7} - 17)/27$. The strict rational comparison follows from $2645751^2 < 7 \cdot 10^{12}$ and $8(7 \cdot 2645751/10^6 - 17)/27 > 11261/25000$. Apply the same argument to $-X$ for the other sign probability. □

Scalar consequences of the common inward budget. Some finite count domains use a stronger absolute-third-moment floor or noise ceiling before their five-dimensional subdivision. The following calculation supplies those scalar restrictions. It uses the sign conditioning of (6).

Reflect only for the scalar calculation, so that $0 < p \le 1/2$, $q = 1 - p$, $v = pq$. This preserves ρ, total conditional variance s and total inward displacement. The count probability, its phase and its original orientation are not reflected. Suppose the inward endpoints obey $a_0 + a_1 \le d$ with $0 < d \le p$. The conditional third-moment inequalities and Cauchy's inequality give

$$\rho \ge I(p,s,d) := \frac{v(1-2v) + (3p-d)s + s^2/(qd)}{(v+s)^{3/2}}. \tag{80}$$

Indeed, writing $b = qs_0$, $c = ps_1$ and $b + c = s$, the absolute-cubic numerator is at least

$$v(1-2v) + 3pb + 3qc - a_0 b - a_1 c + \frac{b^2}{qa_0} + \frac{c^2}{pa_1}.$$

Its linear part is at least $(3p-d)s$, while its quadratic part is at least $(b+c)^2/(q(a_0+a_1)) \ge s^2/(qd)$. Vanishing inward endpoints force the corresponding centered variance to vanish; those terms are handled before division. If $d = 0$, then $s = 0$ directly.

The derivative has the exact numerator

$$(v+s)^{5/2} I_s = v(6p - 3p^2 - d - 3/2) + \left\{ \frac{2p}{d} - \frac{3p-d}{2} \right\} s + \frac{s^2}{2qd}. \tag{81}$$

Consequently, on $p \in [l,u] \subset (0, 1/2]$, the two gates $d \le l$ and $f := 6l - 3l^2 - d - 3/2 \ge 0$ imply

$$\rho \ge \rho_B(p) := \frac{1 - 2p(1-p)}{\sqrt{p(1-p)}} \ge \rho_B(u).$$

The last function is decreasing on this half-interval. If a previously acquired ceiling is $\rho \le R$, a point with

$$R^2 p(1-p) - (1 - 2p(1-p))^2 < 0$$

is impossible. This polynomial increases on $0 < p \le 1/2$, because its derivative is $(1 - 2p)(R^2 + 4 - 8p(1-p)) \ge 0$. Exact bisection may therefore raise the closed lower probability endpoint to a known infeasible point; retaining that endpoint is a conservative enlargement.

When $f > 0$ and $s \le S$, formula (81) also gives

$$I_s \ge \frac{l(1-l)f}{[u(1-u)+S]^{5/2}}.$$

Integration from zero supplies a new variance ceiling. Directed square roots and outward division make every finite update valid separately.

A useful weaker floor survives failure of the monotonicity gate. Assume still $0 < d \le l$ and set

$$a = (3/2 - 6l + 3l^2 + d)_+, \qquad B = 2l/d - 3u/2 \ge 5/4.$$

The only possible decrease of I is bounded below by $-(va - Bs)_+/v^{5/2}$. Integrating this negative triangle gives

$$\rho \ge \rho_B(u) - \frac{a^2}{2B\sqrt{l(1-l)}}. \tag{82}$$

In particular, failure of the monotonicity gate must not be replaced by the unsupported assertion $\rho \ge \rho_B(p)$.

There is a second variance contraction which works with either sign of f. The global Taylor upper bound for $(v+s)^{3/2}$, combined with (80), gives the necessary inequality

$$Gs^2 + fs \le \epsilon\,[u(1-u)+S]^{3/2}, \tag{83}$$

where

$$\epsilon = (R - \rho_B(u))_+, \qquad G = \frac{1}{(1-l)d} - \frac{3(1-2l(1-l))}{8l(1-l)} \ge \frac{5}{8l(1-l)} + \frac{3}{4} > 0.$$

To see this, subtract $\rho_B(p)(v+s)^{3/2}$ from the lower numerator in (80), use the quadratic Taylor upper bound on the subtracted term, and then use $p \ge l$ and $\rho_B(p) \ge \rho_B(u)$. The positive part in ϵ is necessary when $f < 0$. The larger real root of (83), enclosed from above, is a valid new upper bound for s, even for negative f.

Finally suppose the original joint resource has the stronger form

$$d_{\text{actual}} \le d_0(1 - s/S_0), \qquad S_0 = \Lambda/n.$$

To test a proposed upper endpoint $a \in (0, S_0)$, assume $s \ge a$. Then $d_{\text{actual}} \le d_0(1 - a/S_0)$, so the preceding quadratic calculation can be repeated with this smaller inward bound. If its resulting upper bound is strictly less than a, the whole region $s \ge a$ is impossible. Otherwise the probe proves no lower bound on s. A finite sequence of such hypothetical tests is sound without any convergence assertion. Throughout it, the original S_0 remains in the joint budget; a separately improved noise ceiling clips that budget and does not replace its intercept.

7.3 Paired Fourier cancellation

The triangle LOW bound separates the fourth- and fifth-order errors. Near a two-point law their leading imaginary parts partly cancel. We retain that cancellation in one Gaussian norm; all other Taylor and power remainders remain those of Proposition 5.6.

Lemma 7.12 (The cancellation norm and its finite tail bounds)**.** *Put* $I = \sqrt{\pi/2}$ *and*

$$J(z,c) = \int_0^\infty e^{-x^2/2}|zx^2 - cx^4|\,dx.$$

The function J is convex in (z,c) and invariant under simultaneous reflection. Its maximum on a rectangle is therefore bounded by its largest corner value. For $zc \le 0$ it equals $I(|z|+3|c|)$. For $z,c>0$, put $x=z/c$ and define

$$d_{k+1}=1, \qquad d_j = 1+\frac{j}{x d_{j+1}} \quad (j=k,\dots,1), \qquad m_k(x)=d_1^{-1}.$$

Then J is bounded by each of

$$\begin{aligned} U(z,c) &= I|z-3c| + 6\sqrt{zc}\,e^{-x/2}, \\ V(z,c) &= I(z-3c) + \sqrt{zc}\,e^{-x/2}\left(6+\frac{2(3-x)}{x}m_k(x)\right), \end{aligned} \tag{84}$$

where $k=11$ for $x\ge 3$ and $k=12$ otherwise. The parenthesized factor is positive. For negative z,c, apply these formulas to $|z|,|c|$.

Proof. Convexity follows by integrating the absolute value of a linear function; successive endpoint bounds give the rectangle assertion. For $zc>0$, reflection and integration by parts, split at $r=\sqrt{z/c}$, give

$$J = I(z-3c) + 6cre^{-r^2/2} + 2(3c-z)\int_r^\infty e^{-t^2/2}\,dt.$$

Dropping a nonpositive last term, or bounding its integral by I when positive, proves U. To retain the signed tail, set $I_j(r)=\int_0^\infty t^j e^{-rt-t^2/2}\,dt$ and $q_j = I_j/I_{j-1} > 0$. Integration by parts yields

$$I_0 = \frac{1}{r+q_1}, \qquad q_j = \frac{j}{r+q_{j+1}}, \qquad I_0 = e^{r^2/2}\int_r^\infty e^{-t^2/2}\,dt.$$

Truncating at $q_{k+1}=0$ reverses the inequality at every reciprocal: odd truncations are lower bounds, even ones upper bounds. At $r=\sqrt{x}$ their values multiplied by r are exactly $m_k(x)$ above. The tail's multiplier $3c-z$ selects the stated parity and proves V, without requiring convergence of an infinite fraction. For $x<3$ the parenthesized factor exceeds 6; for $x\ge 3$, $0<m_{11}(x)<1$ makes it greater than $6-2(x-3)/x>4$. □

For outward evaluation use an upper enclosure for I in U and in the case $zc\le 0$. In V, use a lower enclosure when $z-3c<0$ and an upper one otherwise; the remaining factor is positive. Thus square roots and $e^{-x/2}$ may be bounded upward. Let J_u be the largest such corner bound, taking the minimum of U,V when applicable.

A single transfer formula. Retain the hypotheses and notation of Proposition 5.6: $N>2$, $1\le\mu_4\le M\le 11.1$, $|\gamma|\le g\le 1.85$, and actual ceilings $|a|\le a_*$, $|z|\le Z$, $\mu_6\le M_6$, where

$$a=(3-\mu_4)/24, \qquad z=(10\gamma-\mathbb{E}X^5)/120, \qquad c=a\gamma/6.$$

Any rectangle containing the actual pair (z,c) may be used above. For $F_N \ge (N/(N-1))^{3/2}$ and $I_u \ge I$, a valid LOW triple is

$$\begin{aligned} L_0 &= F_N J_u + \frac{F_N I_u}{N-1}(35a_*g^3/48 + 35Zg^2/24) + 251g^3/17280, \\ L_H &= F_N I_u\{a_* + 5Zg/[2(N-1)]\} + 251g^2/4800, \\ L_2 &= E/(2a_0^2) + B^2/(2b_0^3), \end{aligned} \tag{85}$$

with the same $E = M_6/720+g^2/72+(1/2+g^2/144)^3/5$, $B = a_*+Z/2+E/4$, $a_0 = (N-1)/(2N)$, and any $0 < b_0 \le (45-M)(1-2/N)/96$.

Indeed, write $b = n-1$ and $v = b\gamma t^3/6$. The leading imaginary part in the common Taylor proof, after division by t^3 and rotation by Ht, is bounded by

$$|zt^2 - atv| + a_*t|\sin v - v| + Zt^2|\cos v - 1| + |H|t\{a_*t + Zt^2|\sin v|\}.$$

Multiply by $ne^{-bt^2/2}$ and integrate. The first term contributes $nJ(z,c)/b^{3/2}$; the others use $|\sin v - v| \le |v|^3/6$, $|\cos v - 1| \le v^2/2$ and $|\sin v| \le |v|$. The recurrence $\int_0^\infty x^{2j}e^{-x^2/2}\,dx = (2j-1)!!I$ gives respectively the coefficients $35/48, 35/24, 1, 5/2$ in (85). The common power/Taylor errors and the signed Gaussian comparator give exactly the unchanged L_2 and the two 251 terms. All inverse-index factors decrease for $n \ge N$; the conclusion is uniform when the law and threshold vary with n.

One moment envelope for the coefficient rectangle. The broad triangle choice is already proved in Section 5.8. With its exact radius-six and actual-radius quantities Z_0, Z_r, it uses

$$120Z = \min\{E_5 + 10g, Z_0, Z_r\}, \qquad a_* = \max(2, M-3)/24, \quad M_6 = RE_5. \tag{86}$$

Here $E_5 \ge \mathbb{E}|X|^5$ and $|X| \le R \le 6$ are previously acquired. The radius-six derivative test and the actual-radius clipped vertex are exactly those in that proof; no moment ceiling is substituted from an unrelated marginal. A positive rectangle gives the stronger paired choice.

Proposition 7.13 (A positive cancellation rectangle)**.** *Suppose* $\mathbb{E}X = 0$, $\mathbb{E}X^2 = 1$, $|X| \le R$, $1 \le \mu_4 \le M < 3$, $0 \le \gamma_l \le |\gamma| \le g$, $g^2 < 2$, $R > 2g$, *and* $M - 1 - \gamma_l^2 \ge 0$. *After reflecting both odd Fourier coefficients if necessary, set*

$$\begin{aligned}
t_* &= \min\{g, (10-2M)/(2(R-g))\},\\
120Z &= \min\{E_5 + 10g, R(M-1) + (10-2M)t_* - (R-g)t_*^2\},\\
z_l &= \max\{-Z, [8\gamma_l - \gamma_l^3 - (R+2\gamma_l)(M-1-\gamma_l^2)]/120\}, \quad z_u = Z,\\
a_* &= (2-\gamma_l^2)/24, \qquad c_l = (3-M)\gamma_l/144, \quad c_u = (2-\gamma_l^2)g/144,\\
M_6 &= \min\{RE_5, 1 + 3g^2 + g^4 + (R^2 + 2Rg + 3g^2 + 2)(M-1-\gamma_l^2)\}.
\end{aligned}$$

These are valid coefficient bounds and $[z_l, z_u] \times [c_l, c_u]$ *contains* (z, c). *If* $3g^2 \le 2$, *one may further decrease* c_u *to* $\min\{c_u, (2g - g^3)/144\}$. *Use the damping constant* $b_0 = (45-M)(1-2/N)/96$*; for* $M \le 2$ *the weaker* $43(1-2/N)/96$ *is also valid.*

Proof. For $Q = X^2 - \gamma X - 1$, the common Pearson identities and one further multiplication give

$$\begin{aligned}
\mathbb{E}Q^2 &= \mu_4 - 1 - \gamma^2, & \mathbb{E}X^5 &= \gamma^3 + 2\gamma + \mathbb{E}[Q^2(X + 2\gamma)],\\
\mu_6 &= 1 + 3\gamma^2 + \gamma^4 + \mathbb{E}[Q^2\{(X+\gamma)^2 + 2\gamma^2 + 2\}]. & &
\end{aligned} \tag{87}$$

Thus $\mu_4 \ge 1 + \gamma^2$, proving a_*, c_l, c_u; the optional upper bound uses increasing $2t - t^3$ on $[0, g]$. The positive-kernel estimates of Section 5.8 give $-Q_R(t) \le 120z \le P_R(t)$ for $t = \gamma \ge 0$. Since $M < 3$, their absolute upper envelope is $R(M-1) + (10-2M)t - (R-g)t^2$, maximized at t_*. Moreover $(-Q_R)'(t) = 10 - 2M + 2Rt + 3t^2 > 0$, proving z_l. In the last identity, bound each nonnegative factor upward using $t \le g$ and $\mathbb{E}Q^2 \le M - 1 - \gamma_l^2$ to obtain M_6. The transfer's domain follows from $M < 3 < 11.1$, $g < \sqrt{2} < 1.85$. □

Finally a probability row must supply a lower bound for the actual ρ, not automatically its Bernoulli counterpart. If $p \in [l, u] \subset (0, 1/2)$, $\theta \le \Theta < 1$, and acquired inner endpoints are at least $a \ge 0$, the conditional cubic identity gives $\rho \ge (1-\theta)^{3/2}\rho_B(p) + 3a\theta$. For outward $r_- \le \rho_B(u)$, $r_+ \ge \rho_B(l)$ this implies

$$\rho \ge r_* := \max\{1, r_- - (3r_+/2 - 3a)_+\Theta\},$$

because ρ_B decreases on $(0, 1/2)$ and $(1-\theta)^{3/2} \ge 1 - 3\theta/2$. The same r_* is retained in all Fourier corners, signed phase prices, conserved budgets and the signed fourth-moment errors (72); the binary-K alternative still requires $1 - \Theta > 2/3$. Reflection here concerns only the auxiliary norm and does not change the actual contact orientation.

7.4 Confinement from the remaining defect budget

Two different support implications are useful. Gaussian contact excludes a point by a strict negative upper bound on its contact expression; closed cells cover the finite exterior and a derivative-sign certificate covers any remaining ray. The shared Gaussian support theorem supplies this callback, including the bounded-threshold radius and coordinate-strip contracts; it is a prior theorem, not a test on moments alone. Noise confinement proves $|U| \le a$ on an already acquired row. It then implies, without a small-noise limit,

$$|X| \le R_* := \frac{\max\{u + a, 1 - l + a\}}{\sqrt{\min_I p(1-p)}}. \tag{88}$$

The denominator uses $\mathbb{E}U^2 \ge 0$, so zero noise is included. The Fourier/moment formulas remain valid for $R \ge 1$; the paired Pearson choice additionally checks $R > 2g$. In particular there is no intrinsic lower radius 2.3: the radial ratios in Lemma 7.7 increase on all $x \ge 0$, and the other formulas use their explicit domains.

Here is the selected proof schedule. Each row names *an application* of Theorem 7.8, not a different analytic theorem.

Node	Input and operation	Output consumed later
I	$N = 47$, $R = 3.84$, $C \le .4247$; 13 signed passes plus export	100 closed probability cells; 35 survive; $\mathcal{R} < 1.885$, $nD < 47(.590)$
C	Central preparation, contact radius 2.5, probability refinement	45 cells $[.38, .47]$, width .002; canonical simultaneous input
L	Complete complementary cover, contact radius 3, refinement	39 cells $[.302, .38]$; all other complementary cells excluded
C1	First confinement from C; radius formula; enclosure	First improved central table
C2	Antichain confinement from C; intersect C1; enclosure	Selected central table: 45 cells, $[.38, .47]$
S	Separate sign-resource pricing on C2 and L	One left cell excluded; 38 left cells remain
L2	Left contact radius 2.7 from S; enclosure	Selected left table: 38 cells, $[.304, .38]$

Every shared endpoint belongs to its adjacent closed cells. The confinement producing C2 consumes C, not C2. The selected central local comparison uses C2; it does not need the optional further central refinement in S. The final cover is exactly $[.304, .47]$.

Corollary 7.14 (Uniform acquired entry)**.** *Let an attained positive maximum have $n \ge 47$, $C_E \le C \le .4247$, $0 \le n(C - C_E) \le .7$, decrement at most .19, and both deleted score caps at most .4253. Given the shared contact support and strip implications, the exact schedule above places its law in one of the 38 selected left rows or 45 selected central rows, with every resource in that row holding simultaneously. This assertion has no upper bound on n.*

Proof. The shared all-threshold support theorem supplies radius 3.84. Apply the finite enclosure theorem in the displayed order, retaining the weighted bounds at every edge. The initialization's stored upper index 98 occurs only through $.4095 + K/98$; here $K = 0$, so it imposes no upper-index restriction. All subsequent index-dependent inequalities are checked for every $n \geq 47$ by their stated monotonicity gates. The accepted closed covers exclude the complementary cells and give the stated 83 rows. □

The local-comparison interface. For each final row supply $p \in I$, $\theta \leq \Theta$, $\lambda \leq \Lambda$, the actual support/amplitude, $d_Y \leq d_0$, both signed phase limits, and all four coherent quadratics bounding (48). On a phase/noise subcell, let T_H be the maximum of each concave quadratic at its clipped vertex, then the maximum over corners and the minimum with the scalar bound. If a proved moment inequality is

$$D \geq \alpha d_Y + b\theta + P, \qquad \alpha > 0, \quad \ell \geq \ell_-, \quad \lambda \geq L_0,$$

then the exact strict resource supplied to the comparison is

$$n d_Y < G := \frac{T_H/\kappa - NP - b\ell_-^2 L_0}{\alpha}. \tag{89}$$

This uses $n\theta = \ell^2\lambda$ and $nP \geq NP$. If $G \leq 0$, the entire closed subcell is excluded. Formula (89) is the same resource as (48), not an additional allowance.

One confinement lemma with two concentration choices. A proved amplitude is used as a support update, never as a premise of its own proof. The following derivation applies to every acquired row, with the same formulas for the central and left probability ranges.

Statement and hypotheses. Let $n \geq N = 47$, $Y = B + U$, $\mathbb{P}(B = 1) = p < 1/2$ and $\mathbb{E}(U \mid B) = 0$. Put $q = 1 - p$, $s = \mathbb{E}U^2$, $\lambda = ns$ and $X = \ell(Y - p)$, where $\mathbb{E}X^2 = 1$. Thus $\ell^2 = (1 - \theta)/(pq)$ and $\theta = \ell^2 s$. The conditional label-zero support is at most $a \geq 0$ and the label-one support is at least $b \leq 1$, where a, b are support points and $b - a = 1 - d$. In particular $0 \leq a \leq d$, $1 - d \leq b \leq 1$. Assume the initial strip $\operatorname{supp} Y \subset (-.55, 1.55)$. Write $\rho = \mathbb{E}|X|^3$, $\beta = \mathbb{E}[X|X|]$ and $\epsilon = \mathbb{E}|U|$.

Select an acquired closed probability panel $[l, h]$. Its fields give upper bounds $\Theta, d_0, \rho_0, \beta_+, T, \Lambda$, a lower bound β_-, and separate bounds H_+, H_- for the two signs of H. Assume

$$\theta \leq \Theta, \quad d \leq d_0, \quad \rho \leq \rho_0, \quad \beta_- \leq \beta \leq \beta_+, \quad 0 \leq \lambda \leq \Lambda, \quad .4 < C \leq C_0 = .4253.$$

For the positive sign $0 \leq H \leq H_+$; for the negative sign $-H_- \leq H \leq 0$. Let $x = np + H/\ell$ and $\nu = \mathcal{L}(x - \sum_{i<n} Y_i)$. Assume a contact majorant Q for $g(y) = \nu([y, \infty))$, with equality at each support point of Y, and derivative

$$\sqrt{n}(-Q'(y)) = \ell\phi(H/\sqrt{n}) + \frac{\ell}{n}\{3C\beta + (H\phi(H/\sqrt{n}) + 3C\rho)t - 3Ct|t|\}, \quad t = \ell(y - p). \tag{90}$$

Both actual deleted convolutions satisfy, separately,

$$Q_j(w) \leq \frac{\ell\phi(0)w + 2C_0\rho}{\sqrt{j}}, \qquad j = n - 1, n - 2, \tag{91}$$

for every interval of width $w \geq 0$.

Here is the precise common resource assumption. Put $\kappa = 997/15000 < \phi(0)/6$, $J = \int_{[a,b)}(y-a)\nu(dy)$, and note that the central gap contributes $m\ell J = (1-\theta)J$ to L. Thus $L \ge (1-\theta)J$; retain $e \ge 0$ and

$$ne\rho + n\phi(0)D/6 + n^{3/2}L < \mathcal{B}(H), \qquad \mathcal{B}(H) = \min\{T, \max_j(A_j + F_{j,\pm}|H| - Q_jH^2)\}. \quad (92)$$

The four quadratics are the table's `quadratics` rows, using `Qbudget`, not `Qraw`. The Gaussian phase contribution has already been included in `Qbudget`.

For each panel let E, F_ℓ, m_0 be respectively strict lower, upper, and lower rational bounds for

$$\sqrt{(1-\Theta)/(h(1-h))}, \qquad 1/\sqrt{l(1-l)}, \qquad \sqrt{(1-\Theta)l(1-l)}.$$

The bounds use the grid 10^{-6}, with an extra unit outward. Define

$$\begin{aligned} c_+ &= 2(5.16)E^2(1-h-d_0)^2, & c_- &= 2(7.16)E^2(l-d_0)^2,\\ b_* &= \min\{3(4.16)E(1-h-d_0), 3(6.16)E(l-d_0)\},\\ P &= m_0E^2(3.5\,\mathrm{dist}([l,h],[.41886,.41887]) - 2.041d_0)_+^2. \end{aligned}$$

The simultaneous moment inequality has the following usable form:

$$D \ge c_+M_+ + c_-M_- + b_*\theta + P, \quad d = (M_+/p + M_-/q)/\ell, \quad \epsilon \le 2(M_+ + M_-)/\ell, \quad M_\pm \ge 0. \quad (93)$$

To derive it, the endpoint expansion of (9) gives $D \ge m(\sqrt{\sigma+1}A - \sqrt{\sigma-1}B)^2 + 2(\sigma+2)B^2M_+ + 2(\sigma+4)A^2M_- + 3(\sigma+1)B\theta_+ + 3(\sigma+3)A\theta_-$. Insert $A \ge E(l-d_0)$, $B \ge E(1-h-d_0)$ and the rational lower coefficients displayed above. For the square, write $A = \ell(p-a)$, $B = \ell(1-p-(1-b))$, with $a+(1-b) = d \le d_0$. Its unshifted coefficient is $(\sqrt{\sigma+1}+\sqrt{\sigma-1})(p-p_*)$; $\sqrt{\sigma+1}+\sqrt{\sigma-1} > 3.5$ and $\sqrt{\sigma+1} < 2.041$ prove the stated lower square P. For completeness, on the positive class put $Z = X - B \ge 0$. Then $T_+ = 2B^2M_+ + 3B\mathbb{E}[Z^2; X>0] + \mathbb{E}[Z^3; X>0]$ and $R_+ = BM_+ + \mathbb{E}[Z^2; X>0]$. Consequently $(\sigma-1)T_+ + 6BR_+ \ge 2(\sigma+2)B^2M_+ + 3(\sigma+1)B\theta_+$, since $\mathbb{E}[Z^2; X>0] \ge \theta_+$. The reflected expansion gives the negative-class bound. The centered absolute deviation on each class is at most twice its mean distance from the inner endpoint, proving the bound on ϵ. All quantities refer to the same marginal.

Concentration from an antichain.

Lemma 7.15 (Strict-gap concentration)**.** *Suppose $Y = U_0 + B(1 + U_1 - U_0)$, where B is Bernoulli(p) independent of the pair (U_0, U_1), and $1 + U_1 - U_0 \ge c > 0$ almost surely. For independent copies $Y_1, \dots, Y_j$ and any interval I of diameter strictly less than c,*

$$\mathbb{P}\left(\sum_{i=1}^{j} Y_i \in I\right) \le \max_{0\le k\le j} \binom{j}{k} p^k(1-p)^{j-k}.$$

The same bound holds after adding any independent random variable.

Proof. Condition on all noise pairs. The sum is a fixed constant plus $\sum B_iw_i$, with every $w_i \ge c$. The family $\mathcal{F}$ of subsets whose weighted sums lie in I is an antichain: distinct comparable subsets have sums differing by at least c, larger than the diameter. A uniformly random permutation of $\{1, \dots, j\}$ defines the maximal chain of its initial subsets. The chain contains a fixed k-subset with probability $1/\binom{j}{k}$, and meets an antichain at most once. Hence

$$\sum_{S\in\mathcal{F}} \binom{j}{|S|}^{-1} \le 1.$$

Multiplying each summand by its binomial level mass proves the bound conditionally. Integration removes the conditioning. Conditioning on an additional independent variable proves the last assertion. This argument includes all atom collisions and any open or closed interval endpoints. □

For the actual marginal we may couple the two conditional noises independently and then sample an independent label. Since $U_0 \le a$ and $U_1 \ge b-1$, the step is at least $b-a = 1-d$. Strict diameter is necessary: a Bernoulli law on $\{0,c\}$ assigns mass one to $[0,c]$, exceeding its largest atom when $0<p<1$.

A rational cap for every remaining index. Put $J_0 = N-2 = 45$ and define

$$r_L = l-(1-l)/J_0, \qquad r_R = h+h/J_0, \qquad v_* = \min\{r_L(1-r_L), r_R(1-r_R)\}.$$

The checker verifies $0<r_L\le r_R<1$. Adjacent binomial-mass ratios show that any mode k at index $j\ge J_0$ satisfies $(j+1)p-1\le k\le (j+1)p$, hence $r=k/j\in[r_L,r_R]$. In particular $1\le k\le j-1$. Finite Stirling bounds give

$$\sqrt{j}\binom{j}{k}p^k(1-p)^{j-k} \le \frac{\exp(1/(12j)-jD(r\|p))}{\sqrt{2\pi r(1-r)}} < \frac{.4}{\sqrt{v_*}(1-1/(12J_0))}.$$

Here $D(r\|p)\ge 0$ follows from Jensen's inequality for the logarithm; $\exp x \le (1-x)^{-1}$ for $0\le x<1$ follows term by term from the power series. The finite Stirling inequalities used are $\sqrt{2\pi m}(m/e)^m < m! < \sqrt{2\pi m}(m/e)^m \exp(1/(12m))$. Choose the strict 10^{-6} grid upper square root M of $.16/[v_*(1-1/(12J_0))^2]$. Thus, whenever $w<1-d$,

$$Q_j(w) < M/\sqrt{j}, \qquad j\ge n-2\ge J_0. \tag{94}$$

This covers the $n-1$ full summands of ν and the $n-2$ full summands remaining after one mark or deletion. An independent remaining conditional noise preserves the cap. We do not replace the latter index by $n-1$.

Either the score-derived cap or the strict-gap cap may be selected on an interval where its hypotheses hold. The first confinement uses the score cap; later central and left schedules use the strict-gap cap. For the left panels replace the conditional cubic coefficient 24 below by 28; in general any c_ρ satisfying $c_\rho^2[l(1-l)]^3 > [3(1-l)+.55]^2$ is valid.

Signed deletion, including atoms. Let $m=n-1$, $Z = x-\sum_{i=1}^m Y_i$, $\mu = x-mp$ and let $K=\sum B_i$. For a fixed removed label put $T_b(I) = \mathbb{P}(x-\sum_{i<m}Y_i - b - U_b \in I)$. Exchangeability and $mp-K = Z-\mu+\sum U_i$ give the exact identities

$$\nu(I)-T_1(I) = \frac{\mathbb{E}[(Z-\mu+\sum U_i)1_I(Z)]}{mp}, \qquad T_0(I)-\nu(I) = \frac{\mathbb{E}[(Z-\mu+\sum U_i)1_I(Z)]}{mq}.$$

There is no absolute value around the sum of noises. For a full removed pair (Y_i,U_i), condition on the pair. The remaining sum gives a function $f(Y_i)\in[0,Q_{m-1}(w)]$. Since $\mathbb{E}U_i=0$,

$$\mathbb{E}[U_i f(Y_i)] \le Q_{m-1}(w)\mathbb{E}(U_i)_+ = \tfrac{1}{2}\epsilon Q_{m-1}(w).$$

This also works when one conditional law is deterministic.

Let independent conditional noises satisfy $\Delta = U_1-U_0 \ge -d$. For $\tau>0$ put $I^+ = I+1+[-d,\tau]$. On $\Delta\le\tau$, the point for T_1 in I maps to the point for T_0 in I^+. On the complement, conditioning on both noises gives error at most $Q_{m-1}(w)\mathbb{P}(\Delta>\tau)$. Moreover

$$\mathbb{P}(\Delta>\tau) \le \min\left\{\frac{s}{s+p\tau^2}, \frac{d}{d+\tau}\right\}. \tag{95}$$

Indeed $\mathbb{E}\Delta = 0$, $\operatorname{Var}\Delta \le s/p$, and $\Delta + d \ge 0$ has mean d. Cantelli and Markov prove the two bounds. The exceptional event is strict, so $\Delta = \tau$ belongs to the good event.

If I, I^+ are disjoint, combine the two signed noise terms before bounding them. For the same remaining-sum law their conditional probabilities f_I, f_J satisfy

$$0 \le f_I \le Q_I, \quad 0 \le f_J \le Q_J, \quad f_I + f_J \le Q_{\text{hull}}.$$

Since $p < q$,

$$\frac{f_I}{p} + \frac{f_J}{q} \le \min\left\{ \frac{Q_I}{p} + \frac{Q_J}{q}, \frac{Q_{\text{hull}}}{q} + \left(\frac{1}{p} - \frac{1}{q}\right) Q_I \right\}.$$

Multiplication by $\mathbb{E}U_+ = \epsilon/2$ proves the combined bound. Closed, open and half-open intervals are all covered by containment in the corresponding closed interval. No density or atom separation is used.

Slope bounds and the two exterior intervals. Split $|H|$ into 16 closed equal cells $[h_l, h_h]$ and λ into 24 closed equal cells $[s_l, s_h]$. On each box put

$$\rho_* = \min\{\rho_0, \rho_B(l) + 24 s_h/N\},$$
$$b_- = \max\{\beta_-, 1 - 2h - 2(1-h)\Theta\}, \qquad b_+ = \min\{\beta_+, 1 - 2l + \Theta\}.$$

Here $\rho_B(p) = (1 - 2p(1-p))/\sqrt{p(1-p)}$. Write $s_i = \mathbb{E}U_i^2$. Conditional centering and the two sign supports give

$$\rho = \ell^3\{pq(1-2pq) + 3pq(s_0 + s_1) + p\mathbb{E}U_1^3 - q\mathbb{E}U_0^3\}.$$

The initial strip and $d_0 < .55$ imply $|U_i| < .55$. Thus $\rho \le \rho_B + \ell^3(3q + .55)s \le \rho_B + 24s$; the last coefficient inequality is checked as $24^2[l(1-l)]^3 > [3(1-l) + .55]^2$. Similarly $\beta = (1-\theta)(1-2p) + \ell^2(ps_1 - qs_0)$, and $|\ell^2(ps_1 - qs_0)| \le \theta$. These identities prove the stated outward bounds on ρ and β. Set $f = .4F_\ell$, $\varphi_- = .39886(1 - h_h^2/(2N)) > 0$. The exact upper bound for $H\phi(H/\sqrt{n})$ used below is

$$v_+ = \begin{cases} .4h_h & H \ge 0, \\ -\varphi_- h_l & H \le 0. \end{cases}$$

For all $t \le 0$, completing the square in (90) gives

$$\mathcal{M}_- = F_\ell((v_+ + 3C_0\rho_*)^2/(12(.4)) - 3(.4)b_-)_+, \qquad L_- = E\varphi_- - \mathcal{M}_-/N.$$

A positive bound $B_* > \sqrt{(1-l)/l}$ gives valid central-gap lower and positive-half-line upper slope bounds

$$\mathcal{M}_+ = F_\ell(.4h_h B_* + 3C_0 B_*(B_* - 1) - 1.2b_-)_+,$$
$$L_g = E\varphi_- - \max(\mathcal{M}_-, \mathcal{M}_+)/N,$$
$$U_+ = f + \frac{F_\ell}{N}\left\{ 3C_0 b_+ + \frac{(.4h_h + 3C_0\rho_*)^2}{12(.4)} \right\}.$$

They follow by completing a square on the positive half-line and using $t(t-1) \le B_*(B_* - 1)$ for $1 \le t \le B_*$; on $0 \le t \le 1$ that term has the favorable sign. Negative values are covered by $\mathcal{M}_-$. All divisions and positivity requirements are explicit gates.

The sharper lower-ray slope. Take $\tau_- = 5r_0/12$, $\eta = 10^{-5}$ and $I = [a-r, a-d-\tau_- -\eta]$, where $r \ge r_0$. Since $a \le d$, the entire interval has $t \le t_* = -E(l+\tau_-+\eta) < 0$. Write $B(t) = b_- + \rho_* t + t^2$ and $B_*^{\min} = B(\min(t_*, -\rho_*/2))$. For the actual C and $t \le t_*$,

$$3C(\beta + \rho t + t^2) + H\phi t \ge 1.2B(t) + v_+ t + 3(C_0 - .4)\min(0, B_*^{\min}).$$

The inequality uses $t < 0$ and separates the possible negative part of $B(t)$ before varying C. Minimize the right side at the clipped vertex $\min(t_*, -(v_+ + 1.2\rho_*)/2.4)$ and call its value q_*. Then

$$L = E\varphi_- - F_\ell \max(0, -q_*)/N$$

is a lower slope on the whole interval, for every $n \ge N$. Positive corrections are discarded. Therefore contact gives

$$\sqrt{n}\nu(I) \ge L(r - d - \tau_- - \eta).$$

This one slope replaces the previous separate bounds on the long and short parts of the interval.

Lower interval geometry. Its translate is $I^+ = [a - r + 1 - d, b - \eta]$, of width $r - \eta$. The initial strip implies $I^+ \subset [a + g, b)$, $g = 1 - .55 - 2d_0 > 0$, hence $\nu(I^+) \le J/g$. The intervals are disjoint throughout $r \le .55 + d_0$, since $\inf I^+ - \sup I = 1 - r + \tau_- + \eta > 0$. At $r = r_0$, concentration and directed distance bounds are

$$\begin{aligned} Q_I &= f(r_0 - \tau_- - \eta) + 2C_0\rho_*, & Q_J &= f(r_0 - \eta) + 2C_0\rho_*, \\ Q_{\text{hull}} &= f(1 + r_0 - \eta) + 2C_0\rho_*, \\ D_I &= (v_H - l - \tau_- - \eta)_+, & D_J &= (v_H + 1 - l - \eta)_+, \end{aligned}$$

where $v_H = -h_l/F_\ell$ for $H \ge 0$ and $v_H = h_h/E$ for $H \le 0$. These signs follow from $\mu = p + H/\ell$.

Upper interval geometry. For a support point $y \ge 1 + r_0$, take $\tau_+ = 7r_0/15$, $I = [a, a + \tau_+]$ and $I^+ = [b, a + 1 + 2\tau_+]$. The strict gates $d_0 + 2\tau_+ < r_0$ and $1 - d_0 - \tau_+ > 0$ give $I^+ \subset [b, y)$ and disjointness. Contact, including the atom at b, and the leakage definition give

$$\sqrt{n}\nu(I) \ge L_g(1 - d) - \sqrt{n}J/\tau_+, \qquad \sqrt{n}\nu(I^+) \le U_+(.55 + d).$$

Use

$$\begin{aligned} Q_I &= f\tau_+ + 2C_0\rho_*, & Q_J &= f(2\tau_+ + d_0) + 2C_0\rho_*, \\ Q_{\text{hull}} &= f(1 + 2\tau_+) + 2C_0\rho_*, \\ D_I &= (d_0 + \tau_+ + h_h/E - l)_+, & D_J &= (1 + d_0 + 2\tau_+ + h_h/E - l)_+. \end{aligned}$$

The upper side deliberately keeps the absolute phase bound.

The strict-gap alternative. If $1 - .55 - 2d_0 > 0$ and the displayed lower intervals have width below $1 - d$ throughout $r \in [r_0, .55 + d_0]$, take $Q_I = Q_J = M$. For the upper intervals the corresponding gate is $1 - 2d_0 - 2\tau_+ > 0$; then again $Q_I = Q_J = M$. Use the same directed distances. Their concentration caps are now independent of r, and use $a_e = F_d M\{1/l + 1/(1-h)\}/2$ in the next formulas. The lower-ray gate reduces to $L > 0$: its contact side increases with r, while the entire error is constant on the acquired finite strip. For the score-derived cap retain the explicit affine slope gate below. No assertion outside that strip is needed.

One budget, two error bounds. Let F_d be the strict grid upper bound for $\sqrt{\max\{(47/46)^3, 47/45\}}$. It simultaneously covers the two concentration factors in (91) after multiplication by $\sqrt{n}$, uniformly for $n \geq 47$. For either pair of intervals set

$$a_e = \frac{F_d}{2} \min\left\{ \frac{Q_I}{l} + \frac{Q_J}{1-h}, \frac{Q_{\text{hull}}}{1-h} + \left(\frac{1}{l} - \frac{1}{1-l}\right) Q_I \right\},$$
$$A = F_d \left\{ Q_I D_I / l + Q_J D_J / (1-h) \right\}, \qquad B = F_d Q_I.$$

The probability denominators are outward because $1/p - 1/(1-p)$ is decreasing on $(0, 1/2)$. Use $a_d = L$, $a_J = 1/g$ on the lower side, and $a_d = L_g + U_+$, $a_J = 1/\tau_+$ on the upper side. For the phase cell, maximize each concave budget quadratic at its vertex clipped to $[h_l, h_h]$, take their maximum, then the minimum with T; call this upper bound T_*. Put

$$R = T_*/\kappa - b_* E^2 s_l, \quad \alpha = \min(c_+ l E, c_-(1-h)E), \quad G = (R - NP)/\alpha.$$

If $G \leq 0$, the strict common budget excludes the entire closed box. Otherwise $nd < G$ and

$$B_\tau = \min\{s_h/(s_h + N l \tau^2), G/(N\tau + G)\}$$

bounds (95). With

$$z = \max\left\{ \frac{a_d/l + 2a_e}{E c_+}, \frac{a_d/(1-h) + 2a_e}{E c_-}, \frac{\kappa a_J}{1 - \Theta} \right\},$$

one valid total error bound is

$$\mathcal{E}_0 = (zR + A)/N - zP + BB_\tau. \tag{96}$$

This follows by multiplying the single budget by z/κ. The inverse-index coefficient $zR + A$ is checked positive.

A stronger bound spends the same resource on the pair event. Let $R_0 = R - NP > 0$ and define the three pairs

$$(v_+, w_+) = \left(\frac{a_d/l + 2a_e}{E c_+} \frac{R_0}{N}, \frac{R_0}{E c_+ l} \right),$$

$$(v_-, w_-) = \left(\frac{a_d/(1-h) + 2a_e}{E c_-} \frac{R_0}{N}, \frac{R_0}{E c_-(1-h)} \right), \qquad (v_J, w_J) = \left(\frac{\kappa a_J}{1-\Theta} \frac{R_0}{N}, 0 \right).$$

They are upper cost/displacement coordinates at the vertices of the simplex with resources nc_+M_+, nc_-M_- and $n^{3/2}L/\kappa$. For any $t \geq 0$,

$$\frac{D}{N\tau + D} \leq \frac{N\tau}{(N\tau + t)^2} D + \frac{t^2}{(N\tau + t)^2} \qquad (D \geq 0).$$

The difference is $N\tau(D-t)^2/[(N\tau + D)(N\tau + t)^2]$. Consequently a further valid error bound is

$$\mathcal{E}_t = \frac{A}{N} + \max_{i \in \{+,-,J\}} \left(v_i + B \frac{N\tau}{(N\tau + t)^2} w_i \right) + B \frac{t^2}{(N\tau + t)^2}.$$

The checker takes the minimum of (96) and these eleven bounds with $t = jG/10$, $j = 0, \ldots, 10$. Each is individually valid; no numerical minimization or derivative of their minimum is assumed.

The lower contradiction is the strict gate $Lr_0 - L(\tau_- + \eta) > \mathcal{E}$. To extend it to every $r \geq r_0$, bound increments directly in the original affine-in-width signed deletion bound for the same law. The budget gives $n\epsilon < E_* = 2T/(\kappa E \min(c_+, c_-))$. The sufficient slope gate is

$$L > \frac{F_d f}{N}\left\{D_I/l + D_J/(1-h) + \frac{E_*}{2}\left(1/l + 1/(1-h)\right)\right\} + F_d f B_\tau.$$

Using the uncoupled increment is safe even when the hull bound was used at r_0. The upper contradiction is $L_g - .55U_+ > \mathcal{E}$. Thus no support point lies below $a - r_0$ or above $1 + r_0$. Since $a \geq 0$, $b \geq 1 - d_0$ and $d_0 < r_0$, the label-zero and label-one supports both satisfy $|U| < r_0$.

The selected schedules. The ordinary, central-antichain, and left-antichain certificates substitute their recorded r_0 into these same gates. Noise-dependent amplitudes are the same theorem with a positive lower noise endpoint in $R = T_*/\kappa - b_* E^2 s_l$. In the central rows the additional amplitudes are $.35, .30, .275$ on $[.05, .08], [.08, .10], [.10, \Lambda]$; on the left they are $.375, .325, .30$. Intersect each interval with its own row's Λ and each amplitude with that row's global amplitude. The fixed acceptance checks both signs of H, all closed phase/noise cells, clipped endpoints, and strict budget exclusions. No proof of LOW, MID, or a binomial score comparison is used by this confinement lemma.

Finite-index substitution in confinement. Here the numerical constants in the confinement proof of Section 7.4 are parameters. More precisely, replace its floor 47 by $N \geq 39$, its initial strip .55 by a number $w \in (0, 1)$, and its conditional cubic coefficient by $c_\rho > 0$. Retain a probability panel $[l, h] \subset (0, 1/2)$ and assume

$$\operatorname{supp} Y \subset (-w, 1 + w), \quad d_0 < w, \quad c_\rho^2 [l(1-l)]^3 > [3(1-l) + w]^2.$$

The strip and the inward bounds $U_0 \leq d_0$, $U_1 \geq -d_0$ together give $|U_i| < w$. The displayed conditional cubic identity gives $\rho \leq \rho_B + \ell^3(3q + w)s \leq \rho_B + c_\rho s$. Use $J_0 = N - 2$ for the binomial mode interval and the deletion factor $\sqrt{\max\{(N/(N-1))^3, N/(N-2)\}}$. In the interval geometry use $g = 1 - w - 2d_0 > 0$ and $r \leq w + d_0$; in the upper contradiction use $L_g - wU_+ > \mathcal{E}$. The other width, disjointness, positive-slope and budget gates are unchanged. These substitutions follow directly through the proof: w enters only the conditional cubic bound and the endpoints of the known strip, while the index enters only the decreasing factors already bounded above. They therefore prove the same confinement implication whenever all the resulting gates pass.

The positive-allowance calculation at actual index 39 uses $N = 39$, $J_0 = 37$ and deletion factor $\sqrt{\max\{(39/38)^3, 39/37\}}$. Its acquired initial widths are $w = .55$ or $.575$; the central and left cubic coefficients are 24 and 30, respectively. All the same explicit gates are checked with these parameters before a contraction is used. The central floor-40 sequence uses $N = 40$, $w = .55$ initially and $c_\rho = 24$. The left rows start with their separately acquired strip $w \in \{.55, .575, .6\}$ and use $c_\rho = 30$. After a strict contraction, the proved amplitude becomes the next strip width. Thus every later use has an already established strip, and the left sequence does not assume the narrower uniform-tail strip.

7.5 The common comparison identity and the nearest count

A law concentrated near two points has a natural integer count and a small centered error. We compare it with its own Bernoulli count. The same exact identity handles every noise regime. There are two useful estimates of its right side: discard all but the nearest outgoing count, or retain all counts through a random cutoff. The estimates share their normalization, moments, and tail bounds; only the final scalar witness changes.

The identity and the common numerical test. Let $0 < p \le 1/2$, $q = 1 - p$, $v = pq$, and let

$$Y = B + U, \qquad \Pr(B = 1) = p, \qquad \mathbb{E}(U \mid B) = 0.$$

The two actual conditional laws U_0, U_1 obey

$$|U_j| \le A, \quad U_0 \le a_0, \quad U_1 \ge -a_1, \quad a_0, a_1 \ge 0, \quad d = a_0 + a_1 \le d_0 < \min(p, q), \quad d_0 \le A. \tag{97}$$

Write $s_j = \mathbb{E}U_j^2$, $s = qs_0 + ps_1$, $\lambda = ns$, $\ell = (v + s)^{-1/2}$, $\theta = s/(v + s)$ and $\rho = \mathbb{E}|\ell(Y - p)|^3$. At the actual threshold choose

$$x = np + H/\ell = k + u, \qquad k \in \{1, \dots, n - 1\}, \quad |u| \le \tfrac{1}{2}, \qquad \delta = k - np = H/\ell - u.$$

Either nearest integer is allowed at a tie. Put

$$\pi_j = \binom{n}{j} p^j q^{n-j}, \quad F_B(j) = \Pr(\mathrm{Bin}(n, p) \le j), \qquad \rho_B = (1 - 2v)/\sqrt{v},$$

with $\pi_j = 0$ outside $[0, n]$. Let W_j be an independent sum of j copies of U_1 and $n - j$ copies of U_0. The signed scores are

$$C = \frac{\sqrt{n}}{\rho}\{\Pr(\textstyle\sum Y_i \le x) - \Phi(H/\sqrt{n})\}, \qquad C_B = \tfrac{\sqrt{n}}{\rho_B}\{F_B(k) - \Phi(\delta/\sqrt{nv})\}.$$

Throughout this section $C \ge c_* > 0$; the selected application uses $c_* = .4$. Define

$$\begin{aligned} f(u) &= \frac{\Phi((\delta + u)/\sqrt{n(v + s)}) - \Phi(\delta/\sqrt{n(v + s)})}{\pi_k}, \\ Q &= \Pr(\textstyle\sum Y_i \le k + u) - F_B(k), \\ \mathcal{R} &= \sqrt{n}\{\Phi(\delta/\sqrt{nv}) - \Phi(\delta/\sqrt{n(v + s)})\} + C(\rho_B - \rho). \end{aligned} \tag{98}$$

Direct subtraction gives the exact identity

$$\rho_B(C - C_B) = \sqrt{n}\pi_k\{Q/\pi_k - f(u)\} + \mathcal{R}. \tag{99}$$

In particular, the phase u, the sign of H, and both conditional laws refer to the same marginal throughout.

Theorem 7.16 (Parameterized local comparison)**.** *Fix $n \ge N$, $\lambda > 0$ and a closed parameter cell on which* (97) *holds. Suppose $\sqrt{n}\pi_k \ge m > 0$. Either of the following witnesses proves $\rho_B(C - C_B) < -\varepsilon\lambda$.*

1. Nearest-count witness. *Valid bounds O, T, R_0, κ satisfy*

   $$\Pr(W_k > u) + f(u) \ge O, \quad \sum_{i \ge 1} \Pr(W_{k+i} \le u - i) \le T, \quad \pi_{k+i}/\pi_k \le R_0, \quad \mathcal{R} \le \kappa\lambda/n.$$

   Only legal counts occur in the sum; adding positive upper bounds for other counts is allowed. If $D = O - R_0T > 0$ and

   $$mD - \kappa\lambda/n > \varepsilon\lambda, \tag{100}$$

   the conclusion follows.

2. Whole-count witness. *Bounds $B(\lambda) < 0$ and $I(\lambda) \ge 0$ satisfy*

   $$\rho_B(C - C_B) \le \sqrt{n}\pi_k\lambda B(\lambda) + \lambda I(\lambda)/n.$$

   It is enough that

   $$mB(\lambda) + I(\lambda)/N < -\varepsilon. \tag{101}$$

The witnesses may differ from cell to cell. A finite closed cover and a separate treatment of $\lambda = 0$ therefore suffice for a complete local comparison.

Proof. Conditioning on the binomial count gives, with all atom conventions intact,

$$Q = -\sum_{i=0}^{k} \pi_{k-i} \Pr(W_{k-i} > u+i) + \sum_{i=1}^{n-k} \pi_{k+i} \Pr(W_{k+i} \le u-i). \tag{102}$$

Dropping the outgoing terms with $i > 0$ gives $Q/\pi_k - f(u) \le -D$. Use (99) and $D > 0$. For the second witness, use $B < 0$ before replacing $\sqrt{n}\pi_k$ by m, and $I \ge 0$ before replacing n by N. These sign requirements are part of the test, not conventions. □

Normalization and the repair, once for every witness. Here is an explicit finite recipe for the constants required above. Let $p \in [p_-, p_+] \subset (0, 1/2)$, $n \ge N$, $\lambda \le \Lambda$, $\theta \le \Theta$, and $|\delta| \le \Delta$, where $\Delta \ge 1$ also includes every binomial mode. Choose $0 < r_- < p_- - \Delta/N$ and $p_+ + \Delta/N < r_+ < 1/2$. Let $v_* > 0$ be a lower bound for $p(1-p)$. Choose positive E, L, R bounding, on $[r_-, r_+]$,

$$|D'''(r\|p)| = \frac{1-2r}{r^2(1-r)^2}, \qquad |(\log(r(1-r)))'| = \frac{1-2r}{r(1-r)}, \qquad \frac{1}{12r(1-r)},$$

respectively, where $D(r\|p) = r\log(r/p) + (1-r)\log((1-r)/q)$. Then, with $Z(t) = Lt/2 + R + Et^3/(6N)$,

$$\sqrt{n}\pi_k = \frac{\phi(0)}{\sqrt{v}} e^{-\delta^2/(2nv)+\zeta}, \qquad n|\zeta| < Z(|\delta|). \tag{103}$$

The finite Stirling estimate $\log(j!) = (j+1/2)\log j - j + \log(2\pi)/2 + r_j$, $0 < r_j < 1/(12j)$, used in the concentration argument above gives this formula on substitution in $n!/(k!(n-k)!)$ and Taylor expansion of relative entropy and the logarithmic prefactor. Here is a finite proof of the estimate. For $a_j = \log(j!) - (j+1/2)\log j + j$ and $z = (2j+1)^{-1}$,

$$0 < a_j - a_{j+1} = \sum_{i \ge 1} \frac{z^{2i}}{2i+1} < \frac{1}{12j(j+1)}.$$

Summation bounds $a_j - \lim a_i$. The integration-by-parts recursion for $\int_0^{\pi/2} \sin^j t\, dt$, and the inequalities between successive integrals, give Wallis's product and $\lim a_i = \log(2\pi)/2$. The entropy Taylor error and factorial error give respectively $E|\delta|^3/(6N)$ and $1/[12r(1-r)]$ in Z.

Consequently a valid mass lower bound is any positive

$$m < \frac{.39886}{\sqrt{p_+(1-p_+)}} \left[1 - \frac{\Delta^2/(2v_*) + Z(\Delta)}{N}\right]. \tag{104}$$

The same formula at a mode proves $\sqrt{n}\max_j \pi_j < 1$ whenever $v_*[1 - Z(\Delta)/N]^2 > .4^2$ and $Z(\Delta) < N$. The rational bounds $.39886 < \phi(0) < .4$ follow, for example, by the alternating arctangent series in Machin's identity.

Differentiating (98) gives

$$\log f' = -\tfrac{1}{2}\log(1 + s/v) + \delta^2/(2nv) - (\delta+u)^2/(2n(v+s)) - \zeta. \tag{105}$$

Thus $n|\log f'| \le B_0\lambda + P(D, a)$ on $|\delta| \le D \le \Delta$, $|u| \le a \le 1/2$, where

$$B_0 > \frac{1}{2v_*} + \frac{\Delta^2}{2Nv_*^2}, \qquad P(D,a) = \frac{2Da + a^2}{2v_*} + Z(D).$$

Choose $0 < b < 1$ above $[B_0\Lambda + P(\Delta, 1/2)]/N$ and put $F = (1-b)^{-1}$. The elementary estimates $e^z \le (1-z)^{-1}$ and $1 - e^{-z} \le z$ imply

$$|f' - 1| \le (B_1\lambda + P_1)/n, \qquad B_1 = FB_0, \quad P_1 = FP(D,a). \tag{106}$$

This supplies both a derivative interval $l_f \le f' \le h_f$ and the finer phase-error bound $|u - f(u)| \le a(B_1\lambda + P_1)/n$.

The exact count variance is

$$t_j = \mathbb{E}W_j^2 = \lambda + (j - np)(s_1 - s_0), \qquad |s_1 - s_0| \le s/p \le c_v s, \quad c_v \ge 1/p_-. \tag{107}$$

It gives $\omega_-\lambda \le t_k \le \omega_+\lambda$ with $\omega_\pm = 1 \pm c_v D/N$. All uses below require the checked condition $\omega_- > 0$. Also $t_{k+i} \le iV_*\lambda$ for $V_* = 1 + c_v(\Delta + 1)/N$. The binomial recursion gives, for every legal $i \ge 1$,

$$\frac{\pi_{k+i}}{\pi_k} \le R_0, \qquad R_0 > 1 + \frac{p_+ + \Delta - 1}{Nv_*[1 - (\Delta - 1)/(Np_-)]}, \tag{108}$$

provided the denominator is positive and $p_+ + \Delta < 2$. Indeed the first ratio has this bound and all subsequent rising factors are at most one. No count outside $[0, n]$ is used in this recursion.

Finally (97) fixes the signs inside the absolute cubic. Expansion and centering give

$$\mathbb{E}|Y - p|^3 = v(1 - 2v) + 3pq(s_0 + s_1) + p\mathbb{E}U_1^3 - q\mathbb{E}U_0^3 \ge v(1 - 2v) + (3p - d)s.$$

The inequality $(1 + z)^{3/2} \le 1 + 3z/2 + 3z^2/8$ for $z \ge 0$ yields

$$(v + s)^{3/2}(\rho - \rho_B) \ge s\left(6p - 3p^2 - \tfrac{3}{2} - d - \frac{3(1 - 2v)s}{8v}\right). \tag{109}$$

Let $Q_* > 0$ be a lower bound for the bracket and $m_c \le c_*/[p_+(1 - p_+) + \Lambda/N]^{3/2}$. The normal variance correction is nonpositive for $\delta \le 0$; for $\delta > 0$ its derivative in s is at most $.4\delta/(2v^{3/2})$. Therefore

$$\mathcal{R} \le (P_*D_+ - m_cQ_*)\lambda/n, \qquad P_* \ge \frac{.4}{2v_*^{3/2}}, \quad D_+ \ge \delta_+. \tag{110}$$

One may take $D_+ = D$, but retaining the positive part is often decisive. All bounds concern $v + s$, which can exceed $1/4$.

Two elementary tools for probabilities.

Lemma 7.17 (A shifted quartic)**.** *Suppose W is a sum of independent centered variables bounded in absolute value by A, and $t = \mathbb{E}W^2$. Let $K_0 \ge (9 + 6\sqrt{3})/4$ and $a > 0$. For $r \ge 0$ and $\sigma \in \{-1, 1\}$,*

$$\Pr(W + \sigma r > 0) \ge \frac{(3a^2 - A^2)t - 3t^2 - 4Art - 6r^2t - r^4 + 3a^2r^2 + 2\sigma a^3 r}{K_0 a^4}. \tag{111}$$

Proof. The polynomial $h_a(z) = (-z^4 + 3a^2z^2 + 2a^3z)/(K_0a^4)$ is below $\mathbf{1}_{\{z>0\}}$: it is nonpositive for $z \le 0$, and its maximum for $z > 0$ is at $z/a = (1 + \sqrt{3})/2$ and is at most one. Independence gives $|\mathbb{E}W^3| \le At$ and $\mathbb{E}W^4 \le A^2t + 3t^2$. Expand $\mathbb{E}h_a(W + \sigma r)$ and apply these two inequalities. The value at the threshold is zero, so atoms require no limiting argument. □

Lemma 7.18 (One common tail bound)**.** *If independent centered $X_i \le A$ have total variance at most V, then, for $b > 0$, their sum has upper tail bounded by the minimum of one and*

$$\frac{V}{V + b^2}, \qquad \mathcal{B}(A, V, b) = e^{b/A}\left(\frac{V}{V + Ab}\right)^{b/A + V/A^2}, \qquad \frac{\mathbb{E}(P_\mu - j)_+^3}{(b/A + \mu - j)^3}, \tag{112}$$
$$\mu = V/A^2, \qquad 0 \le j < b/A + \mu.$$

where P_μ is Poisson. The same holds for a lower tail when $|X_i| \le A$. All bounds apply to closed events, hence also to strict events.

Proof. Cantelli's bound follows by applying Markov to $(\sum X_i + V/b)^2$. Taylor's integral remainder gives $\mathbb{E}e^{zX_i} \le \exp\{\mathbb{E}X_i^2(e^{zA} - 1 - zA)/A^2\}$ for $z \ge 0$. Multiply and use $z = A^{-1}\log(1 + Ab/V)$ to get $\mathcal{B}$. The zero-variance case is immediate.

For the third bound, smooth $f(x) = (x - t)_+^3$ by a nonnegative smooth compactly supported density. Then $f'', f''', f'''' \ge 0$. For $\mathcal{A}f(x) = f(x + A) - f(x) - Af'(x)$, the Taylor quotient

$$\frac{f(x + y) - f(x) - yf'(x)}{y^2} = \int_0^1 (1 - r)f''(x + ry)\, dr$$

is nondecreasing in y. A centered $X \le A$ of variance v consequently satisfies $\mathbb{E}f(x + X) \le f(x) + (v/A^2)\mathcal{A}f(x)$. For $\mathcal{P}_z f(x) = \mathbb{E}f(x + A(P_z - z))$, $\partial_z \mathcal{P}_z f = \mathcal{P}_z \mathcal{A}f$ and

$$\partial_z^2 \mathcal{P}_z f = \mathcal{P}_z \mathcal{A}^2 f \ge 0, \qquad \mathcal{A}^2 f(x) = \int_0^A \int_0^A (A - r)(A - w)f''''(x + r + w)\, dr\, dw.$$

Thus $\mathcal{P}_{v/A^2} f \ge f + (v/A^2)\mathcal{A}f$. Replace the independent variables successively; translations and expectations preserve the derivative signs and Poisson rates add. Increasing the total variance to V is permitted by Jensen after adding independent centered Poisson noise. Removing the smoothing is justified by bounded actual upper supports and finite Poisson third moments. Markov at $t = A(j - \mu)$ proves the displayed bound. □

These bounds have finite rational evaluations. For example,

$$\mathbb{E}(P_\mu - j)_+^3 = \mu^3 + 3(1 - j)\mu^2 + (1 - 3j + 3j^2)\mu - j^3 + e^{-\mu} \sum_{h=0}^{\lfloor j \rfloor} \frac{\mu^h (j - h)^3}{h!}. \tag{113}$$

Replacing $e^{-\mu}$ by $(\sum_{h=0}^{24} \mu^h/h!)^{-1}$ enlarges the nonnegative correction. Alternatively sum the positive Poisson series through 32; the remaining tail is at most its first term divided by $1 - \mu(34 - j)^3/[34(33 - j)^3]$, provided $j < 33$ and that ratio is below one. For Bennett, put $x = Ab/V$ and $L = [(b/A + \mu)\log(1 + x) - b/A]$. The positive series $\log(1 + x) = 2\sum_{h \ge 0} z^{2h+1}/(2h+1)$, $z = x/(2+x)$, gives lower bounds for L; any positive lower exponential series gives an upper bound for e^{-L}. Rounding a positive lower L down before exponentiation is also valid. These are choices of rational witnesses, not optimization claims.

The nearest-count witness, including vanishing noise. For $u = -r \le 0$, add $-h_f r$ to (111) with $\sigma = 1$; for $u = r \ge 0$, add $l_f r$ with $\sigma = -1$. Both bound $\Pr(W_k > u) + f(u)$ from below. For a cell $t \in [t_-, t_+]$, $r \in [r_-, r_+]$, this expression is concave in t (second derivative $-6/(K_0 a^4)$), so only t_-, t_+ are needed. At each endpoint, bound a monomial with positive coefficient at r_- and one with negative coefficient at r_+. Any rational $a > 0$ is a valid witness. Other valid choices are

$$\frac{r_-^2}{t_+ + r_-^2} - h_f r_+ \quad (u \le 0), \qquad l_f r_- \quad (u \ge 0). \tag{114}$$

They follow from Cantelli and nonnegativity, respectively. Their maximum or one selected witness is valid without an optimality proof.

For incoming counts use $t_{k+i} \le iV_*\lambda$. On $u \in [-r_+, -r_-]$ the first distance is $1 + r_-$ and every later distance is at least i. On $u \in [r_-, r_+]$ they are $1 - r_+$ and $i(1 - r_+/2)$ for $i \ge 2$. Let b, b_o be these first and later distances. On $\lambda \in [L, U]$, let T_1 be any bound from Lemma 7.18 at (A, V_*U, b). Since $\mathcal{B}(A, iV, ib) = \mathcal{B}(A, V, b)^i$, all later counts total at most $B^2/(1 - B)$ for $B = \mathcal{B}(A, V_*U, b_o) < 1$. Thus

$$D = O - R_0\{T_1 + B^2/(1 - B)\} > 0, \qquad mD - \kappa U/N > \varepsilon U \tag{115}$$

proves the entire cell when $\kappa \geq 0$. The selected MID normalizers explicitly verify this sign. This is the common central and left MID proof.

A cell touching $\lambda = 0$ requires estimates proportional to λ; a positive lower endpoint may not be substituted. The following rescaling provides them. Put $r = zt$, $t_* = \omega_+ U$ and $d_a = K_0 a^4$. The shifted quartic plus br, divided by t, becomes

$$F_{\sigma,b}(z,t) = \frac{3a^2 - A^2 - 3t}{d_a} + \left(\frac{2\sigma}{K_0 a} + b - \frac{4At}{d_a}\right) z + \frac{3a^2 t - 6t^2}{d_a} z^2 - \frac{t^3 z^4}{d_a}. \tag{116}$$

It extends to $t = 0$ and is concave in t. On $0 \leq z \leq z_0$, replacing z^4 by $z_0^2 z^2$ gives a lower quadratic at $t = t_*$; at $t = 0$ it is linear. Endpoints and a clipped quadratic vertex therefore give an exact lower bound $J(\sigma, b, z_0)$ for the whole rectangle.

For negative phase choose $z_-, c_- > 0$ and verify

$$J(1, -h_f, z_-) > c_-, \quad \frac{z_-^2}{1 + z_-^2 t_*} - h_f z_- > c_-, \quad \frac{2z_-}{(1 + z_-^2 t_*)^2} > h_f, \tag{117}$$
$$\frac{1}{1 + 4t_*} - \frac{h_f}{2} > c_- t_*, \qquad z_- t_* < 1/2.$$

The quartic handles $r \leq z_- t$. Beyond that point use $r^2/(t + r^2) - h_f r$. Its derivative at $r = z_- t$ is positive by the third gate; its second derivative has just one sign change, so any later critical point is a maximum. Its minimum is at $z_- t$ or $1/2$, covered by the other gates. Hence the own count exceeds $c_- t$. For positive phase choose $z_+, c_+ > 0$ with

$$J(-1, l_f, z_+) > c_+, \qquad l_f z_+ > c_+, \qquad z_+ t_* < 1/2. \tag{118}$$

The quartic handles $r \leq z_+ t$ and the normal term handles the rest. The own count exceeds $c_+ t$ and also exceeds $l_f r$.

For incoming tails proportional to λ, fix a Poisson anchor j. A positive series term of index h divided by $\lambda(b/A + \mu - j)^3$ has logarithmic derivative, multiplied by λ,

$$h - 1 - \mu - \frac{3\mu}{b/A + \mu - j}, \qquad \mu = V_* \lambda / A^2.$$

If $b/A - j > 0$ and this is nonnegative at $\lambda = U$ for the first positive index, it is nonnegative for every term and all $0 < \lambda \leq U$. Therefore the Poisson bound divided by λ is largest at U. Likewise $B^2/[\lambda(1 - B)]$ is increasing whenever $\lambda B'/B \geq 1/2$. Differentiating the Bennett formula gives

$$\lambda B'/B = b/A - \mu \log(1 + b/(A\mu));$$

The function $g(\mu) = b/A - \mu \log(1 + b/(A\mu))$ decreases because $\log(1+x) > x/(1+x)$ for $x > 0$. Moreover the logarithmic derivative of $B^2/[\lambda(1-B)]$, multiplied by λ, is $2g - 1 + gB/(1-B) \geq 0$ when $g \geq 1/2$. Thus checking the gate at U proves the whole interval. A rational upper bound for the logarithm verifies that gate. On cells using the fixed positive bound $l_f r_-$ instead, unscaled endpoint tails suffice: after a positive bracket is checked, divide it by U to obtain a lower bound per λ. These arguments, followed by (100), prove TINY uniformly down to arbitrary positive noise, including a deterministic conditional component.

7.6 Whole-count comparison

Assume $c = 1 - d_0 > 1/2$. Couple all $2n$ conditional variables independently and set $W_j = \sum_{i \leq j} U_{1,i} + \sum_{i > j} U_{0,i}$, extending it constantly outside $[0, n]$. Every increment of $j + W_j$ is at least c. Define

$$K = \max\{j \in \mathbb{Z} : j + W_j \leq k + u\}, \qquad R = K - k.$$

An independent binomial selector has the correct actual count laws, so

$$\Pr(\textstyle\sum Y_i \le k+u) = \mathbb{E}F_B(k+R). \tag{119}$$

Let $\chi = (n+1)p - k < 2$ and $m_B = \max_j \pi_j$. For every integer r,

$$F_B(k+r) - F_B(k) - r\pi_k \le \frac{m_B}{(n+1)v}\left[(\chi-1)_+ r_+ + \chi_+\binom{r_-}{2} + \binom{r_-}{3}\right]. \tag{120}$$

Indeed, the $\mathrm{Bin}(n+1,p)$ mass is $\pi_j^+ = q\pi_j + p\pi_{j-1} \le m_B$. Factorial cancellation gives

$$\pi_j - \pi_{j-1} = \frac{(n+1)p - j}{(n+1)v}\,\pi_j^+ \qquad (j \in \mathbb{Z}),$$

including the exterior zero masses. Thus the positive part at $j = k+i$ is at most $m_B(\chi - i)_+/((n+1)v)$. Double summation gives (120): for $r > 0$ only $i = 1$ can contribute; for $r = -h < 0$ use $(\chi+i)_+ \le \chi_+ + i$ and sum the multiplicities $h-1-i$. Zero extension makes the same calculation valid outside the legal count range.

Write $t = \mathbb{E}W_k^2$. Crossing monotonicity and Cauchy–Schwarz give

$$\mathbb{E}R_+ \le \frac{t}{(c-u)^2}, \quad \mathbb{E}\binom{R_-}{2} \le \frac{t}{[c+\min(u,c/3)]^2}, \quad \mathbb{E}\binom{R_-}{3} \le \frac{t\sqrt{A^2+3t}}{(2c-1/2)^3}. \tag{121}$$

For the first, $R = i \ge 1$ implies $-W_k \ge ci - u$ and $(ci-u)^2 \ge i(c-u)^2$. For $R = -h \le -2$, the next count is outside the closed event, so $W_k > c(h-1) + u$. Put $j = h-1$. The function $j(j+1)/(2(cj+u)^2)$ decreases for $j \ge 1$ if $u \le c/3$; for larger u reduce to that endpoint. Also $(j^3-j)/(6(cj-1/2)^3)$ decreases for $j \ge 2$, since its derivative has the sign of $2cj - (3j^2-1)/2 < 0$. These give the last two bounds using $\mathbb{E}(W_k)_+^3 \le \sqrt{\mathbb{E}W_k^2\mathbb{E}W_k^4}$.

The marked crossing. Let $D_i = U_{1,i} - U_{0,i}$, $\vartheta = \mathbb{E}(D_i)_+ = \mathbb{E}(D_i)_-$, $0 < \eta \le 1$, and $R_f = \mathbb{E}\{u + \eta - W_k\}$, where $\{x\} = x - \lfloor x \rfloor$ is the fractional part. The two deleted laws are

$$A_0 \overset{d}{=} kU_1 + (n-k-1)U_0, \qquad A_1 \overset{d}{=} (k-1)U_1 + (n-k)U_0.$$

Then

$$\begin{aligned} &\mathbb{E}R \le u + \eta - R_f + J, \\ &J = \frac{\vartheta}{\eta}\sum_{i\ge 1}\{\Pr(A_0 \le u + d_0 - ci) + \Pr(A_1 > u + \eta + i - A)\}. \end{aligned} \tag{122}$$

Here are the pathwise details. Put $z = u - W_k$ and mark

$$Z_f = \sum_{i=1}^{n-k}(D_{k+i})_-\mathbf{1}_{\{W_k \le u - ci\}}, \quad Z_b = \sum_{i=1}^{k}(D_{k-i+1})_+\mathbf{1}_{\{W_k > u+\eta+i\}}.$$

If $R > z + \eta$, the crossing activates the marks through R: for $R > 0$ use $z \ge cR$, while for $R < 0$ use $z < R - \eta$. In either case $R \le z + Z_f + Z_b$. Hence if $h = R - \lfloor z + \eta \rfloor > 0$, $Z_f + Z_b > h - 1 + \eta \ge \eta h$. Thus $R \le \lfloor z + \eta \rfloor + (Z_f + Z_b)/\eta$ on every path. On an active forward mark $U_0 > U_1 \ge -d_0$; deleting its pair implies $A_0 \le u + d_0 - ci$. On a backward mark delete $U_1 \le A$ to obtain the strict event for A_1. The remaining sum is independent of the entire pair, so the expectations factor. Extending to all $i \ge 1$ adds only nonnegative bounds. Finally $|R| \le 1 + (|W_k| + |u|)/c$ proves integrability and permits expectations. No coupling has replaced either deleted marginal.

A finite bound for every mark. Fix $\lambda \in [L, U]$, $L > 0$, and $t \le \omega_+ \lambda$; put $T_L = \omega_+ L$, $T_U = \omega_+ U$. On a phase subcell $u \in [v_l, v_h]$, for $i = 1, 2$ let $b_1 = ci - d_0 - v_h > 0$ and $b_2 = i - A + v_l > 0$. Lemma 7.18 gives a bound $\omega_+ \lambda K_i^{\mathrm{ind}}$, where

$$K_i^{\mathrm{ind}} = \sum_{j=1}^{2} \min\left\{ \frac{1}{b_j^2 + T_L}, \frac{\overline{P}(A, T_U, b_j)}{T_L} \right\}.$$

The denominator T_L is a lower bound for the chosen upper envelope $\omega_+ \lambda$, not for any actual variance. Alternatively couple $A_0 = Z + U_1$, $A_1 = Z + U_0$ with $Z \stackrel{d}{=} (k-1)U_1 + (n-k-1)U_0$. The two events project into $Z \le -\alpha$ and $Z > \beta$, where $\alpha = ci - 2d_0 - v_h > 0$, $\beta = i - A - d_0 + v_l > 0$. For any $z \in (-\alpha, \beta)$ their sum is at most $(\omega_+ \lambda + z^2)/\min(z + \alpha, \beta - z)^2$. Consequently a second kernel is

$$K_i^{\mathrm{com}} = \frac{1 + z^2/T_L}{\min(z + \alpha, \beta - z)^2}, \qquad z = \max\{-T_L/\beta, \min((\beta - \alpha)/2, T_L/\alpha)\}.$$

This center is inside the interval; differentiating the two quadratic ratios on either side of the midpoint proves its optimality, although any interior rational center suffices. Take the smaller kernel on each phase subcell and the maximum over their closed cover. Denote it K_i. For $i \ge 3$ use fourth moments and the convex midpoint integral for $(cx - b)^{-4}$. On $u \in [u_l, u_h]$ this gives

$$K = K_1 + K_2 + \frac{A^2 + 3T_U}{3c} \left\{ \frac{1}{((5/2)c - A - u_h)^3} + \frac{1}{((5/2)c - A + u_l)^3} \right\}, \qquad J \le \frac{\vartheta}{\eta} \omega_+ \lambda K. \tag{123}$$

All denominators must be positive. Indeed, the tail fourth moment is at most $A^2 t + 3t^2$, and integration from $5/2$ to infinity covers every remaining integer, including the harmless added illegal counts.

If the same-law resource supplies $nd < G$ and $G^2 < c_v N L$, then

$$\vartheta \le \frac{c_v G \lambda}{c_v n \lambda + G^2}. \tag{124}$$

To prove this, for centered $X \ge -b$, $\mathbb{E}X^2 \le V$ and $b^2 \le V$, the quadratic $b(X - V/b)^2/(b + V/b)^2$ majorizes X_- on $[-b, \infty)$. Its expectation is at most $bV/(V + b^2)$. Apply this to $D_i \ge -d$, using $\mathrm{Var}(D_i) = s_0 + s_1 \le c_v \lambda / n$, and monotonicity in $b \le \sqrt{V}$. The case $d = 0$ is immediate. With $\eta = z\lambda$, $z > 0$, $zU \le 1$, (123) becomes

$$\frac{J}{\lambda} \le \frac{\omega_+ K c_v G}{z(c_v N \lambda + G^2)}. \tag{125}$$

The fractional-part gain. If each centered summand has support span at most $b < 1/2$ and the sum has variance t, then, for any real w and $0 < r_* \le 2\sin^2(\pi b)/b^2$,

$$\mathbb{E}\{w - W\} \ge \frac{5}{23}(1 - e^{-r_* t}). \tag{126}$$

Indeed $1 - \cos(2\pi x) \le (23/5)x$ for $0 \le x \le 1$. A finite proof of this sole scalar inequality is recorded below. For an independent copy Z' of one summand, concavity of sine on $[0, \pi b]$ gives

$$|\mathbb{E}e^{2\pi i Z}|^2 = 1 - 2\mathbb{E}\sin^2 \pi(Z - Z') \le e^{-4\sin^2(\pi b)\,\mathrm{Var}(Z)/b^2}.$$

Multiply square roots over the independent summands and use periodicity. This argument includes zero variances and integer collisions. Here the conditional spans are at most $A + d_0$, so this is the b to use.

For a reproducible scalar proof, $x \ge 10/23$ follows from $1-\cos \le 2$. On $[0,10/23]$ use the degree-ten alternating upper polynomial for $1-\cos(2\pi x)$, with directed coefficients:

$$Q(x)=\sum_{j=1}^{5}\frac{(-1)^{j+1}(2\pi_j x)^{2j}}{(2j)!},\qquad \pi_j=\begin{cases}3.141592653589794, & j \text{ odd},\\ 3.141592653589793, & j \text{ even}.\end{cases}$$

Put $P(x)=23/5-Q(x)/x$. On the eight intervals with left endpoint $a_j=5j/92$ and width $w=5/92$, its degree-nine Bernstein coefficients are

$$b_{jk}=\sum_{i=0}^{k}\frac{\binom{k}{i}}{\binom{9}{i}}\sum_{h=i}^{9}[x^h]P(x)\binom{h}{i}a_j^{h-i}w^i,\qquad 0\le j<8,\quad 0\le k\le 9.$$

The 80 rational coefficients are positive; the retained scalar acceptor reconstructs them. Expansion in powers of the unit-interval coordinate proves the formula; Bernstein weights are nonnegative and sum to one. The alternating remainder and the stated rational bounds on π follow from the same elementary series above. This completely specifies the finite scalar obligation. A rational rate follows by replacing $\sin(\pi b)$ with $x-x^3/6+x^5/120-x^7/5040$, $x=3.14159b$, on the checked alternating-series domain.

Set $r=r_*\omega_-$, require $rU<3$, and define

$$g=\frac{5}{23}\left(\frac{r^3L^2}{6}-\frac{r^4L^3}{24}+\frac{r^5L^4}{120}-\frac{r^6L^5}{720}\right). \tag{127}$$

The sixth-order Taylor lower bound for $1-e^{-r\lambda}$ gives $R_f/\lambda \ge 5r/23-5r^2\lambda/46+g$. The discarded higher-order part is increasing as a function of λ for $r\lambda<3$: its derivative has the sign of $48-18x+(24/5)x^2-x^3>0$ on $[0,3]$. For instance this polynomial has positive degree-three Bernstein coefficients $48,30,132/5,51/5$ there. Thus freezing it at L is valid.

One final expression. On a signed phase cell find an interval $[x_l,x_h]$ containing H/ℓ by its four endpoint quotients, using positive bounds for ℓ. Let

$$D=\max(|x_l-u_h|,|x_h-u_l|),\quad D_+=\max(0,x_h-u_l),\quad a=\max(|u_l|,|u_h|),\quad b_\chi=p_+-x_l.$$

The same formula allows coarser upper bounds D,D_+,b_χ, as used by the central certificate. Check $D\le\Delta$ and $b_\chi+u_h<2$. With $\omega_\pm=1\pm c_vD/N$, the curvature coefficient is

$$V_0=\frac{\omega_+}{v_*}\left[\sup_{u_l\le u\le u_h}\left\{\frac{(b_\chi+u-1)_+}{(c-u)^2}+\frac{(b_\chi+u)_+}{[c+\min(u,c/3)]^2}\right\}+\frac{\sqrt{A^2+3T_U}}{(2c-1/2)^3}\right]. \tag{128}$$

On a rational phase subcell, both numerators are bounded at the right endpoint; the first denominator is smallest there, and the second is smallest at the left endpoint. This proves the supremum bound, including both kinks. Equations (119)–(121), $\sqrt{n}m_B\le 1$, and (99) now give the whole-count witness with

$$B(\lambda)=z+\frac{\omega_+Kc_vG}{z(c_vN\lambda+G^2)}-\frac{5r}{23}+\frac{5r^2\lambda}{46}-g, \tag{129}$$
$$I(\lambda)=V_0+aB_1+\frac{aP_1}{\lambda}+P_*D_+-m_cQ_*.$$

We used $|u-f(u)|\le a(B_1\lambda+P_1)/n$ and $\sqrt{n}\pi_k\le 1$. Every constant is fixed before λ varies. Both B and $mB+I/N$ are convex on $\lambda>0$; I is decreasing. Thus their two endpoint gates $B<0$, $I>0$, $mB+I/N<-.001$ prove the entire closed noise cell. This proves all three LOW variants by exactly the same argument.

7.7 Signed refinements and the complete local covers

The retained tables contain 45 central probability intervals covering $[.38, .47]$ and 38 left intervals covering $[.304, .38]$. Each row supplies its own Θ, signed H bounds, d_0, Λ, and global amplitude; the confinement profiles may improve the amplitude with λ. All these are hypotheses of the present section, acquired earlier without using its conclusion. The shared resource has the form

$$nD_{\mathrm{mom}} < \frac{15000}{997} T_H, \qquad D_{\mathrm{mom}} \geq \alpha d + \beta\theta + P.$$

Consequently a noise cell $[L, U]$ has

$$nd < G = \{(15000/997)T_H - NP - \beta \ell_-^2 L\}/\alpha.$$

Here T_H is the maximum of the acquired concave quadratic budget on the closed signed H cell, further bounded by its scalar ceiling. It already contains the phase charge. If $G \leq 0$, the entire cell is impossible and receives no score test; the exclusion is strict even at $G = 0$.

For positive noise the complete cover is the following. Each interval is intersected with the row ceiling Λ; shared endpoints are included.

Noise	Central phase	Left phase	Witness
$(0, .05]$	all	all	nearest count (scaled)
$[.05, .06]$	$\lvert u\rvert \geq .09$	$\lvert u\rvert \geq .09$	nearest count
$[.06, .08]$	$\lvert u\rvert \geq .13$	$\lvert u\rvert \geq .14$	nearest count
$[.08, .10]$	$\lvert u\rvert \geq .18$	$\lvert u\rvert \geq .19$	nearest count
$[.10, .12]$	$\lvert u\rvert \geq .29$	$u \leq -.29$	nearest count
$[.10, .12]$	—	$u \in [.29, .5]$	whole count
$[.05, \Lambda]$	remaining phase	remaining phase	whole count

Beyond .12 the last row covers the full phase interval. The three LOW objects are one central cover, one left cover with phase slices, and one extra positive left cover. Their distinction is only the closed domain and the chosen outward constants. Signed H cells meet at zero; nearest count ties meet at $u = \pm 1/2$. The fixed coverage checker also checks every row-ceiling singleton and every entirely excluded LOW row.

The common TINY parameters are

$$N = 47, \quad p \in [.304, .47], \quad |H| \leq 1.393, \quad \Theta = .0154, \quad d_0 = .0181, \quad A = .425, \quad \Lambda = .05.$$

For its scaled own witness use $a = .4784$, $K_0 = 9/(4 \cdot .4641)$, $(z_-, c_-) = (1.66, .514)$ and $(z_+, c_+) = (1.2, 1.036)$. Its six phase cells are $[-.5, 0], [0, .1], \ldots, [.4, .5]$. The first two use scaled Poisson anchors $1.25, 1$, respectively; the other four use the fixed positive normal bound. The exact normalizers and gates specified in the accompanying certificate map verify these constants. The central MID amplitudes are $.35, .35, .30, .275$ in its four rows. The left ones are $.375, .375, .325, .30$, each further reduced by the row's global amplitude. No interpolation of the tables is used.

All six accepted comparisons give the same strict conclusion

$$\rho_B(C - C_B) < -.001\lambda.$$

Their proofs have now reduced to Theorem 7.16, the shared normalization, and the two explicitly derived witnesses. The arithmetic objects still have their original domains and targets; no certificate branch has been silently deleted or relabeled.

If $\lambda = 0$, conditional centering and $p, q > 0$ imply $U_0 = U_1 = 0$ almost surely. The law is exactly Bernoulli. Schulz's strict finite-binomial bound [7, Theorem 1] states, for every $0 < p < 1$, finite n and real x,

$$\sqrt{n}\,|\mathbb{P}(\mathrm{Bin}(n,p) \le x) - \Phi((x-np)/\sqrt{npq})| < C_{\mathrm{E}}\rho_B.$$

It applies directly to the actual threshold, including atoms. At positive noise the same theorem closes $C < C_B < C_{\mathrm{E}}$. No estimate divides by zero or replaces a uniform small-noise bound with a fixed-law limit.

A triangle of allowed noise and displacement. Introduce Bernoulli coordinates

$$X = \ell(B_0 + U - p), \quad B_0 = \mathbf{1}_{\{X>0\}}, \quad \ell = \bar{A} + \bar{B} = \frac{m}{pq}, \quad \mathbb{E}[U \mid B_0] = 0.$$

Then

$$\ell^2 = \frac{1-\theta}{pq}, \quad m\ell = 1-\theta, \qquad d = \frac{\Delta_A + \Delta_B}{\ell}, \quad \lambda = n\mathbb{E}U^2 = \frac{npq\theta}{1-\theta}.$$

Theorem 7.19 (Joint inward/noise constraint)**.** *Suppose $n \ge N > 0$, and bounds for this same law give*

$$nD \le Nd_D, \quad \theta \le \Theta < 1, \quad pq \le v_*, \quad A + B \ge S_* > 0,$$

and

$$D \ge b + 6mS_*(\Delta_A + \Delta_B) + K\theta, \qquad b \ge 0, \quad K > 0.$$

Set $r = d_D - b$. If $r > 0$, define

$$d_* = \frac{r}{6(1-\Theta)S_*}, \qquad \Lambda = \frac{Nv_*r}{K(1-\Theta)}.$$

Then

$$\boxed{\frac{d}{d_*} + \frac{\lambda}{\Lambda} \le 1.} \tag{130}$$

If $r < 0$, the premises are impossible. If $r = 0$, they force $d = \lambda = 0$.

Proof. Multiply the assumed joint lower bound for D by n. Since all coefficients being weakened are nonnegative,

$$\begin{aligned} Nd_D \ge nD &\ge nb + 6n(1-\theta)S_*d + K\frac{(1-\theta)\lambda}{pq} \\ &\ge Nb + 6N(1-\Theta)S_*d + \frac{K(1-\Theta)}{v_*}\lambda. \end{aligned}$$

Subtract Nb. For $r > 0$, division by Nr proves (130). For $r \le 0$, positivity of both coefficients gives the stated alternatives without dividing by zero. □

Apply this with the binary charge in (11), $S_* = A_l + B_l$ and $K = \min\{3(\sigma+1)B_l, 3(\sigma+3)A_l\}$.

Corollary 7.20 (Closed noise bands and outward rounding)**.** *If $0 \le L \le \lambda \le U \le \Lambda$, then*

$$d \le d_*(1 - L/\Lambda).$$

Both intercepts in (130) *may first be replaced by larger positive numbers. The same conclusion then holds using those enlarged intercepts and the enlarged noise ceiling.*

Proof. Rearrange (130) and use $\lambda \ge L$. Increasing either denominator weakens its nonnegative summand, so it preserves the joint inequality. This also proves the rounding assertion. □

The coefficient must be bounded in the correct direction. Suppose $n \ge N$, $p \in [p_-, p_+] \subset (0, 1/2)$, $0 \le d \le d_0$, and $0 \le \lambda \le \Lambda$. Choose $0 < v_* \le p_-(1-p_-)$ and put

$$V^* = p_+(1-p_+) + \Lambda/N, \qquad Q_* = 6p_- - 3p_-^2 - \frac{3}{2} - d_0 - \frac{3(1-2v_*)\Lambda}{8v_*N}.$$

Then $Q \ge Q_*$: the first term increases with p, and $(1-2v)/v$ decreases with v. In particular, Q_* may be negative. Already with $d = s = 0$ its sign changes at $p = 1 - 1/\sqrt{2}$.

Let C be the score multiplying the moment correction, and assume $0 < c_* \le C \le c^*$. Choose positive multipliers satisfying

$$0 < m_- \le \frac{c_*}{(V^*)^{3/2}}, \qquad m_+ \ge \frac{c^*}{v_*^{3/2}},$$

and define the signed gain

$$G_* := \min\{m_- Q_*, m_+ Q_*\}. \tag{131}$$

Proposition 7.21 (Signed gain and nonnegative repair)**.** *Let Φ denote the standard normal distribution function. For an actual count displacement δ, set*

$$\mathcal{R} = \sqrt{n}\left\{\Phi\left(\frac{\delta}{\sqrt{nv}}\right) - \Phi\left(\frac{\delta}{\sqrt{n(v+s)}}\right)\right\} + C(\rho_B - \rho).$$

If $D_+ \ge \max(\delta, 0)$ and $P_* \ge .4/(2v_*^{3/2})$, then

$$\mathcal{R} \le (P_* D_+ - G_*)\frac{\lambda}{n} \le \kappa\frac{\lambda}{N}, \qquad \kappa = \max\{0, P_* D_+ - G_*\}. \tag{132}$$

Proof. Write $A = C/(v+s)^{3/2}$. The hypotheses give $0 < m_- \le A \le m_+$. By (109), $C(\rho - \rho_B) \ge sAQ \ge sAQ_*$. If $Q_* \ge 0$, $AQ_* \ge m_- Q_*$; if $Q_* < 0$, $AQ_* \ge m_+ Q_*$. This proves $C(\rho - \rho_B) \ge sG_*$ in both cases. One must use the *upper* multiplier in the negative case.

The normal part is nonpositive when $\delta \le 0$. For $\delta > 0$, differentiation with respect to an intermediate variance $t \in [0, s]$ gives

$$\frac{d}{dt}\left[-\sqrt{n}\,\Phi\left(\frac{\delta}{\sqrt{n(v+t)}}\right)\right] = \frac{\delta}{2(v+t)^{3/2}}\phi\left(\frac{\delta}{\sqrt{n(v+t)}}\right) \le \frac{.4\delta}{2v_*^{3/2}} \le P_* D_+.$$

Integrate over $[0, s]$, and combine with the moment bound. Finally $s = \lambda/n$ and $\kappa \ge 0$, so $(P_* D_+ - G_*)\lambda/n \le \kappa\lambda/n \le \kappa\lambda/N$. The clamp at zero is needed before replacing n by its lower bound. □

A binomial mass bound on the actual phase cell. Suppose the finite Stirling argument has supplied positive constants E, L_0, R on a fixed segment containing every k/n under consideration, with

$$\sqrt{n}\,\pi_k = \frac{\phi(0)}{\sqrt{v}}\exp\{-\delta^2/(2nv) + \zeta\}, \qquad n|\zeta| \le Z_N(|\delta|), \qquad Z_N(t) = \frac{L_0 t}{2} + R + \frac{Et^3}{6N}, \quad n \ge N.$$

The constants may have been computed on a much wider global phase range. Let $v_- \le v \le v_+$ and let $q_+ \ge \sqrt{v_+}$.

Lemma 7.22 (Mass localization)**.** *On any smaller closed cell where $|\delta| \le D$,*

$$\sqrt{n}\,\pi_k \ge \frac{.39886}{q_+}\left[1 - \frac{D^2/(2v_-) + Z_N(D)}{N}\right].$$

The maximum of this expression and any previously proved mass lower bound is also a valid mass lower bound. There is no requirement $D \ge 1$.

Proof. Set $A_* = D^2/(2v_-) + Z_N(D) \ge 0$. The logarithmic formula gives $\sqrt{n}\pi_k \ge (.39886/q_+)e^{-A_*/n}$. Now $e^{-z} \ge 1 - z$ and $n \ge N$. If the bracket is negative the displayed inequality still holds, and the previously positive bound can be retained. The global Stirling segment, rather than the smaller value D, continues to justify E, L_0, R. □

A sharper elementary harmonic constant.

Lemma 7.23. *For every $x \ge 0$,*

$$1 - \cos(2\pi x) \le \frac{4553}{1000}x.$$

Proof. Put $K = 4553/1000$. For $x \ge 2/K$ the assertion follows from $1 - \cos \le 2$. On $[0, 2/K]$ take the degree-fourteen alternating upper Taylor polynomial $Q(x)$ for $1 - \cos(2\pi x)$. Its remainder is nonpositive: the first omitted term is negative, and the remaining term magnitudes decrease, since $(4\pi/K)^2 < 16 \cdot 15$.

Use the directed endpoints

$$\pi_- = 3.141592653589793 < \pi < 3.141592653589794 = \pi_+.$$

In each positive coefficient of Q replace π by π_+; in each negative coefficient replace it by π_-. This only increases Q. The exact checker expands $P(x) = K - Q(x)/x$, a polynomial of degree thirteen, on each of the 64 intervals $[2j/(64K), 2(j+1)/(64K)]$. If $P(x) = \sum_{h=0}^{13} p_h x^h$, its Bernstein coefficients on $[a, a+w]$ are

$$b_k = \sum_{i=0}^{k} \frac{\binom{k}{i}}{\binom{13}{i}} \sum_{h=i}^{13} p_h \binom{h}{i} a^{h-i} w^i, \qquad k = 0, \dots, 13.$$

All 896 rational coefficients are positive; the smallest is greater than .00012312. Thus $P > 0$ on the full interval. The value at zero follows by continuity (or directly).

For completeness the stated π endpoints are checked using alternating series for $\arctan(1/5)$ (32 terms) and $\arctan(1/239)$ (12 terms). The tangent addition formula gives $\tan(4\arctan(1/5) - \arctan(1/239)) = 1$. The angle lies between zero and $4/5 < \pi/2$, so it equals $\pi/4$. The positivity follows already from $4(1/5 - 1/(3 \cdot 5^3)) - 1/239 > 0$; the upper bound follows from $\arctan z < z$. This proves Machin's identity on the correct branch. The full rational recipe is `check_harmonic4553.py`; no sampled positivity is used. □

If the original characteristic-function argument gives $|\mathbb{E}e^{2\pi i W}| \le e^{-rt}$, Lemma 7.23 upgrades its fractional part estimate from $(5/23)(1 - e^{-rt})$ to

$$\frac{1000}{4553}(1 - e^{-rt}).$$

Consequently every occurrence of $5/23$ and $5/46$ in that harmonic gain is replaced by $1000/4553$ and $500/4553$, respectively. The existing degree-six Taylor gain and its monotonicity argument remain unchanged.

The signed Gaussian correction in nearest-count comparison. Let $n \ge N$, $v = p(1-p) > 0$, $s \ge 0$, $\lambda = ns$, and let the actual nearest count and phase satisfy

$$\delta = k - np = h - u, \qquad f(t) = \frac{\Phi((\delta + t)/\sqrt{n(v+s)}) - \Phi(\delta/\sqrt{n(v+s)})}{\pi_k}.$$

The actual k and δ are held fixed when t varies. Assume the proved finite Stirling formula on this count range is

$$\sqrt{n}\,\pi_k = \frac{\phi(0)}{\sqrt{v}}\exp\{-\delta^2/(2nv)+\zeta\}, \qquad n|\zeta| \le Z_N(D), \quad |\delta| \le D.$$

Write $v \ge v_- > 0$ and $Z = Z_N(D) \ge 0$.

Lemma 7.24 (Negative phase)**.** *Suppose $u = -r$, $0 \le a \le r \le b$, and $h \le h_*$. Define*

$$t_* = \max(0, \min(b, h_* + b)), \qquad G_- = 2(h_* + b)t_* - t_*^2, \qquad \eta_- = G_-/(2v_-) + Z.$$

If $D^2 \le Nv_-$ and $\eta_- < N$, then

$$f'(-t) \le \frac{1}{1-\eta_-/N} \quad (0 \le t \le r), \qquad u - f(u) \le \frac{b\eta_-}{n(1-\eta_-/N)}.$$

Proof. Direct differentiation and the Stirling formula give

$$\log f'(-t) = -\tfrac{1}{2}\log(1+s/v) + \frac{\delta^2 s}{2nv(v+s)} + \frac{2\delta t - t^2}{2n(v+s)} - \zeta.$$

Since $\log(1+x) \ge x/(1+x)$ for $x \ge 0$, the first two terms are at most

$$\frac{s}{2(v+s)}\left(\frac{\delta^2}{nv} - 1\right) \le 0.$$

This cancellation removes the variance term from the upper derivative bound. Because $\delta = h + r$, $r \le b$ and $0 \le t \le r \le b$,

$$2\delta t - t^2 \le 2(h_* + b)t - t^2 \le G_-.$$

The last maximum is the concave quadratic maximum on $[0, b]$, achieved at the displayed projection t_*. It is nonnegative because $t = 0$ is available. Therefore $\log f'(-t) \le \eta_-/n$. Using $e^x \le (1-x)^{-1}$ for $0 \le x < 1$ gives

$$f'(-t) - 1 \le \frac{\eta_-/n}{1-\eta_-/n} \le \frac{\eta_-}{n(1-\eta_-/N)}.$$

Integrate $u - f(u) = \int_0^r (f'(-t) - 1)\,dt$. □

Equivalently, $G_- = b^2 + 2h_*b$ for $h_* \ge 0$, and $G_- = \max(0, h_* + b)^2$ for $h_* < 0$. This sharper vertex is used in the selected finite left LOW implementation.

Lemma 7.25 (Positive phase)**.** *Suppose $u \in [a, b] \subset [0, 1/2]$, $h \le h_*$, and define*

$$G_+ = \max\left(0, \max_{z\in[a,b]}(2h_*z - z^2)\right).$$

Then

$$u - f(u) \le \frac{b}{n}\left(\frac{\lambda}{2v_-} + \frac{G_+}{2v_-} + Z\right).$$

The inner maximum is at the projection of h_* onto $[a, b]$.

Proof. The exact derivative identity and $\log(1+s/v) \le s/v$ give

$$\log f'(t) \ge -\frac{\lambda + 2\delta t + t^2}{2nv} - \zeta.$$

For fixed actual u, the function $2\delta t + t^2$ is convex on $[0, u]$. Its maximum is at an endpoint, and $\delta = h - u$ gives

$$2\delta t + t^2 \le \max(0, 2hu - u^2) \le G_+.$$

Use $e^{-x} \ge 1 - x$ and integrate from zero to u. □

Whole-count insertion and interval coverage. The whole-count identity is split as $Q/\pi_k - f(u) = (Q/\pi_k - u) + (u - f(u))$. Thus one needs an upper bound on the signed correction $u - f(u)$; an absolute-value estimate is unnecessary. After division by λ, the new coefficients replacing the old Gaussian correction are

$$\begin{aligned} u \geq 0: &\quad \frac{b}{2v_-} + \frac{b\{G_+/(2v_-) + Z\}}{\lambda}, \\ u \leq 0: &\quad \frac{b\eta_-}{(1-\eta_-/N)\lambda}. \end{aligned}$$

On each fixed phase cell these have the form $A + B/\lambda$ with $A, B \geq 0$. They therefore preserve the original convex endpoint check on each closed noise interval. Every phase cell is confined to one half-line; zero is a shared closed endpoint. The count displacement bound D comes from the same actual $h - u$ cell, and the Stirling constants continue to come from the already proved global count segment.

The other all-index steps are unchanged: the event bracket is checked negative before substituting a binomial mass lower bound, and the entire inverse-n correction is checked positive before replacing n by N. The inverse correction is $I(\lambda) = K + A + B/\lambda$, hence decreases with λ; positivity at both endpoints (in particular the upper endpoint) licenses the all-index step throughout the interval. The final score is convex in λ: its event fraction is a positive constant divided by an affine positive function, its B/λ coefficient is nonnegative, and its remaining terms are affine or constant. These are independent gates in the exact kernel. The formulas apply to both signs of h without reflecting the law or changing the count during integration. They also supply an improved upper derivative for the negative-phase MID inequalities.

Full-phase MID below the whole-count threshold. For $\lambda \in [.05, .08]$, the selected shifted-quartic MID comparisons are checked on every closed phase interval of length .001 in $[0, 1/2]$, for both phase signs. The shifted-quartic polynomial and nearest-count identity are defined at phase zero, so the two closed covers include $u = 0$ directly. This permits the whole-count LOW comparison to start at $\lambda = .08$. The four MID bands use phase starts $0, 0, .19, .27$ at noise bands $[.05, .06]$, $[.06, .08]$, $[.08, .10]$, and $[.10, .12]$, respectively.

Positive allowances and finite inverse-index corrections. For actual index 39 put $K = 741/4000$. The target is $C < C_B + K/39$, so the scalar tests differ from the zero-allowance tests in Section 7.5. All of them follow from the same identity (99). Write $b = \sqrt{39}\pi_k$ and retain the verified two-sided bound $m \leq b \leq 1$. A nearest-count bracket D may now have either sign; consequently

$$bD \geq \min\{mD, D\}.$$

If the repair is at most $r\lambda/39$, with $r \geq 0$, a noise band $[a, b_\lambda]$ is excluded by the intermediate-noise test

$$\min\{mD, D\} - rb_\lambda/39 + K/39 > 0.$$

Indeed the identity gives $\rho_B(C - C_B) < K/39$. The elementary inequality $\rho_B \geq 1$ follows from $(1-2v)^2 - v = (1-v)(1-4v) \geq 0$, $0 < v \leq 1/4$. Thus $C < C_B + K/39 \leq C_{\mathrm{E}} + K/39$.

At arbitrarily small positive noise the rescaled quartic and tail estimates give the same inequality per unit λ. The tiny-noise test adds the allowance

$$\frac{K}{39\max\{.05, \Lambda\}}$$

to that per-noise reserve. Multiplying by $0 < \lambda \leq \min\{.05, \Lambda\}$ again gives a bound strictly below $K/39$. No positive lower noise cutoff is inserted.

For the whole-count test, write

$$S(\lambda) = \max\{mB(\lambda), B(\lambda)\} + I(\lambda)/39.$$

Both the bracket and the inverse correction may have either sign; the first maximum uses $m \le b \le 1$, while the division by 39 uses the actual integer. Hence $\rho_B(C - C_B) \le \lambda S(\lambda)$. The saved gate on $[a, b_\lambda]$ is

$$K\rho_{B,l}/39 - b_\lambda S(\lambda) > 0,$$

where $\rho_{B,l}$ is a positive lower bound for ρ_B. If $S < 0$ the conclusion is immediate; otherwise it follows by $\lambda \le b_\lambda$ and $\rho_B \ge \rho_{B,l}$. The formula is convex in positive λ, since its only nonaffine terms are positive multiples of $(\alpha\lambda + \beta)^{-1}$ and $1/\lambda$, and maxima of convex functions preserve convexity. Both endpoint gates therefore cover the whole noise cell.

At zero noise, the strict finite-binomial theorem gives $C < C_{\mathrm{E}}$, so every positive allowance is excluded there as well. For a bounded integer band $N \le n \le L$, a signed inverse correction has the valid bound $I/n \le \max(I/N, I/L)$. If $b = \sqrt{n}\pi_k \in [b_0, 1]$, a signed bracket satisfies $bJ \le \max(b_0 J, J)$ and $bJ \ge \min(b_0 J, J)$. These two-sided rules replace the sign restrictions needed by the infinite-index test. The actual index remains in every binomial slack calculation. With a uniform binomial reserve $\epsilon_p > 0$ and an upper bound S on $\rho_B(C - C_B)/\lambda$, it is enough to verify $\rho_{B,l}\epsilon_p > U \max(S, 0)$ on a noise cell $\lambda \le U$. No negative inverse correction is enlarged by replacing its denominator with the lower index.

The local exclusion, including equality.

Proposition 7.26 (Finite local exclusion)**.** *For an integer* $42 \le n \le 48$*, no attained positive ratio maximum satisfies*

$$0 < n(C_n - C_{\mathrm{E}}) \le 54/100, \qquad n(C_{n-1} - C_n)_+ \le 19/100,$$
$$C_n < 4226/10000, \qquad C_{n-1}, C_{n-2} < 4253/10000.$$

Proof. Apply the support polynomial with $(N, L, K, q, R) = (42, 48, 0, .19, 3.84)$. In particular its favorable fifth-order term is evaluated at 48; the infinite-range support row cannot be substituted for this finite band. The subsequent bounded-threshold tests use excess allowance .54, loss .19 and own-score cap .4226. The resource invariant is $\mathcal{R}_n \le r$, $nD \le 42d$. The transfer argument of Section 7.2 applies at floor 42, including the signed curvature, retained constant penalties and actual-noise triangle. Confinement uses the already acquired strips and the substitutions in Section 7.4, with deletion floor 40. The left radius first contracts to 3.025; contact then proves the coordinate strip $(-.575, 1.575)$ before confinement is invoked. On the four noise bands $[.05, .08]$, $[.08, .10]$, $[.10, .12]$ and $[.12, \Lambda]$, the finite central amplitude ceilings are $.375, .325, .275, .25$, and the left ceilings are $.40, .35, .30, .275$. Each ceiling is intersected with its row's independently acquired global amplitude. These finite profiles use the same proved confinement gates with their own parameters; the uniform-tail profiles are not substituted for them.

The accepted acquisition cover has 184 closed cells: 92 exclusions, 45 left rows covering $[.29, .38]$ and 47 central rows covering $[.38, .474]$. Thus the moment and probability complements are excluded before the comparison. Every row retains its own resource and both threshold signs. The central positive-noise cover uses TINY on $(0, .05]$; above .05 the whole-count phase radii are .09,.14,.19,.29,.5 on bands separated by .06,.08,.10,.12, with the complementary nearest-count phases through .12. The left cover uses TINY on $(0, .05]$, full-phase MID on $[.05, .08]$, then nearest count outside and whole count inside phase radii .19 and .27 on $[.08, .10]$ and $[.10, .12]$; above .12 the whole-count test covers every phase. Each

band is intersected with its actual acquired noise ceiling. The 78 central and 74 left signed whole-count rows, and the nearest-count covers, include every seam.

The signed cubic multipliers, noise-dependent inward bound, localized binomial mass, 4553/1000 harmonic inequality and signed Gaussian correction proved above give the strict comparison on these entire closed cells. The two complete LOW families retain their inverse-index sign gates before any floor substitution. Zero noise is Bernoulli. The strict local comparisons and Schulz's bound contradict $C_n > C_{\mathrm{E}}$. This proposition has the stated upper index 48; it is separate from the floor-40 comparison used only at actual indices $40, 41, 42$. □

Proposition 7.27. *There is no attained positive closed-event ratio maximizer satisfying*

$$n \geq 47, \qquad 0 \leq n(C_n - C_{\mathrm{E}}) \leq 7/10, \qquad n(C_{n-1} - C_n)_+ \leq 19/100,$$
$$C_n < 4247/10000, \qquad C_{n-1}, C_{n-2} < 4253/10000.$$

Proof. The support, acquisition and confinement implications proved above and their accepted closed numerical covers place the same law in one of the 45 central or 38 left rows. Their six local tests cover every positive noise, both signs of H and every phase, and give $\rho_B(C_n - C_B) < -.001\lambda$. Schulz's theorem gives $C_B < C_{\mathrm{E}}$. For zero noise the law is exactly Bernoulli, and the same finite-binomial theorem gives a strict bound at every threshold. Both cases contradict $C_n \geq C_{\mathrm{E}}$. The initial score interval is $[.4095, .4247]$, the native barrier is zero, and the resource identity uses $e = C_n - C_{\mathrm{E}}$ only as a nonnegative summand. Strict numerical reserves are independent of $e > 0$. The stored value 98 in initialization occurs only in $.4095 + K/98$ with $K = 0$; the support's last band and all later monotonicity gates have no upper index limit. □

Corollary 7.28. *Once $C_j \leq C_{\mathrm{E}}$ has been proved at every index, $C_n < C_{\mathrm{E}}$ for every $n \geq 47$, in particular for every $n \geq 700$.*

Proof. If $C_n = C_{\mathrm{E}}$, ratio attainment supplies the positive closed-event maximum. The predecessor loss and scaled excess are zero, and both deleted bounds are at most C_{E}. All hypotheses of the proposition hold. Attainment is essential here: strictness for each individual law alone would not rule out a nonattained supremum. □

8 Completion of the proof

We now put together the finite induction and the two local exclusions. Each exclusion concerns one attained ratio maximizer. The induction supplies its predecessor hypotheses before any acquisition or comparison is applied to that law.

Proof of Theorem 1.1. The finite induction, Theorem 6.12, gives $C_j \leq C_{\mathrm{E}}$ for $1 \leq j \leq 42$. If a later strict failure exists, choose its least index n. Both deleted constants are then at most C_{E}, and Corollary 3.3 yields

$$0 < n(C_n - C_{\mathrm{E}}) < \frac{3003}{10250} < .3, \qquad C_n < \frac{73}{175} < .4226, \qquad n(C_{n-1} - C_n)_+ = 0.$$

Also $C_{n-1}, C_{n-2} \leq C_{\mathrm{E}} < .4253$. For $43 \leq n \leq 48$, these are the hypotheses of the finite local exclusion, Proposition 7.26. For $n \geq 49$ they are the hypotheses of the uniform exclusion, Proposition 7.27. In each case attainment supplies the maximizer and the local comparison rules it out. Thus $C_n \leq C_{\mathrm{E}}$ at every index.

Now suppose $C_n = C_{\mathrm{E}}$ for some $n \geq 47$. The just-proved upper bound makes the two deleted constants at most C_{E} and the predecessor loss zero. The scaled excess is zero as well. The uniform local exclusion includes this boundary case, so attainment again gives a

contradiction. This proves the strict assertion about the supremum. Finally, Proposition 1.3 supplies a fixed Bernoulli law whose normalized discrepancies converge to C_{E}. Hence every smaller universal coefficient fails. □

There is no assumption of a sharp tail in this induction. The numerical finite base uses previously established smaller indices and its stated loose ceilings. The finite local exclusion is proved at its actual indices, and the uniform exclusion uses analytic estimates valid for every index above its floor. Their certificates establish numerical inequalities on those independently justified domains. This order is also why equality is considered only after the weak all-index bound has been proved.

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